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Proof-of-Collaboration: A Novel Consensus Algorithm with Highly Efficient Power Consumption

إثبات التعاون: خوارزمية توافق جديدة ذات كفاءة عالية في استهلاك الطاقة

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Abstract

Blockchain systems secured by Proof-of-Work (PoW) provide open participation and well-studied security properties, but continuous competitive block-header hashing can consume substantial energy. This thesis proposes Proof-of-Collaboration (PoCol), a coordinated proof-of-work protocol that retains a hash-target puzzle and cumulative-work chain structure while coordinating candidate-header evaluation. In each PoCol mining round, participating miners use one immutable canonical block template and one nonce-free header template. Candidate headers are instantiated only by varying the nonce field. A deterministic seed-based procedure assigns mutually disjoint portions of the permitted nonce domain to registered miners. Under the common-template and honest disjoint-assignment assumptions, no two honest miners evaluate the same serialized candidate-header input within that round. The protocol specification also defines miner registration and leaving, inactivity handling, a proposed collaborative reward-distribution mechanism, and verification of the successful miner's range membership. Open participation, fairness, incentive compatibility, Sybil resistance, and PoW-equivalent chain security remain design objectives rather than experimentally established guarantees.

Chapter 5 provides the complete protocol specification for the common-template PoCol design. The empirical implementation, however, is intentionally narrower: an idealized discrete-event abstraction of collaborative search, disjoint nonce allocation, and operating-policy behavior was implemented in an extended BlockSim simulator with explicit work and energy instrumentation.

In the experimental design, 20 logical miners were used, with 20% designated as reserve miners, resulting in 4 reserve miners and 16 active primary miners. This configuration was selected because it provides sufficient reserve capacity to exercise standby, activation, reassignment, and handoff mechanisms while allowing the 1,600-nonce domain to be divided equally among the 16 active miners. Each active miner therefore receives an initial range of 100 nonce values, corresponding to four batches of 25 nonces. The best throughput-oriented PoCol policy retained 90.39% of the accepted-block output of the no-floor PoCol control while achieving a 55.05% calculated energy reduction relative to the within-run power-null reference. In the matched same-template comparison, PoCol also reduced modeled total energy relative to the active-capacity-matched PoW control; however, this reduction was accompanied by lower accepted-block production and substantially greater round duration. The results therefore demonstrate an energy-service trade-off under the evaluated configurations.

Keywords: Blockchain, Consensus Algorithms, Proof of Work, Proof of Collaboration, Coordinated Mining, Energy-Aware Consensus, Sustainability, BlockSim.

الملخص

توفر أنظمة البلوكشين المؤمنة بواسطة إثبات العمل (PoW) مشاركة مفتوحة وخصائص أمنية مدروسة جيدًا، لكن التجزئة التنافسية المستمرة لرؤوس الكتل قد تستهلك طاقة كبيرة. تقترح هذه الأطروحة إثبات التعاون(PoCol) وهو بروتوكول منسق لإثبات العمل يحافظ على لغز هدف التجزئة وهيكل سلسلة العمل التراكمي، مع تنسيق تقييم رؤوس الكتل المرشحة. في كل جولة تعدين باستخدام بروتوكول PoCol، يستخدم المُعدِّنون المشاركون قالب كتلة قياسيًا واحدًا غير قابل للتغيير وقالب رأس خاليًا من قيمة الـ nonce ويتم إنشاء نسخ من رؤوس الكتل المرشحة فقط عن طريق تغيير حقل الرقم العشوائي. ويقوم إجراء حتمي قائم على البذرة بتخصيص أجزاء متباينة تمامًا من نطاق الرقم العشوائي المسموح به للمعدنين المسجلين. وبناءً على افتراضات القالب المشترك والتخصيص المتباين النزيه، لا يقوم أي اثنين من المعدنين النزيهين بتقييم نفس المدخلات المسلسلة لرؤوس الكتل المرشحة ضمن تلك الجولة. تحدد مواصفات البروتوكول أيضًا تسجيل المُعدِّنين وخروجهم، والتعامل مع عدم النشاط، وآلية مقترحة لتوزيع المكافآت التعاونية، والتحقق من عضوية المُعدِّن الناجح في النطاق. تظل المشاركة المفتوحة، والإنصاف، وتوافق الحوافز، ومقاومة هجمات Sybil وأمن السلسلة المكافئ لـ PoW أهدافًا تصميميةً وليس ضمانات ثابتة تجريبيًّا.

يقدم الفصل 5 المواصفات الكاملة للبروتوكول لتصميم PoCol القائم على القالب المشترك. ومع ذلك، فإن التنفيذ التجريبي أضيق نطاقًا عن قصد: فقد تم تنفيذ تجريد مثالي للأحداث المنفصلة للبحث التعاوني، وتخصيص أرقام عشوائية غير متداخلة، وسلوك سياسة التشغيل في محاكي BlockSim الموسع مع أدوات قياس صريحة للعمل والطاقة.

في التصميم التجريبي، استُخدم 20 مُعدِّنًا منطقيًا، خُصص 20% منهم كمُعدِّنين احتياطيين، أي 4 مُعدِّنين احتياطيين و16 مُعدِّنًا أساسيًا نشطًا. وقد اختير هذا التكوين لأنه يوفر عددًا كافيًا من المُعدِّنين الاحتياطيين لاختبار آليات الاستعداد والتفعيل وإعادة التخصيص والتسليم، وفي الوقت نفسه يسمح بتقسيم نطاق مكوّن من 1,600 قيمة للـ nonce بالتساوي بين المُعدِّنين النشطين الستة عشر، بحيث يحصل كل مُعدِّن على نطاق أولي مكوّن من 100 قيمة، أي أربع دفعات من 25 قيمة. وقد احتفظت أفضل سياسة PoCol الموجهة نحو الإنتاجية بنسبة 90.39% من إنتاج الكتل المقبولة مقارنةً بنظام PoCol الضابط من دون حد أدنى، مع تحقيق انخفاض محسوب في استهلاك الطاقة بنسبة 55.05% مقارنةً بالمرجع الداخلي لنفس مسار التشغيل الذي يفترض عدم الاستفادة من حالات القدرة المنخفضة (within-run power-null reference). وفي المقارنة المتطابقة القائمة على القالب نفسه، خفّض PoCol أيضًا إجمالي استهلاك الطاقة المحسوب مقارنةً بضابط PoW المتطابق في القدرة النشطة؛ إلا أن هذا الانخفاض ترافق مع انخفاض في إنتاج الكتل المقبولة وزيادة كبيرة في زمن الجولة. ولذلك تُظهر النتائج مقايضةً بين استهلاك الطاقة ومستوى الخدمة ضمن التكوينات التي جرى تقييمها.

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List of Abbreviations

AI	Artificial Intelligence
ASIC	Application-Specific Integrated Circuit
BFT	Byzantine Fault Tolerance
CBNSI	Cambridge Blockchain Network Sustainability Index
CO_2	Carbon Dioxide
DAG	Directed Acyclic Graph
DLBC	Deep Learning-Based Consensus
DPoS	Delegated Proof of Stake
EVM	Ethereum Virtual Machine
FL	Federated Learning
IoT	Internet of Things
PBFT	Practical Byzantine Fault Tolerance
P2P	Peer-to-Peer
PoA	Proof of Authority
PoB	Proof of Burn
PoC	Proof of Contribution
PoCol	Proof of Collaboration
PoET	Proof of Elapsed Time
PoFL	Proof of Federated Learning
PoH	Proof of History
PoR	Proof of Reputation
PoS	Proof of Stake
PoSpace	Proof of Space
PoW	Proof of Work
PPT	Probabilistic Polynomial Time
PUE	Power Usage Effectiveness
RQ	Research Question
SHA-256	Secure Hash Algorithm 256-bit
TEE	Trusted Execution Environment
TPS	Transactions per Second
UTXO	Unspent Transaction Output
VPP	Virtual Power Plant
VRF	Verifiable Random Function

Chapter 1
Introduction

Chapter 1
Introduction

1.1. Background

Initially designed by Nakamoto as a peer-to-peer electronic-payment system, blockchain technology is now used in banking, supply-chain management, e-government, healthcare, and education. These applications rely on decentralized, tamper-evident ledgers and auditable state transitions. Representative studies include circular supply chains (Huang et al., 2022), electronic voting (Farooq et al., 2022), real-estate smart contracts (Ullah & Al-Turjman, 2023), and secure blockchain–IoT health-record monitoring (Alam et al., 2023). Blockchain technology is also used in energy systems and virtual power plants for coordination, trading, and traceability.

Consensus protocols are the mechanisms by which distributed participants establish an accepted ordering of state transitions under a defined fault and adversary model. Blockchain consensus draws on both classical distributed-systems research, including Byzantine fault tolerance (Castro & Liskov, 1999; Lamport et al., 1982), and resource- or stake-based mechanisms developed for open or economically governed networks. Representative families include Proof-of-Work (PoW), Proof-of-Stake (PoS), Delegated Proof-of-Stake (DPoS), Proof-of-Authority (PoA), Proof-of-Elapsed-Time (PoET), storage-based proofs, and contribution- or reputation-based designs. These families differ in trust assumptions, participation rules, finality, communication cost, hardware dependence, governance, and energy demand (Lashkari & Musilek, 2021).

Consensus algorithms, initially developed for the cryptocurrency sector, are now being repurposed to address challenges in smart cities, power markets, renewable-energy distribution, traceability, and data-intensive applications. Numerous studies report that the substantial energy consumption of Proof of Work (PoW) blockchains affects operational sustainability (Cambridge Centre for Alternative Finance, n.d.; Zhang et al., 2023). Comparable historical findings were reported for Ethereum before its transition to Proof of Stake (Cortes-Goicoechea et al., 2021; McDonald, 2021). Researchers have therefore investigated efficient SHA-256 hardware (Koutra & Tenentes, 2024; Pham et al., 2020), energy-aware policies such as Green-PoW (Lasla et al., 2022), useful-work approaches including Proof-of-Learning (Bravo-Marquez et al., 2019), DLBC (Li et al., 2019), and PoFL (Qu et al., 2021), and lower-computation consensus mechanisms (Platt et al., 2021; Sapra et al., 2023). This thesis introduces PoCol to coordinate candidate-header evaluation through one immutable template and disjoint nonce ranges. The design eliminates exact duplicate inputs under those assumptions and enables lower calculated energy only when an explicit operating policy creates lower-power residence; fairness, decentralization, and security equivalence remain objectives for further validation.

1.2. What the Problem Is and Why It Exists

Proof-of-Work creates an important sustainability challenge because security is obtained through sustained computational expenditure in an open competitive environment. Bitcoin demonstrates that a permissionless hash-based ledger can operate for long periods under adversarial conditions, but its security and performance depend on assumptions about adversarial hash power, network propagation, miner incentives, and chain-selection behavior (Garay et al., 2015; Gervais et al., 2016; Nakamoto, 2008). Network-scale electricity use is correspondingly substantial and depends on aggregate hash rate, mining-hardware efficiency, market conditions, and the electricity mix (Cambridge Centre for Alternative Finance, n.d.; de Vries, 2018; Zhang et al., 2023).

Alternative consensus mechanisms reduce or replace continuous PoW hashing, but they introduce different trust, governance, hardware, and economic assumptions. PoS and DPoS replace hash-rate-based leader competition with stake- or delegate-based selection (Bachani & Bhattacharjya, 2022; BitShares, n.d.; Ge et al., 2022; King & Nadal, 2012); PoA uses governed validator identities (Barinov et al., 2018; Manolache et al., 2022); PoET relies on trusted execution environments (Hyperledger Sawtooth (Intel), n.d.); PoSpace uses storage commitments (Dziembowski et al., 2015); Proof of Burn uses irreversible economic sacrifice (Karantias et al., 2020); Proof of Contribution and Proof of Reputation use contribution- or reputation-based selection rules (Xue et al., 2018; Zhuang et al., 2019). These mechanisms should therefore be compared as distinct design families rather than treated as interchangeable low-energy substitutes for PoW.

Research has also explored energy-aware and useful-work approaches. Green-PoW reduces active mining under selected conditions (Lasla et al., 2022); Proof-of-Learning, DLBC, and Proof of Federated Learning redirect or couple computation to machine-learning tasks (Bravo-Marquez et al., 2019; Li et al., 2019; Qu et al., 2021); and hardware research improves the efficiency of SHA-256 mining itself (Koutra & Tenentes, 2024; Pham et al., 2020). These approaches address different subsets of the sustainability, security, performance, and governance design space. The literature does not support a single mechanism as simultaneously optimal across all four dimensions.

Against this background, the thesis investigates a narrower question: whether coordinated evaluation of one immutable candidate-header template can eliminate exact duplicate serialized inputs and whether explicit post-range operating policies can reduce modeled energy while retaining useful block production. PoCol does not assume that nonce partitioning alone saves energy, nor does this thesis claim formal equivalence with deployed PoW in security, decentralization, or incentives. Those properties are treated as design objectives and limitations requiring separate formal and empirical validation.

1.3. Questions and Goals for the Research

This thesis centers on the design, implementation, and bounded evaluation of Proof-of-Collaboration (PoCol). The empirical objective is to quantify exact-input duplication, calculated energy, physical work, block production, round latency, and the operational costs of floor management and reassignment under clearly identified simulator controls.

The research is guided by the following four questions:

RQ1: Does coordinated disjoint nonce evaluation eliminate exact-input duplication under one common immutable template?

RQ2: How much calculated energy reduction is associated with low-power-state residency within PoCol?

RQ3: How much block production is retained relative to a no-floor PoCol control?

RQ4: How does PoCol compare with a matched same-template PoW control in energy, physical work, block production and round latency?

The thesis seeks to accomplish the following specific objectives to address these inquiries:

1. O1: To conduct a structured comparative evaluation of representative blockchain consensus mechanisms with respect to functionality, security assumptions, decentralization characteristics, and energy implications (Lashkari & Musilek, 2021; Platt et al., 2021).
2. O2: To specify a coordinated Proof-of-Work protocol in which miners derive a common immutable template, receive deterministic disjoint nonce assignments, and use a pre-committed collaborative reward rule, while explicitly identifying unresolved contribution-verification, Sybil, and incentive requirements (Lasla et al., 2022; Nakamoto, 2008).
3. O3: To implement reproducible PoCol and matched-control simulation paths and evaluate three evidence layers: macro-scale fixed-horizon accounting experiment, within-PoCol operating-policy experiment, and matched same-template PoW comparison. Quantitative reward fairness, decentralization, complete distributed template agreement, and formal chain-security equivalence remain outside the completed empirical evaluation (Alharby & van Moorsel, 2019; Paulavičius et al., 2021).

This thesis focuses on the consensus and block-production layers of public or permissionless blockchain systems, in which any node may join, propose blocks, and participate in consensus subject to the protocol rules. Smart-contract languages, application-layer protocols, and detailed token-economic design are outside its scope. Case studies and motivating examples may derive from areas including financial transactions (Nakamoto, 2008; Truby, 2018), supply chains (Hu et al., 2019; Huang et al., 2022), e-voting (Farooq et al., 2022), real estate management (Ullah & Al-Turjman, 2023), education (Smolenski, 2021), and energy systems (Ge et al., 2024; Taherdoost, 2024a), however, the PoCol design aims to be universal and application-agnostic.

1.4. Summary of the Research Methodology

The research methodology combines a structured comparative review, protocol specification, analytical modeling, deterministic discrete-event simulation, and preregistered paired experiments. The methodology was deliberately revised during the research so that work-efficiency claims, electrical-energy claims, service outcomes, and unvalidated security or incentive objectives are reported separately.

The first phase conducted a structured comparative review of blockchain consensus mechanisms, energy-aware variants, and representative sustainability studies (Lashkari & Musilek, 2021; Sapra et al., 2023). The review provides qualitative evidence synthesis across heterogeneous protocol families. (Page, McKenzie, et al., 2021; Page, Moher, et al., 2021).

The second phase specified the PoCol protocol, including its threat model, common-template invariant, deterministic nonce-domain allocation, round lifecycle, block verification, reward-distribution rule, and energy-accounting boundaries. Security, liveness, Sybil resistance, contribution verification, and incentive compatibility are discussed as preliminary analytical objectives rather than proved properties (Garay et al., 2015; Pass et al., 2017).

The third phase implemented and validated three complementary simulation paths. First, the macro-scale matrix comprised 63 experimental configurations across 30 master seeds, giving 1,890 physical executions at a fixed 10,000-second horizon. Two preregistered reduced-scale experiments then evaluated operating policies with 20 logical miners over 300 seconds: the second used four PoCol scenarios and 12 paired seeds, whereas the third used five scenarios and 12 paired seeds, including two same-template PoW controls. The later target-coupled engine evaluates SHA-256 (header||nonce) against a fixed target and records physical, unique, duplicate, and post-round evaluations.

The methodology combines protocol specification, deterministic simulation, and preregistered statistical analysis. Hardware measurement, production deployment, full distributed template agreement, formal chain-security proofs, and incentive validation remain future work.

1.5. Important Contributions

The contributions of this thesis are organized as conceptual, methodological, and empirical contributions, with explicit boundaries between the protocol specification and the subset implemented in the simulator:

1. C1: A structured comparative examination of representative blockchain consensus mechanisms across functionality, security assumptions, deployment context, and energy implications. The review is intended as decision-oriented qualitative synthesis rather than a universal quantitative ranking.
2. C2: A coordinated Proof-of-Collaboration protocol specification that integrates one immutable canonical template, deterministic disjoint nonce ownership, locally verifiable range assignment, collaborative reward rules, and explicit post-range operating policies. Under honest common-template execution, the disjoint ranges eliminate exact serialized candidate-header duplication.
3. C3: An analytical and measurement framework that separates physical work, unique and duplicate evaluations, wall-clock power-state energy, accepted-block service, and round latency.
4. C4: A reproducible simulation testbed comprising the macro-scale accounting matrix, the within-PoCol policy experiment, and the matched same-template PoW comparison. The empirical contribution quantifies the energy/work-efficiency versus service/latency trade-off, preserves the outcomes of preregistered decision criteria, including criteria that were not simultaneously met, and records matched comparators and integrity checks. The evidence is simulation-based and is not a production deployment, hardware measurement, fairness experiment, or formal security proof.

1.6. The Structure of the Thesis

The remainder of the thesis is organized as follows.

Chapter 2 introduces the technical foundations required for the study, including blockchain architecture, cryptographic primitives, network models, Byzantine fault tolerance, Nakamoto-style Proof of Work, alternative consensus mechanisms, and the distinction between power and energy metrics.

Chapter 3 presents a structured comparative literature review of blockchain consensus mechanisms, with particular attention to functionality, security assumptions, energy consumption, sustainability, and directly related collaborative Proof-of-Work approaches.

Chapter 4 defines the system model, adversary assumptions, performance and energy metrics, design goals, and formal problem formulation for Proof of Collaboration (PoCol).

Chapter 5 specifies the PoCol consensus protocol, including canonical block-template construction, deterministic nonce-domain allocation, collaborative mining, miner-set management, reward distribution, and preliminary security and robustness analysis. It also compares PoCol conceptually with representative consensus mechanisms such as PoW.

Chapter 6 describes the BlockSim-based prototype, the idealized PoCol abstraction implemented for evaluation, the experimental design, parameter settings, data-collection pipeline, reproducibility procedure, and threats to validity.

Chapter 7 reports the authoritative stages work, energy, service, and operating-policy results. Numerical comparisons with PoS and PBFT are not claimed because no matched implementations of those protocols were executed.

Chapter 8 summarizes the findings and contributions, discusses the extent to which the research questions were addressed, identifies practical limitations, and outlines directions for future work.

Chapter 2
Technical Background

Chapter 2
Technical Background

2.1. Blockchain Fundamentals

A blockchain is a distributed ledger in which participating nodes maintain and validate an ordered record of state transitions according to a consensus protocol (Lashkari & Musilek, 2021; Nakamoto, 2008). Rather than relying on a single database administrator, blockchain systems distribute validation authority among participating nodes, and accepted state changes are governed by protocol rules (Lashkari & Musilek, 2021; Sunny et al., 2022). Cryptographic linking and replicated validation make previously accepted records tamper-evident, although the degree of immutability depends on the consensus model and its security assumptions. Blockchain technology has been applied in financial services, supply-chain management, electronic voting, real estate, healthcare, energy systems, and education (Sunny et al., 2022; Taherdoost, 2024a). This section introduces the blockchain concepts required for the subsequent analysis of consensus mechanisms and energy consumption.

2.1.1. Distributed ledger and blocks

Blockchain systems organize validated state transitions in an append-only ledger whose integrity is protected by cryptographic commitments and consensus rules. In conventional linear blockchains, transactions are grouped into blocks, and each block header commits to the preceding block and to consensus-relevant metadata, such as the previous-block hash, timestamp, transaction commitment, and difficulty-related fields (Lamport et al., 1982; Nakamoto, 2008). Modifying an accepted historical block changes its cryptographic digest and invalidates subsequent links unless the attacker can also replace the affected chain according to the protocol's consensus rule. Node responsibilities differ: archival nodes may retain the complete history, full validating nodes verify consensus rules and maintain the state required for validation, validators or miners participate in block production, and light clients verify selected claims using block headers and cryptographic proofs. Bitcoin represents ledger state through unspent transaction outputs (UTXOs), where transactions consume previously unspent outputs and create new outputs assigned to recipients (Christidis & Devetsikiotis, 2016; Del Monte et al., 2020). Permissioned systems may instead authorize a known validator set and employ Byzantine-fault-tolerant protocols such as PBFT and its variants (Castro & Liskov, 1999; Na et al., 2022; Putz et al., 2019). Many blockchain systems can be modelled as replicated state machines: correct replicas apply the same agreed sequence of valid transactions and consequently reach the same state (Castro & Liskov, 1999; Lamport et al., 1982). Ledger structures may be linear, Directed Acyclic Graph (DAG)-based, sharded, or consortium-oriented, and these designs make different trade-offs in throughput, latency, fault tolerance, storage, and decentralization (Del Monte et al., 2020; Lashkari & Musilek, 2021).

2.1.2. Network model and nodes

Blockchain data are disseminated over a peer-to-peer overlay, commonly through gossip-based propagation of transactions and blocks (Lashkari & Musilek, 2021; Nakamoto, 2008). Full validating nodes verify consensus rules and maintain the state required to validate new transactions and blocks, mining or validator nodes additionally participate in block production, and light clients use block headers and cryptographic proofs without storing or validating the complete ledger. The network is commonly modelled as partially synchronous: after an unknown global stabilization time, message delays are bounded, although the bound itself need not be known in advance (Castro & Liskov, 1999; Lamport et al., 1982). Bitcoin is a permissionless system in which participation is open to nodes that follow the protocol and meet the resource requirements of block production (Nakamoto, 2008; Sapra et al., 2023). By contrast, permissioned systems such as Hyperledger Fabric and consortium networks restrict validation to approved organizations or identities (Barinov et al., 2018; Manolache et al., 2022). Other designs weight participation by stake, organizational authority, reputation, or hybrid combinations of these factors (Bano et al., 2019; Barinov et al., 2018; Kleinrock et al., 2020; OpenEthereum, n.d.; Zhuang et al., 2019).

Related abstractions have also been applied to energy and cyber-physical systems. Xu et al. proposed a blockchain-based decentralized scheduling model in which distributed participants coordinate flexible loads and renewable generation through a shared ledger (Xu et al., 2023). Li et al. examined blockchain-based management and trading of power-data assets (Li et al., 2023). Other proposals use distributed ledgers for renewable-energy traceability, peer-to-peer energy trading, and profit distribution in virtual power plants (Machado et al., 2024). These applications typically rely on permissioned or consortium networks with explicit trust assumptions, bounded participant sets, and application-specific requirements for network latency, privacy, and governance.

2.1.3. Transactions and Smart Contracts

Transactions encode authorized state changes. In UTXO-based systems, a transaction consumes one or more unspent outputs and creates new outputs that may be spent by their designated owners. Account-based systems update balances or contract state associated with accounts. Before inclusion in a block, a transaction must satisfy syntactic validity rules, authorization requirements such as digital signatures, and any protocol-specific state-transition constraints.

Smart contracts are programs deployed on a blockchain platform and executed deterministically by validating nodes when transactions invoke their functions. Some platforms support Turing-complete languages, whereas others deliberately restrict expressiveness to simplify verification or reduce risk. Contract code and consensus-critical state are committed to the ledger, while application data may also be stored or referenced off-chain depending on the platform. Smart contracts can reduce reliance on centralized intermediaries for processes whose rules and inputs can be represented and validated on-chain, but they do not eliminate governance, legal, or operational dependencies. Prior studies apply smart contracts to circular supply chains, electronic voting, real-estate transactions, smart-city services, agriculture, healthcare, and digital credentials. In these applications, transactions invoke contract logic, and the resulting state updates are replicated across validating nodes.

2.2. Consensus Mechanisms in Distributed Systems

Distributed systems use consensus protocols so that non-faulty participants can agree on an ordered history despite failures and, depending on the model, adversarial behavior (Castro & Liskov, 1999; Lamport et al., 1982). In blockchain systems, the consensus design influences validation authority, fault tolerance, finality, throughput, communication overhead, governance, and resource consumption. This section summarizes classical Byzantine-fault-tolerant protocols, Nakamoto-style Proof-of-Work, Proof-of-Stake families, and selected alternatives relevant to the thesis.

2.2.1. Classical consensus (BFT)

Classical Byzantine-fault-tolerance research predates blockchain and typically assumes a known replica set together with authenticated communication. The Byzantine Generals Problem formalizes agreement in the presence of participants that may fail or behave arbitrarily (Lamport et al., 1982). These models provide the foundation for later replicated-state-machine protocols and many permissioned blockchain designs.

Practical Byzantine Fault Tolerance (PBFT), introduced by Castro and Liskov, provides state-machine replication that tolerates fewer than one-third Byzantine replicas in the standard $3f + 1$ setting. Its normal-case protocol uses pre-prepare, prepare, and commit phases, with view change to replace a faulty primary. Safety follows from quorum intersection under the Byzantine fault bound, whereas liveness additionally requires suitable network-timing assumptions (Castro & Liskov, 1999). Subsequent work has extended PBFT with proactive recovery and alternative structures for permissioned deployments (Castro & Liskov, 2002; Na et al., 2022).

BFT-style protocols have been adapted to many domain-specific settings, including IoT and consortium systems (Mišić et al., 2024; Xu et al., 2021; Zhang et al., 2023). Their attraction is deterministic agreement under explicit fault bounds, but classic PBFT-style all-to-all communication grows quadratically with the replica count in the normal case, which limits practical scaling of a single committee. Modern BFT research therefore focuses heavily on reducing communication, parallelizing ordering, or structuring validator participation (Hu et al., 2025; Lyu et al., 2025; Xu et al., 2024).

2.2.2. Nakamoto-style consensus (Proof-of-Work)

Nakamoto's Bitcoin protocol introduced a permissionless consensus design in which miners repeatedly hash serialized candidate block headers and seek a digest below a target (Nakamoto, 2008). The 32-bit nonce is only one mutable field; miners can generate additional candidate headers by changing the coinbase extra nonce, the Merkle root, timestamps within permitted bounds, and other allowed header fields. A valid proof of work is therefore a property of the complete serialized header, not of a nonce value in isolation. Chain selection is based on cumulative proof of work, and rollback risk decreases with confirmation depth under the protocol's assumptions, although attacks such as selfish mining and double spending remain possible and the precise security threshold depends on the network and adversary model (Garay et al., 2015; Gervais et al., 2016; Nakamoto, 2008).

PoW offers permissionless entry without requiring a pre-existing identity or stake position, but sustained hash computation creates substantial operating energy demand. Network energy use is determined by aggregate mining activity, hardware efficiency, operating time, and infrastructure overhead rather than by a nonce count alone (de Vries, 2018; Koutra & Tenentes, 2024; Pham et al., 2020). This distinction is central to the thesis: eliminating duplicate candidate inputs can improve work coverage, but it does not by itself reduce electrical energy. Energy reduction occurs only when active power, active duration, or active participation is reduced.

2.2.3. Proof-of-Stake and its variants (DPoS, etc.)

Proof of Stake (PoS) replaces hash-rate-based Sybil resistance with economic stake. Early systems such as PPCoin selected validators according to stake-related rules, while later PoS protocols added randomized committee selection, slashing, checkpointing or finality mechanisms, and staking pools (Ge et al., 2022; King & Nadal, 2012). PoS protocols generally require far less continuous computation than PoW, but their security and liveness depend on protocol-specific assumptions about stake distribution, validator behavior, network timing, and recovery from long-range threats. Delegated Proof of Stake (DPoS) further restricts block production to elected delegates, trading broader direct participation for operational efficiency (Bachani & Bhattacharjya, 2022; BitShares, n.d.).

Other mechanisms allocate block-production authority using governed identity, trusted hardware, reputation, or other resources. Proof of Authority relies on an approved validator set (Barinov et al., 2018; Manolache et al., 2022). Proof of Elapsed Time uses trusted execution environments to support randomized leader waiting, which introduces a hardware-trust assumption and has been studied for multi-certificate attacks (Chen et al., 2017; Hyperledger Sawtooth (Intel), n.d.; Wang et al., 2022). Reputation-based protocols assign influence from reputation records (Aluko & Kolonin, 2021; Zhuang et al., 2019). These mechanisms are not security-equivalent substitutes for PoW; each changes the adversary model and governance boundary.

Storage-based constructions such as Proofs of Space replace continuous hash competition with proofs tied to committed storage capacity (Ateniese et al., 2014; Dziembowski et al., 2015), while Proof of Burn ties influence to irreversible asset destruction (Karantias et al., 2020). Other proposals reward useful or application-specific contributions (Xue et al., 2018). Across these Proof-of-X families, the principal analytical point is not that one mechanism dominates, but that different resources and trust assumptions produce different security, performance, governance, and sustainability trade-offs.

2.3. Energy Consumption in Blockchain Systems

The way blockchain consumes energy, particularly in proof-of-work (PoW) public networks (Alofi et al., 2024; Sedlmeir et al., 2020), has become a central concern in debates over the environmental impact and operating costs of blockchain technology. Comparing consensus mechanisms therefore requires precisely defined power and energy metrics, an understanding of how these metrics are measured empirically, and an assessment of how the environmental and economic effects of blockchain deployment arise.

2.3.1. Energy and power basics (definitions, metrics)

Power is the rate of energy transfer or conversion and is measured in watts (W); energy is power integrated over time and is measured in joules (J), watt-hours (Wh), kilowatt-hours (kWh), or larger units such as terawatt-hours (TWh). Blockchain studies report network power, total energy over a stated horizon, energy per block, energy per transaction, and hardware efficiency such as joules per terahash. These metrics are meaningful only when the measurement boundary, hardware assumptions, workload, active duration, and security context are stated explicitly (de Vries, 2018; Zhang et al., 2023).

De Vries and related studies estimate network power demand using bottom-up models based on observed hash rate, mining-hardware efficiency, and data-center Power Usage Effectiveness (PUE) (de Vries, 2018; Truby, 2018). Cambridge indices report estimates of electricity consumption and associated emissions for major blockchain networks (Cambridge Centre for Alternative Finance, n.d.; Stoll et al., 2019). Zhang et al. modelled climate-related effects of cryptocurrency energy use under alternative regulatory and technological scenarios (Zhang et al., 2023). Other studies examined Ethereum's energy profile before and after the transition from Proof of Work to Proof of Stake at the Merge (Ibañez & Rua, 2023; McDonald, 2021).

2.3.2. Empirical measurements of PoW/PoS energy consumption

Empirical estimates indicate that global PoW mining can consume electricity on the scale of national power systems, although the result depends on network activity, hardware efficiency, electricity sourcing, and modelling assumptions (Cambridge Centre for Alternative Finance, n.d.; Khosravi & Säämäki, 2023). De Vries argued that fixed power-per-hash estimates may understate consumption when they omit the effects of market price, capital investment, hardware turnover, and expansion of mining capacity (de Vries, 2018). Sapra et al. reviewed how application workload, network topology, and hardware selection affect the energy consumption of PoW-based blockchain systems (Sapra et al., 2023). Truby analyzed legal and policy options for reducing Bitcoin's environmental impact while noting possible tensions with innovation and financial inclusion (Truby, 2018).

Ethereum's transition from PoW to PoS provides a prominent comparison of consensus-related energy demand. Studies of pre-Merge Ethereum primarily attribute its high energy consumption to competitive PoW mining, while full-node operation accounts for a much smaller component (McDonald, 2021). Measurements of validators and full nodes in PoS-based Ethereum indicate substantially lower operational energy use than PoW mining, although PoS introduces different communication, hardware, and validator-management requirements (Cortes-Goicoechea et al., 2021; Ibañez & Rua, 2023). Platt et al. compared the energy characteristics of PoW, PoS, PBFT, and other mechanisms, noting that permissioned BFT protocols can have relatively low energy demand at moderate validator-set sizes but face scalability constraints as communication overhead grows (Platt et al., 2021).

Alternative mechanisms may reduce or repurpose the resource consumption associated with PoW. Green-PoW modifies mining participation and competition to reduce active computation under its specified policy (Lasla et al., 2022). Proof-of-Learning and DLBC couple block-production incentives to machine-learning tasks (Bravo-Marquez et al., 2019; Li et al., 2019), while Proof of Federated Learning integrates federated-learning work into a consensus design (Qu et al., 2021). Storage-based proofs replace much of the continuous hashing burden with storage commitment and proof generation, but storage hardware manufacture, operation, maintenance, and proof computation still carry resource costs (Ateniese et al., 2014; Dziembowski et al., 2015).

2.4. Summary

This chapter has provided the technical context required to assess the impact of energy usage on blockchain consensus mechanisms. It first reviewed the fundamentals of blockchain technology - the distributed ledger, peer-to-peer (P2P) networking, and the role of transactions and smart contracts across application areas. It then examined the consensus mechanisms used by distributed systems, including classical Byzantine Fault Tolerance (BFT) protocols, Nakamoto-style Proof of Work (PoW), and a range of Proof of Stake and other Proof-of-X (PoX) protocols, each based on different security assumptions and resource requirements for maintaining the ledger. Finally, it summarized empirical and theoretical work on the energy consumption of blockchains, including the associated environmental and economic impacts, at both the protocol level and in specific energy-system applications.

Chapter 3
Literature Review and Related Work

Chapter 3
Literature Review and Related Work

3.1. Introduction

The literature contains a wide range of consensus families developed for different trust and resource models. PoW uses competitive computation, PoS uses economic stake, BFT protocols use quorum-based replica agreement, DPoS delegates block production, PoA relies on governed validator identity, and other mechanisms use storage, trusted hardware, reputation, or measurable contribution. These mechanisms cannot be ranked meaningfully without specifying the deployment model and the dimension being compared (Bhutta et al., 2021; Islam et al., 2023; Wang et al., 2019; Xiao et al., 2020).

Existing surveys provide valuable taxonomies and protocol descriptions, but their scopes and comparison criteria differ. Some emphasize architecture or taxonomy, others focus on security, performance, or particular application settings (Bhutta et al., 2021; Islam et al., 2023; Wang et al., 2019; Xiao et al., 2020). This chapter therefore does not claim that prior reviews failed to address comparison; instead, it contributes a structured synthesis that places functionality, security assumptions, and energy implications in one decision-oriented framework and makes the limits of qualitative cross-family comparison explicit.

Accordingly, this chapter presents a structured comparative literature review of representative blockchain consensus mechanisms across three analytical dimensions: functionality, security, and energy consumption. The objective is evidence-based qualitative synthesis rather than a universal benchmark or meta-analysis.

The review has two principal roles in the thesis. First, it establishes the terminology and trade-off dimensions used later to motivate PoCol. Second, it positions collaborative and partitioned Proof-of-Work approaches as the most relevant architectural comparators for the proposed design. The resulting tables are qualitative synthesis tools, not standardized measurements of protocol performance or security.

The structured comparative literature review addresses the following research questions:

RQ3.1: Which principal functional characteristics serve to differentiate blockchain consensus algorithms with respect to their validation model, finality, fault tolerance, throughput, and scalability? (Xiao et al., 2020)

RQ3.2: In what ways do prominent consensus algorithms diverge in their resilience to security threats, including Byzantine faults, majority attacks, Sybil attacks, collusion, and pressures toward centralization? (Wang et al., 2019; Xiao et al., 2020)

RQ3.3: How does energy consumption differ across the various consensus families, and which design decisions exert the greatest influence on the sustainability profile of blockchain systems? (Bhutta et al., 2021; Islam et al., 2023)

RQ3.4: What trade-offs emerge among functionality, security, and energy efficiency in the selection of a consensus mechanism for a given blockchain application? (Wang et al., 2019; Xiao et al., 2020)

RQ3.5: Which consensus mechanisms are most appropriate for distinct deployment contexts, such as public blockchain platforms, consortium networks, enterprise environments, and resource-constrained systems? (Islam et al., 2023; Xiao et al., 2020)

Recent work continues to extend blockchain consensus design in new directions. Ladder, for example, is a convergence-based structured-DAG design that facilitates parallel processing of blocks, achieving increased throughput and lower confirmation latency (Hu et al., 2025). Similarly, Orthrus accelerates Multi-BFT consensus through concurrent partial ordering of transactions, reducing latency while maintaining a global ordering of all transactions (Lyu et al., 2025). Finally, Crackle is a fast sector-based BFT consensus protocol with sublinear communication complexity, improving scalability relative to communication-heavy BFT methods (Xu et al., 2024). While these works do not replace the consensus families evaluated in the current survey, they are representative of the fact that a significant amount of current work on this topic focuses on addressing long-standing problems with throughput, latency, and/or communication overhead, particularly in high-performance and BFT-based applications (Hu et al., 2025; Xu et al., 2024).

3.2. Methodology

This chapter uses a structured comparative literature-review method informed by PRISMA reporting principles (Page, McKenzie, et al., 2021; Page, Moher, et al., 2021). It is not presented as a full systematic review because the thesis does not report a complete PRISMA flow diagram, database-level record counts, duplicate-removal counts, or a formal risk-of-bias or study-quality assessment. The purpose is therefore transparent, reproducible qualitative synthesis rather than exhaustive evidence identification or quantitative meta-analysis.

3.2.1. Review Design

The review process was conducted in five stages:

1. formulation of the review objectives and research questions,
2. identification of relevant studies through database searching,

3. title and abstract screening,
4. full-text eligibility assessment, and
5. structured data extraction and qualitative synthesis.

3.2.2. Information Sources and Search Period

The structured database search covered publications from 2008 through January 2025 in IEEE Xplore, ScienceDirect, and the ACM Digital Library, supplemented by backward reference checking. Targeted update searches were subsequently conducted through August 2026 for directly relevant collaborative-PoW, auditable-work, and recent consensus studies. The original structured window and later targeted updates are therefore reported separately, and no claim of exhaustive coverage after January 2025 is made (IEEE Professional Communication Society, 2019; Kitchenham & Charters, 2007).

Table (3.1): Inclusion and exclusion criteria

Criterion Type	Criterion
Inclusion	Study proposes, analyzes, or compares a blockchain consensus mechanism
Inclusion	Study discusses functionality, security, scalability, or energy/energy consumption of consensus algorithms
Inclusion	Study is foundational, highly cited, or directly relevant to the historical evolution of consensus mechanisms
Inclusion	Study provides recent advances relevant to one or more selected consensus families
Exclusion	Study focuses on blockchain applications without substantial treatment of consensus design
Exclusion	Study lacks sufficient technical detail for comparative analysis
Exclusion	Study is duplicated or superseded by a more complete version
Exclusion	Study is non-scholarly, promotional, or lacks reliable technical grounding
Exclusion	Study addresses unrelated distributed-systems topics without direct blockchain-consensus relevance

3.2.3. Search Strategy

The search strings combined general blockchain-consensus terms with mechanism-specific and evaluation-oriented keywords. Representative search expressions included:

- “blockchain consensus” AND survey
- “consensus algorithm” AND blockchain
- “Proof of Work” OR PoW OR “Proof of Stake” OR PoS

- PBFT OR BFT OR DPoS OR PoA OR PoET OR PoH OR “Proof of Space”
- blockchain AND security AND consensus
- blockchain AND energy consumption AND consensus
- blockchain AND scalability AND consensus

Search syntax was adapted to each database. Priority was given to peer-reviewed journal and conference publications and to foundational primary technical sources when they defined a protocol. White papers, standards, technical documentation, theses, and preprints were retained only where they were directly relevant and were identified as such in the bibliography. This source hierarchy is important because the review combines mature consensus families with recent proposals that do not yet have equivalent peer-reviewed evidence (IEEE Professional Communication Society, 2019; Kitchenham & Charters, 2007).

3.2.4. Eligibility Criteria

Table 3.1 summarizes the criteria used for searching. Studies were included if they satisfied at least one of the following criteria:

1. they proposed, analyzed, or compared a blockchain consensus mechanism,
2. they discussed consensus from the perspective of functionality, security, scalability, or energy consumption,
3. they provided foundational or widely cited technical insights relevant to the evolution of blockchain consensus mechanisms, or
4. they introduced recent advances directly relevant to the selected consensus families considered in this survey.

Studies were excluded if they:

1. focused on blockchain applications without meaningful treatment of consensus design,
2. lacked sufficient technical detail for comparative analysis,
3. were duplicates or shorter versions of more complete studies,
4. were non-scholarly opinion pieces or promotional material, or

5. addressed unrelated distributed-systems topics without direct relevance to blockchain consensus (IEEE Professional Communication Society, 2019; Kitchenham & Charters, 2007).

3.2.5. Study Selection Procedure

After duplicate removal, titles and abstracts were screened against the eligibility criteria, followed by full-text assessment of potentially relevant studies. Publications retained in the synthesis were those judged to contribute directly to the review questions or to the technical history of the selected consensus families.

The selection procedure was informed by PRISMA principles of transparent identification, screening, eligibility assessment, and inclusion (Page, McKenzie, et al., 2021; Page, Moher, et al., 2021). Because the thesis does not provide the complete record counts and flow diagram required for a full PRISMA report, the resulting chapter is described consistently as a structured comparative review rather than a systematic review.

3.2.6. Data Extraction

Table 3.2 summarizes the data extraction framework used in this survey. For each included study, the following information was extracted in a structured manner:

- consensus mechanism or protocol family,
- validation logic and participation model,
- trust assumptions and deployment setting,
- fault tolerance characteristics,
- scalability and performance implications,
- security properties and reported attack surfaces,
- energy or power-related observations,
- representative application contexts.

The extraction framework is aligned with the review questions and supports comparison across heterogeneous consensus families. It records protocol family, validation logic, participation model, trust assumptions, fault model, representative attack surfaces, performance and scalability implications, energy observations, and deployment context (IEEE Professional Communication Society, 2019; Kitchenham & Charters, 2007).

3.2.7. Selection of Consensus Algorithms

The selected mechanisms represent major design directions rather than an exhaustive catalogue of every published consensus variant. The set includes classical BFT/PBFT, PoW, PoS, DPoS, PoA, PoET, PoB, PoH as an ordering/time primitive used within a broader consensus architecture, PoSpace, PoR, and PoC. This framing avoids treating mechanisms with different roles as strictly interchangeable consensus algorithms while retaining them for comparative context.

Table (3.2): Data extraction framework used in this survey

Extracted Item	Description
Consensus family	PoW, PoS, BFT, PBFT, DPoS, PoA, PoET, PoH, PoSpace, PoB, PoR, PoC
Validation logic	Computational work, economic stake, delegated validation, trusted authority, storage commitment, etc.
Participation model	Permissionless, permissioned, consortium, hybrid
Trust assumptions	Open adversarial, bounded-trust, authority-based, reputation-based, contribution-based
Fault tolerance	Degree and type of fault or adversarial tolerance reported in the literature
Security properties	Representative attack surface, collusion exposure, majority-control susceptibility, centralization pressure
Performance and scalability	Throughput tendency, communication overhead, validator-set growth constraints
Energy consumption	Relative computational and infrastructural intensity of the validation process
Application suitability	Public blockchain, enterprise, IoT, consortium, high-throughput DApps, etc.

3.2.8. Qualitative Comparative Coding Framework

The reviewed studies use heterogeneous terminology, assumptions, metrics, and implementation settings, so a single quantitative scale cannot be applied defensibly across all consensus families. The comparison tables therefore use qualitative synthesis indicators such as low, moderate, high, limited, good, and very good. These labels summarize recurring architectural tendencies reported in the selected literature; they are not calibrated numerical thresholds, probabilities, or measured effect sizes. Tables 3.3 and 3.4 state the coding logic used to make those judgments explicit (IEEE Professional Communication Society, 2019; Kitchenham & Charters, 2007).

3.2.9. Methodological Scope and Limitations

This review has three principal limitations. First, it is a structured comparative review rather than a statistical meta-analysis or a fully reported systematic review. Second, consensus studies differ substantially in terminology, trust assumptions, workloads, hardware, network size, and evaluation metrics, which limits direct one-to-one comparison. Third, the qualitative labels in the comparison tables are evidence-synthesis indicators rather than standardized quantitative benchmarks. These limitations are stated explicitly to prevent false precision.

Table (3.3): Qualitative coding rubric for the security comparison table

Dimension	Coding Logic
Representative Security Exposures	Assigned from the most frequently reported attack surfaces in the reviewed literature for each mechanism, including Sybil attacks, majority-control attacks, collusion, long-range attacks, centralization pressure, network partitioning, insider threats, and validator compromise
Qualitative Risk Indication: Low	Limited architectural exposure under the mechanism's normal operating assumptions, with strong resistance in the intended deployment setting
Qualitative Risk Indication: Medium	Noticeable exposure to one or more relevant attack classes, but with mitigating architectural or governance controls reported in the literature
Qualitative Risk Indication: Medium to High	Recurrent exposure to multiple attack surfaces or strong dependence on concentration-sensitive assumptions such as stake, delegates, authority, reputation, or hardware trust
Majority-Control Attack Susceptibility	Marked "Yes" only when the literature or protocol structure supports an equivalent majority-control pathway, whether through hash power, stake concentration, delegate collusion, validator compromise, or reputation manipulation

3.2.10. Clarification of Comparative Interpretation

The comparative framework does not define a universal numerical score for blockchain consensus. Labels such as low, medium, high, good, and very good are contextual synthesis indicators whose interpretation depends on the protocol's trust model, validator composition, hardware, network scale, implementation details, and deployment environment (Kitchenham & Charters, 2007; Page, McKenzie, et al., 2021).

Accordingly, no single composite score is assigned across functionality, security, and energy consumption. Public permissionless systems may prioritize adversarial resilience and open participation, whereas consortium or enterprise systems may prioritize deterministic finality, governance control, bounded membership, or low operational overhead. The tables should therefore be read as context-dependent decision support rather than as a global ranking of consensus mechanisms (IEEE Professional Communication Society, 2019; Page, Moher, et al., 2021).

Table (3.4): Qualitative coding rubric for the comparative summary table

Dimension	Coding Logic
Deployment Suitability	Assigned according to the typical governance and participation model reported for the mechanism: permissionless, permissioned, consortium, enterprise-controlled, or hybrid
Fault Tolerance	Summarized from the mechanism's commonly reported tolerance to faulty or adversarial nodes, taking into account whether resilience depends on hash power, stake concentration, bounded Byzantine assumptions, authority trust, or delegated honesty
Energy consumption: Very Low	No continuous mining competition and low validation overhead, typically found in authority-based, permissioned, or lightweight delegated settings
Energy consumption: Low	Limited computational burden under normal operation, with validator selection based on stake, delegation, reputation, or contribution rather than persistent computation
Energy consumption: Low to Moderate	Lower than PoW, but still involving non-trivial storage, hardware, or infrastructure overhead
Energy consumption: Moderate	Reduced computational intensity compared with PoW, but involving resource sacrifice, storage burden, or persistent auxiliary overhead
Energy consumption: Very High	Continuous computation-intensive validation, typically associated with mining competition as in PoW
Scalability: Limited	High communication overhead, low throughput tendency, or significant degradation with validator-set growth
Scalability: Good	Better validator efficiency or throughput than classical mining- or communication-heavy protocols, but still constrained by trust, governance, or infrastructure assumptions
Scalability: Very Good	High throughput tendency and relatively low coordination overhead in the mechanism's intended deployment environment
Security Profile	Assigned as a qualitative synthesis of adversarial resilience, trust assumptions, concentration risk, and exposure to representative attack surfaces in the reviewed literature

3.3. Background

3.3.1. Definition of Consensus Algorithms in Blockchain

A blockchain consensus protocol defines how eligible participants propose, validate, order, and accept state transitions under a stated trust and fault model. Different protocol families use different sources of authority or scarcity: PoW uses computational work, PoS uses economic stake, BFT protocols use authenticated quorum agreement, and other designs use delegated authority, trusted hardware, storage, reputation, or measurable contribution. Consensus design therefore determines not only block production, but also finality semantics, fault tolerance, participation requirements, and governance boundaries.

These mechanisms are design alternatives rather than successive improvements along a single scale. Changes that reduce computation or increase throughput may introduce stronger trust assumptions, smaller validator sets, different economic incentives, or new attack surfaces. For this reason, the review evaluates each family within its intended operating assumptions rather than treating energy efficiency, decentralization, security, and scalability as universally comparable scalar properties (Wang et al., 2019; Xiao et al., 2020).

3.3.2. Consensus Algorithms in Blockchain

Proof-of-Work was the first widely deployed permissionless blockchain consensus mechanism, but its sustained computational competition creates high operating energy demand (de Vries, 2018; Nakamoto, 2008). Subsequent mechanisms reduce or replace continuous hashing in different ways. Their lower computational demand should not be interpreted as identical security or decentralization because each design changes how validation authority is allocated and how adversarial behavior is constrained.

Security exposure is also mechanism-specific. PoW is sensitive to concentration of hash power and strategic mining; PoS protocols are sensitive to stake concentration and protocol-specific long-range or equivocation threats; BFT protocols depend on explicit Byzantine fault thresholds and network assumptions; and delegated or authority-based systems concentrate validation in smaller governed sets (Bano et al., 2019; Wang et al., 2019; Xiao et al., 2020).

Energy demand can also be reduced at the implementation layer. For PoW, hardware architecture and ASIC efficiency affect joules per hash and therefore network power for a given hash rate (Koutra & Tenentes, 2024; Pham et al., 2020). Protocol-level and hardware-level efficiency should be distinguished because they change different parts of the energy model.

3.3.3. Importance of Consensus Algorithms

Consensus is central to blockchain because it determines which proposed state transitions become part of the accepted ledger when participants may fail, disagree, or act strategically. Its design influences security and finality, but also validator participation, communication overhead, throughput, governance, and resource consumption (Bano et al., 2019; Xiao et al., 2020).

Accordingly, consensus should be evaluated as a system-design choice rather than only as a computational primitive. A protocol suitable for an open adversarial network may be inappropriate for a small governed consortium, and a mechanism optimized for low operational energy may rely on stronger identity, stake, hardware, or governance assumptions.

Table (3.5): Historical orientation of the principal consensus mechanisms and supporting primitives considered in this review, using representative publication or deployment milestones and primary references.

No.	Consensus Mechanism / Primitive	Representative Platforms / Context	Representative Publication / Deployment Milestone	Primary Reference
1	Byzantine Fault Tolerance (BFT)	Early distributed consensus models, conceptual foundation for permissioned consensus systems	1982	(Lamport et al., 1982)
2	Practical Byzantine Fault Tolerance (PBFT)	Permissioned/replicated-state-machine systems	1999	(Castro & Liskov, 1999)
3	Proof of Work (PoW)	Bitcoin; later PoW blockchains	2008/2009	(Nakamoto, 2008)
4	Proof of Stake (PoS)	Peercoin/PPCoin; later stake-based public blockchains	2012	(King & Nadal, 2012)
5	Proof of Space (PoSpace)	Proof-of-space protocols; later storage-based blockchains	2015 peer-reviewed	(Dziembowski et al., 2015)
6	Proof of Burn (PoB)	Proof-of-burn constructions and specialized blockchain designs	2020 formalization	(Karantias et al., 2020)
7	Delegated Proof of Stake (DPoS)	BitShares and other delegated-validator systems	2014	(BitShares, n.d.)
8	Proof of Authority (PoA)	Governed/permissioned validator networks	2017	(Barinov et al., 2018; Szilágyi, 2017)

9	Proof of Elapsed Time (PoET)	Hyperledger Sawtooth / trusted-hardware leader selection	2017	(Hyperledger Sawtooth (Intel), n.d.)
10	Proof of History (PoH)	Solana ordering/time primitive used within a broader consensus architecture	2018	(Yakovenko, 2018)
11	Proof of Contribution (PoC)	Contribution-based consensus proposals	2018	(Xue et al., 2018)
12	Proof of Reputation (PoR)	Reputation-based consensus proposals	2019	(Zhuang et al., 2019)

Note: The dates in Table 3.5 are representative publication or deployment milestones rather than claims about a single universally agreed 'year of invention'. PoH is included as an ordering/time primitive used within a broader consensus architecture, and the listed platforms are illustrative rather than exhaustive.

3.3.4. Historical Development of Consensus Mechanisms

The historical development of blockchain consensus builds on earlier distributed-agreement research. Byzantine fault tolerance formalized agreement in the presence of arbitrary faults (Lamport et al., 1982), and PBFT demonstrated a practical state-machine-replication protocol for the standard fewer-than-one-third Byzantine setting (Castro & Liskov, 1999). These ideas later informed many permissioned and committee-based blockchain designs.

Bitcoin then demonstrated a deployed permissionless Proof-of-Work design in which chain selection is tied to cumulative computational work (Nakamoto, 2008). Subsequent research explored alternative scarcity or authority models, including PoS (King & Nadal, 2012), DPoS (BitShares, n.d.), PoA (Barinov et al., 2018), PoET (Hyperledger Sawtooth (Intel), n.d.), Proofs of Space (Dziembowski et al., 2015), Proof of Burn (Karantias et al., 2020), Proof of Contribution (Xue et al., 2018), and Proof of Reputation (Zhuang et al., 2019). Proof of History is treated here more precisely as a cryptographic ordering/time primitive used in the Solana architecture rather than as a complete standalone consensus protocol (Shoup, 2022; Yakovenko, 2018).

Table 3.5 is therefore intended only as historical orientation. Later sections compare the mechanisms by explicit functionality, security-assumption, energy, and deployment criteria; the table itself does not rank them.

3.4. Security Aspects of Consensus Algorithms

Consensus security depends on the protocol's trust assumptions, adversary model, network model, and validation-resource distribution. Safety and liveness cannot be summarized by one universal 'security level' across PoW, PoS, BFT, delegated, authority-based, storage-based, and reputation-based designs because the relevant failure and attack conditions differ by family (Abellán Álvarez et al., 2024; Bano et al., 2019; Xiao et al., 2020).

The comparative analysis therefore distinguishes architectural exposure from deployment-specific risk. A mechanism may have a well-understood fault threshold yet still be insecure in a deployment with concentrated resources, weak identity controls, poor incentives, or severe network partitioning. Conversely, a mechanism with centralized governance may be appropriate for a bounded-trust enterprise setting even though it does not satisfy permissionless-decentralization goals.

Table 3.6 summarizes representative attack surfaces and concentration pathways reported for the selected families. Its qualitative labels are synthesis indicators, not measured attack probabilities. They should not be interpreted as a claim that, for example, one family is numerically 'safer' than another without a common adversary model and implementation boundary.

Different consensus families obtain resilience in different ways. PoW ties influence to computational work and therefore makes large-scale history replacement economically and computationally costly, subject to hash-power concentration and network assumptions (Gervais et al., 2016; Nakamoto, 2008). BFT protocols use quorum intersection under explicit Byzantine thresholds (Castro & Liskov, 1999). PoS and delegated protocols depend on stake distribution, validator incentives, and protocol-specific accountability rules (Ge et al., 2022; Saad et al., 2021). Authority-, reputation-, and contribution-based systems introduce their own identity, governance, and measurement assumptions. These are different security models rather than points on one common scale.

Representative attack surfaces likewise differ. PoW is exposed to majority hash-power control, selfish mining, block withholding, and network-layer attacks (Gervais et al., 2016). PoS families may additionally face protocol-specific long-range, equivocation, and stake-concentration problems (Deb et al., 2024; Ge et al., 2022). DPoS and PoA can obtain low coordination overhead from smaller validator sets, but this can increase exposure to collusion, governance capture, or validator compromise (Barinov et al., 2018; BitShares, n.d.; Wang et al., 2022). PoET replaces continuous mining with a trusted-hardware assumption and has been studied for multi-certificate attacks (Chen et al., 2017; Wang et al., 2022).

Operational efficiency must therefore be interpreted together with security assumptions. Classical PBFT-style protocols can provide deterministic agreement for bounded replica sets, but their communication cost grows quickly with validator count (Castro & Liskov, 1999; Castro & Liskov, 2002). Other mechanisms improve throughput or reduce validation cost by limiting block producers, relying on stake or authority, or introducing trusted hardware. No such change is 'free': it alters the trust, governance, or resource assumptions under which security is obtained.

3.4.1. Security Features and Vulnerabilities

A consensus protocol's security profile follows from how validation authority is distributed and from the conditions under which conflicting histories can be accepted. PoW security depends on adversarial hash power, mining strategy, and network propagation (Gervais et al., 2016; Nakamoto, 2008); PoS security depends on the particular stake protocol and its economic and network assumptions (Abellán Álvarez et al., 2024; Ge et al., 2022); and PBFT-style protocols tolerate fewer than one-third Byzantine replicas in the standard 3f+1 model (Castro & Liskov, 1999).

DPoS and PoA improve operational efficiency by restricting block production to elected or governed validators, which may increase exposure to collusion or governance capture (Barinov et al., 2018; BitShares, n.d.; Wang et al., 2022). Storage-, reputation-, and contribution-based protocols similarly introduce resource-concentration, measurement, or identity risks specific to their design. Accordingly, Table 3.6 is best read as an architectural exposure summary, not as a universal risk ranking.

BFT and PBFT-based systems have the capabilities of providing a high level of fault tolerance as long as the percentage of malicious nodes remains below the protocol threshold, thus, they are typically better suited for environments where there are partially trusted and/or permissioned participants rather than public ones (Pennekamp et al., 2020; Zarkesh et al., 2024). Their effectiveness is also dependent on stable communication and bounded participation, in addition to which message complexity typically limits their ability to be deployed successfully at large scale. DPoS and PoA are differentiated from other consensus algorithms because the validators are either selected in advance or voted onto the role by a participant using some other form of authority, but this restriction ultimately increases the risks of collusion, insider manipulation, and governance capture (EL Azzaoui et al., 2022; Vairagade & Hanumantha, 2022). Ultimately, there is no single consensus algorithm that can be declared to be universally "safe", rather, each individual consensus algorithm will have unique attributes that define its relative resiliency against attacks and structural concentration.

Table (3.6): Comparative summary of representative security exposures, qualitative exposure indicators, and representative concentration/control pathways across selected blockchain mechanisms.

No.	Consensus Algorithm	Representative Security Exposures	Qualitative Exposure Indicator	Representative Concentration / Control Pathway
1	BFT	Byzantine faults, message delay, network partitioning, adversarial coordination	Protocol-dependent	Threshold violation / quorum compromise
2	PBFT	Byzantine behavior, network partitioning, replica coordination failures	Low to Medium	>f Byzantine replicas or network-timing failure
3	PoW	51% attack, Sybil attack, mining centralization	Medium	Hash-power concentration / strategic mining
4	PoS	Majority-stake attack, nothing-at-stake, long-range attack, stake concentration	Medium to High	Stake concentration / protocol-specific capture
5	PoSpace	Majority-resource concentration, Sybil attack, grinding attack	Medium	Storage-resource concentration
6	PoB	Majority-influence concentration, centralization risk, economic manipulation	Medium	Economic-influence concentration
7	DPoS	Delegate collusion, voter apathy, governance capture, centralization	Medium to High	Delegate collusion / governance capture
8	PoA	Validator compromise, centralization, identity abuse, single-point trust dependency	Medium	Validator compromise / authority capture
9	PoET	Insider attacks, malicious SGX enclave behavior, trusted-hardware dependence	Medium	TEE compromise / multi-certificate manipulation

10	PoH	Sequencing-layer dependence, centralization risk, validator concentration	Architecture-dependent	Depends on surrounding consensus/validator design
11	PoC	Contribution falsification, insider manipulation, concentration of influence	Protocol-dependent	Contribution-measure manipulation / concentration
12	PoR	Reputation abuse, collusion, Sybil attack, concentration through reputation control	Protocol-dependent	Reputation manipulation / governance concentration

Note: The qualitative exposure indicator is an evidence-based synthesis rather than a quantitative risk probability. The final column identifies the resource, authority, or coordination pathway through which control may concentrate; it does not imply that all mechanisms share an identical “51% attack” threshold. Deployment-specific risk depends on the protocol implementation, validator or miner distribution, governance, incentives, and network assumptions.

3.5. Energy Consumption in Consensus Algorithms

Energy comparisons across consensus mechanisms require an explicit measurement boundary. Reported power or energy can vary with hardware, validator-set size, network topology, workload, communication pattern, data-centre overhead, electricity source, and whether the system is public, permissioned, or consortium-based. This review therefore uses qualitative cross-family energy characterization and does not claim a controlled numerical benchmark across heterogeneous protocols (Bhutta et al., 2021; Islam et al., 2023; Sedlmeir et al., 2020).

Among widely deployed public consensus families, PoW is operationally energy-intensive because network security is coupled to sustained competitive hashing. Network-scale estimates such as the Cambridge index are derived from observed or inferred mining activity and hardware assumptions rather than from a protocol-level constant (Cambridge Centre for Alternative Finance, n.d.; de Vries, 2018). Consequently, real-world PoW energy demand cannot be inferred from an abstract nonce count alone; aggregate hash rate, hardware efficiency, utilization, market incentives, and infrastructure all matter.

PoS and DPoS generally require much less continuous computation than competitive PoW because validator selection and block production are not based on network-wide hash races. Ethereum's transition from PoW to PoS is a prominent example, although the reported reduction depends on the system boundary, client population, hardware, and infrastructure assumptions (Cortes-Goicoechea et al., 2021; Ibañez & Rua, 2023). Lower operational electricity demand does not remove lifecycle, infrastructure, governance, or concentration concerns.

The same caution applies to other low-computation mechanisms. Authority-based, reputation-based, trusted-hardware, and storage-based systems may reduce continuous hashing but still consume energy through validator infrastructure, storage, networking, redundancy, and hardware manufacture. Energy classification therefore describes operating tendency, not complete environmental impact.

Per-transaction energy is also context dependent. Many consensus systems expend energy to secure the ledger over time rather than in direct proportion to the number of transactions included in a particular block. Per-transaction values can therefore change simply because transaction throughput changes. This review uses such metrics only when their denominator and measurement boundary are explicit and does not treat them as a universal cross-protocol sustainability metric (Sedlmeir et al., 2020).

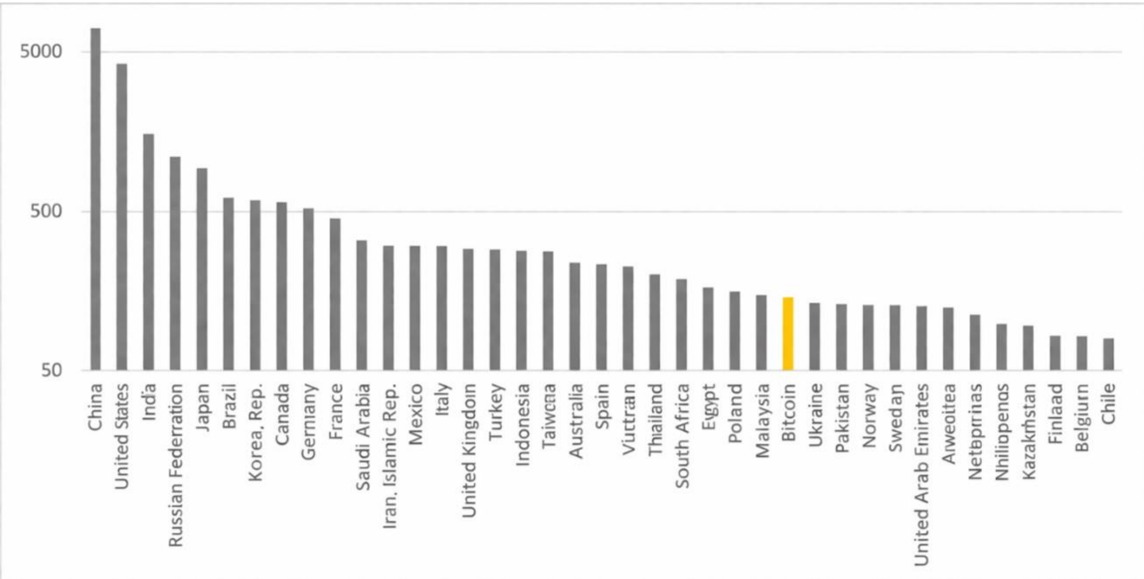

Figure (3.1): Illustrative electricity consumption trend of PoW-based blockchain operation, with Bitcoin used as a representative example of energy-intensive consensus (Annual electricity consumption in TWh) (Cambridge Centre for Alternative Finance, n.d.).

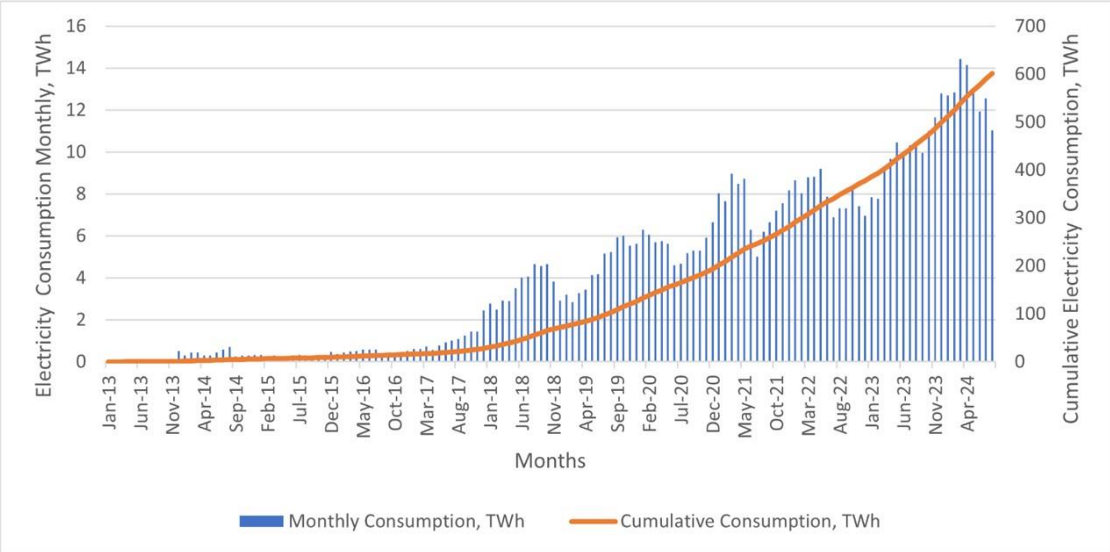

Figure (3.2): Monthly and cumulative electricity consumption associated with Bitcoin mining, illustrating the broader sustainability implications of computation-intensive PoW validation (Cambridge Centre for Alternative Finance, n.d.).

Figures 3.1 and 3.2 provide contextual evidence of the scale and historical evolution of Bitcoin electricity consumption. They are not experimental baselines for PoCol, do not imply that other PoW implementations share the same absolute consumption, and are not used to infer numerical energy differences for PoS, DPoS, or PoCol. Their purpose is to motivate why energy-aware consensus research has become important (Cambridge Centre for Alternative Finance, n.d.; Kohli et al., 2023).

3.6. Overview of the Selected Consensus Algorithms

This section summarizes the selected protocol families and supporting primitives using the same three lenses employed throughout the review: functionality, security assumptions, and resource demand. The mechanisms are not assumed to perform equivalent roles. In particular, BFT/PBFT are agreement protocols for bounded replica sets, PoW and PoS are Sybil-resistance and chain/validator-selection families, and PoH is treated as an ordering/time primitive used with a surrounding consensus architecture (Bano et al., 2019; Castro & Liskov, 1999; Yakovenko, 2018).

The selected mechanisms can be grouped by the resource or authority on which validation depends: computational work (PoW), economic stake or delegation (PoS/DPoS), governed identity (PoA), trusted execution (PoET), storage commitment

(PoSpace), irreversible economic sacrifice (PoB), reputation or contribution (PoR/PoC), and Byzantine quorum agreement (BFT/PBFT). These categories overlap in real systems and should be interpreted as analytical families rather than mutually exclusive implementations (Bano et al., 2019; Xiao et al., 2020).

Their evolution reflects repeated trade-offs among open participation, adversarial resilience, finality, throughput, coordination overhead, governance, and resource cost. PoW offers open resource-based participation but high operating energy; PoS and delegated designs reduce continuous computation while introducing stake and governance assumptions; BFT protocols provide deterministic agreement under explicit Byzantine thresholds but incur communication costs; and authority- or hardware-based systems gain efficiency by accepting stronger trust assumptions (Barinov et al., 2018; Castro & Liskov, 1999; Chen et al., 2017; King & Nadal, 2012).

Other designs alter the resource being committed rather than simply reducing computation. PoSpace uses storage commitment (Ateniese et al., 2014; Dziembowski et al., 2015); PoB uses irreversible economic sacrifice (Karantias et al., 2020); PoR and PoC tie influence to reputation or measurable contribution (Xue et al., 2018; Zhuang et al., 2019). PoH, by contrast, supplies a verifiable ordering/time structure that can reduce coordination ambiguity but does not by itself replace the surrounding consensus protocol (Shoup, 2022; Yakovenko, 2018).

The consequence is that consensus selection is inherently context dependent. The following subsections summarize each family within its intended assumptions, and the later comparison tables should be read as qualitative synthesis rather than as a universal league table of consensus mechanisms.

3.6.1. Byzantine Fault Tolerance (BFT)

Byzantine Fault Tolerance (BFT) is a property and family of protocols that enable replicated systems to maintain agreement despite a bounded number of participants behaving arbitrarily (Lamport et al., 1982). BFT protocols differ in synchrony assumptions, quorum structure, leader selection, communication pattern, and fault threshold. In blockchain practice, BFT-style protocols are most commonly used where validator membership is known or governed, because authenticated quorum communication among a bounded set is part of the operating model.

3.6.2. Practical Byzantine Fault Tolerance (PBFT)

Practical Byzantine Fault Tolerance (PBFT) is a concrete state-machine-replication protocol for the standard 3f+1 model: safety tolerates at most f Byzantine replicas among 3f+1, and liveness additionally depends on suitable timing assumptions (Castro & Liskov, 1999). Its pre-prepare, prepare, commit, and view-change procedures provide deterministic agreement but incur substantial communication as the replica set grows. PBFT is therefore influential in permissioned and consortium settings, while large committees generally require additional scaling techniques (Castro & Liskov, 1999; Castro & Liskov, 2002).

3.6.3. Proof of Work (PoW)

Proof of Work (PoW) ties block-production probability to computational work. Miners evaluate candidate block headers against a target, and the accepted chain is selected by cumulative work (Nakamoto, 2008). PoW supports permissionless participation without a pre-existing identity or stake position, but sustained competitive hashing creates high operating energy demand and can be affected by hash-power concentration, strategic mining, and propagation delay (de Vries, 2018; Gervais et al., 2016). Its security and latency properties are therefore functions of both resource distribution and network conditions.

3.6.4. Proof of Stake (PoS)

Proof of Stake (PoS) replaces continuous hash-rate competition with validator influence derived from economic stake (Ge et al., 2022; King & Nadal, 2012). Modern PoS protocols differ substantially in leader selection, finality, slashing, checkpointing, and randomness, so their security properties must be evaluated protocol by protocol. They generally use much less operational energy than PoW, but introduce stake-concentration, governance, and protocol-specific long-range or equivocation concerns (Deb et al., 2024; Ge et al., 2022; Saad et al., 2021).

3.6.5. Delegated Proof of Stake (DPoS)

Delegated Proof of Stake (DPoS) allows token holders to elect a limited set of block producers (Bachani & Bhattacharjya, 2022; BitShares, n.d.). Restricting active producers can support high throughput and low validation overhead, but it also concentrates operational authority and can increase exposure to delegate collusion, voter apathy, or governance capture. DPoS therefore represents a different governance trade-off rather than a strictly superior form of PoS.

3.6.6. Proof of Authority (PoA)

Proof of Authority (PoA) assigns block-production authority to a governed set of identified validators (Barinov et al., 2018; Manolache et al., 2022). This can provide fast confirmation and low computational overhead in enterprise or consortium settings, but security depends on validator governance, identity protection, and resistance to collusion or order manipulation (Szilágyi, 2017; Wang et al., 2022). PoA therefore sacrifices permissionless openness in exchange for explicit accountability and operational control.

3.6.7. Proof of Elapsed Time (PoET)

Proof of Elapsed Time (PoET) uses trusted execution environments to generate or attest randomized waiting periods for leader selection (Chen et al., 2017; Hyperledger Sawtooth (Intel), n.d.). It avoids continuous PoW-style competition and can therefore reduce computation, but its security rests on the integrity and uniqueness assumptions of the trusted hardware. Research on multi-certificate attacks illustrates that compromising or multiplying trusted-enclave credentials can undermine those assumptions (Wang et al., 2022).

3.6.8. Proof of Burn (PoB)

Proof of Burn (PoB) allocates influence through irreversible destruction of cryptocurrency, using economic sacrifice rather than continuous hash competition as the scarce resource (Karantias et al., 2020). Its operating energy can therefore be lower than PoW, but its security and governance depend on the burn rule, asset distribution, and the ability of participants to acquire and destroy value. Concentration and fairness concerns are economic rather than directly hash-rate based.

3.6.9. Proof of History (PoH)

Proof of History (PoH) is more precisely described as a cryptographic ordering and time-verification primitive. The Solana white paper presents PoH as a proof for verifying the order and passage of time between events and uses it within a broader high-performance blockchain architecture (Yakovenko, 2018). Consequently, PoH should not be interpreted in this review as a standalone replacement for PoW, PoS, or PBFT; its security and fault tolerance depend on the surrounding consensus and validator protocol (Shoup, 2022).

3.6.10. Proof of Space (PoSpace)

Proofs of Space use committed storage capacity as the resource demonstrated by the prover rather than sustained PoW-style hashing (Ateniese et al., 2014; Dziembowski et al., 2015; Fisch, 2019). This can reduce continuous computation, but practical sustainability also depends on storage hardware manufacture, operation, replacement, proof generation, and concentration of storage capacity. Storage-based consensus therefore changes the resource profile rather than eliminating resource cost.

3.6.11. Proof of Reputation (PoR)

Proof of Reputation (PoR) assigns validation influence using a reputation measure derived from prior behavior or other evidence (Aluko & Kolonin, 2021; Zhuang et al., 2019). This can reduce dependence on raw computation or stake, but it introduces difficult questions about reputation initialization, update rules, collusion, manipulation, and entry barriers for new participants. Energy use is generally lower than continuous PoW hashing, but security depends strongly on the integrity of the reputation system.

3.6.12. Proof of Contribution (PoC)

Proof of Contribution (PoC) allocates influence according to a defined contribution measure, such as computational, data, or application-specific work (Buyukates et al., 2023; Xue et al., 2018; Zhou et al., 2023). Its central challenge is measurement integrity: the contribution metric must be observable, difficult to manipulate, and aligned with protocol security. PoC therefore shifts the problem from raw resource expenditure to verifiable valuation of contribution.

3.7. Collaborative PoW Approaches

3.7.1. Parallel and Partitioned Mining

Hazari and Mahmoud proposed Parallel Proof-of-Work as an approach for distributing mining computation among cooperating participants through a coordinating manager (Hazari & Mahmoud, 2020). The approach retains the conventional hash-target Proof-of-Work puzzle while allocating mining tasks among multiple miners to improve block-generation performance and reduce duplication of assigned work. It also considers coordination and reward allocation within the parallel mining structure.

Parallel Proof-of-Work is the closest architectural predecessor to PoCol because both approaches coordinate work among multiple miners rather than leaving all candidate selection unstructured (Hazari & Mahmoud, 2020). The comparison must nevertheless remain bounded: conventional Bitcoin miners can vary extraNonce, Merkle root, timestamp, and other permitted fields, so overlapping numeric nonce values do not necessarily imply duplicate serialized headers. PoCol's experimentally established duplication result applies specifically to the common-template comparison model.

PoCol does not claim to invent parallel mining, nonce partitioning, collaborative rewards, or miner idling independently. Its proposed novelty is the integration of immutable-template agreement, deterministic disjoint nonce ownership, collaborative reward precommitment, and explicit post-range operating policies, together with a reproducible evaluation that separates work efficiency from wall-clock energy and service outcomes.

3.7.2. Collaborative and Team-Based Proof-of-Work

StrongChain introduces weak Proof-of-Work headers that make non-winning computational contributions visible and rewardable (Szalachowski et al., 2019). Instead of preventing overlapping work before mining, StrongChain recognizes useful evidence of mining effort after it has been generated. Its main contribution lies in reducing reward variance and making mining participation more transparent.

Collaborative Proof-of-Work forms dynamic mining groups based on participant capabilities and emphasizes secure contribution verification and fair reward distribution (Haque et al., 2025). This is directly relevant to PoCol because both protocols seek to reward miners participating in collaborative block production. Collaborative Proof-of-Work, however, focuses primarily on group formation, contribution verification, and reward fairness, while PoCol focuses on deterministic separation of the nonce ranges searched under a common immutable candidate-block template.

Proof of Team Sprint is a recent team-based collaborative consensus algorithm intended to reduce the energy inefficiency of conventional Proof-of-Work (Yonezawa, 2026). It organizes miners into teams that collaborate in solving the cryptographic mining puzzle and evaluates the effects of cooperation on energy use and reward fairness. Proof of Team Sprint is therefore the most recent directly related peer-reviewed collaborative consensus approach considered in this thesis.

PoCol differs from these approaches by defining collaboration at the level of deterministic nonce ownership bound to one common template. At the same time, StrongChain, Collaborative Proof-of-Work, and Proof of Team Sprint highlight two issues that PoCol has not yet solved completely: verifying non-winning miners' contribution and designing rewards that remain fair and Sybil-resistant. Registration or a claim of range completion is not, by itself, proof that the assigned search was performed (Haque et al., 2025; Szalachowski et al., 2019; Yonezawa, 2026).

3.7.3. Energy-Aware Mining Policies

Green-PoW reduces energy consumption by restricting mining participation during selected rounds to a smaller group of eligible miners (Lasla et al., 2022). Its energy-saving mechanism is based primarily on reducing the number of simultaneously active miners. Proof of Team Sprint follows a team-oriented approach in which collaborative execution is intended to reduce the amount of active computation attributed to individual participants (Yonezawa, 2026).

PoCol adopts a different energy-management principle. At the beginning of a collaborative round, all eligible miners may participate and receive deterministic, mutually disjoint nonce ranges. Under the proposed energy-saving policy, a miner that exhausts its assigned range without finding a valid candidate header stops active hashing and enters a low-power idle state until the deadline of the current collaborative mining round.

The energy consumed by miner i under this policy is expressed as:

$$E_i = P_i^{\text{active}}\, t_i^{\text{active}} + P_i^{\text{idle}}\, t_i^{\text{idle}} + E_i^{\text{coord}}$$

where $P_i^{(active)}$ is active mining power, $P_i^{(idle)}$ is idle power, and $E_i^{(coord)}$ represents communication, coordination, and state-transition overhead. Accordingly, nonce-range partitioning alone does not guarantee energy reduction. Energy savings arise only when miners spend a non-zero period in a power state that consumes less energy than active hashing.

PoCol may alternatively operate under a continuous-performance policy. In this mode, miners that complete their assigned ranges continue processing newly assigned non-overlapping work rather than entering the idle state. If aggregate hash rate, hardware efficiency, and experiment duration are equal to those of a continuously operating PoW baseline, this mode is expected to consume approximately the same energy. It may, however, improve accepted-block production or transaction throughput under particular evaluated configurations. Such performance improvement must be established experimentally and must not be treated as an automatic consequence of nonce partitioning.

Green-PoW and PoCol therefore reduce active mining through different mechanisms. Green-PoW reduces the number of miners allowed to participate, whereas PoCol reduces the active hashing duration of miners after they complete their assigned work. This distinction allows PoCol to preserve wider initial participation while exposing an explicit trade-off between lower energy use and continued mining performance.

3.7.4. Auditable Mining Contribution

A central challenge in collaborative and partitioned mining is verifying that a miner actually performed its assigned computation. A dishonest miner could claim that it exhausted its range, enter the idle state, and later request a share of the collaborative reward without having evaluated the assigned candidate headers.

Auditable Proof-of-Work provides a mechanism through which miners can produce probabilistic evidence that specified regions of the nonce domain were searched (Lerner, 2026). Although APoW is primarily directed toward detecting work omission and block-withholding behavior in mining pools, its auditing principle is relevant to PoCol. APoW is available as a 2026 arXiv preprint rather than a peer-reviewed journal or conference publication.

A strengthened PoCol design could associate each assigned nonce range with cryptographic commitments, sampled shares, or auditable search evidence. Such mechanisms would help distinguish miners that genuinely completed their assignments from miners that falsely claimed completion, thereby reducing free-riding, false idling claims, and reward manipulation.

3.7.5. Comparison with PoCol and Identified Research Gap

The reviewed approaches address different parts of the problem targeted by PoCol. Parallel Proof-of-Work distributes mining computation and provides the closest architectural precedent for parallel work allocation (Hazari & Mahmoud, 2020). StrongChain improves contribution visibility by recording weak Proof-of-Work solutions (Szalachowski et al., 2019). Collaborative Proof-of-Work emphasizes dynamic group formation, contribution verification, and reward fairness (Haque et al., 2025). Proof of Team Sprint organizes miners into collaborative teams and connects teamwork with energy efficiency and fairness (Yonezawa, 2026). Green-PoW reduces energy consumption by limiting the number of active miners in selected rounds (Lasla et al., 2022). APoW provides a possible mechanism for proving that assigned nonce regions were actually searched (Lerner, 2026).

No reviewed approach was found to combine, in the same specified design, all of the following elements: one identified immutable candidate-block template, deterministic mutually disjoint nonce ownership, a pre-committed collaborative reward transaction, explicit post-range idle and continuous operating policies, and a mechanism intended to verify completion of assigned work. The final item remains incomplete in the present PoCol implementation and is therefore a research requirement rather than a completed contribution.

PoCol addresses this combined design space by integrating the first four elements and by explicitly exposing the unresolved contribution-auditing requirement. Its contribution is therefore architectural integration and measured trade-off characterization, not a claim to have independently invented each component or to have solved all incentive and verification problems.

Table (3.7): PoCol vs. Collaborative PoW Approaches

Characteristic	**Parallel PoW**	**StrongChain**	**Collaborative PoW**	**Proof of Team Sprint**	**Green-PoW**	**APoW**	**PoCol**
Conventional hash-target puzzle	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Parallel or collaborative mining	Yes	Yes	Yes	Yes	Limited by round	Auditing-oriented	Yes
Immutable shared candidate template	Coordinated work	Not central	Not central	Not central	Not central	Not central	Explicit
Deterministic nonce-range allocation	Work allocation	No	Not central	Not central	No	Audited regions	Yes
Explicit mutually disjoint nonce ranges	Implementation-dependent	No	Not central	Not central	No	No	Yes
Contribution visibility or verification	Limited	Weak-header evidence	Explicit	Team-based	Runner-up evidence	Probabilistic audit	Requires extension
Energy-saving mechanism	Not primary	Not primary	Efficiency-oriented	Team cooperation	Fewer active miners	Not primary	Reduced active time
Post-range idle policy	No	No	No	Not explicit	Non-selected miners stop	No	Explicit
Continuous-performance policy	Yes	Yes	Yes	Team-dependent	No	No	Explicit
Main strength	Parallel work allocation	Contribution visibility	Fair collaboration	Team cooperation	Direct active-miner reduction	Auditable search evidence	Integrated work partitioning and operating policies

Main limitation	Manager dependence	Does not prevent overlap	Pool complexity	Team assumptions	Restricted participation	Not a complete consensus protocol	Contribution proof and complete distributed implementation remain incomplete

The comparison does not imply that PoCol is universally superior to the examined protocols. Parallel PoW provides the closest architectural baseline, Green-PoW provides the most relevant energy-policy baseline, Proof of Team Sprint provides the most recent peer-reviewed team-based collaborative baseline, and APoW provides a relevant contribution-auditing direction.

3.8. Comparative Analysis and Research Question Synthesis

This section synthesizes the selected consensus families in a context-dependent manner. A mechanism that is appropriate for a public adversarial network may be unsuitable for a governed consortium, and a mechanism that reduces computation may rely on stronger identity, stake, hardware, or governance assumptions. The purpose of the synthesis is therefore to identify trade-offs and deployment fit rather than to produce a universal ranking (Baboi, 2023; Wang et al., 2019; Xiao et al., 2020).

3.8.1. Comparative Analysis

Meaningful consensus comparison requires explicit attention to validator selection, fault assumptions, participation model, finality, communication cost, resource demand, and governance. These variables interact: changing the mechanism used to obtain Sybil resistance or finality typically changes both the attack surface and the operating cost. Accordingly, no consensus family can be considered optimal in all deployment contexts (Bano et al., 2019; Wang et al., 2019; Xiao et al., 2020).

PoW is well suited to open resource-based participation and has extensively studied adversarial properties, but it incurs sustained computational energy and remains sensitive to hash-power concentration and network effects (de Vries, 2018; Gervais et al., 2016). PoS greatly reduces continuous computation but depends on stake distribution, randomness, validator incentives, and protocol-specific accountability (Ge et al., 2022; Saad et al., 2021). PBFT-style protocols provide deterministic agreement for bounded replica sets under explicit Byzantine thresholds, but communication cost limits straightforward scaling (Castro & Liskov, 1999; Castro & Liskov, 2002).

DPoS and PoA can obtain high operational efficiency by limiting active block producers to elected or governed validators, but that design choice increases dependence on governance and may increase collusion or capture risk (Barinov et al., 2018; BitShares, n.d.; Wang et al., 2022). PoB, PoR, PoC, PoET, and PoSpace move the scarce resource or trust anchor to economic sacrifice, reputation, contribution, trusted hardware, or storage. These mechanisms expand the design space, but each introduces its own measurement, concentration, or trust assumptions.

The principal conclusion is therefore conditional: consensus design is a multi-objective choice among openness, security assumptions, finality, performance, governance, and resource demand. Improving one dimension does not mechanically require degrading another, but practical designs commonly achieve gains by changing assumptions elsewhere. The appropriate mechanism depends on which assumptions and trade-offs are acceptable for the intended application.

3.8.1.1. Security, Scalability, and Energy Trade-offs

The security-scalability-energy relationship should therefore be described as a family of context-dependent trade-offs rather than a universal trilemma. PoW can provide robust open participation at high operating energy; PoS reduces continuous computation but introduces stake and incentive assumptions; PBFT-style designs provide deterministic agreement under explicit fault bounds but incur communication costs; and storage-, authority-, hardware-, reputation-, contribution-, and burn-based designs change the resource or trust anchor. These differences are precisely why cross-family labels must remain qualitative (Castro & Liskov, 1999; de Vries, 2018; Dziembowski et al., 2015; Ge et al., 2022; Karantias et al., 2020).

3.8.1.2. Comparative Summary

Taken together, the reviewed evidence supports treating consensus mechanisms as context-dependent coordination designs rather than interchangeable algorithms. PoW, PoS, BFT/PBFT, delegated, authority-based, storage-based, and contribution/reputation-based mechanisms each occupy different points in the design space. The purpose of Tables 3.7 and 3.8 is to summarize these tendencies for decision support, not to establish a universal ranking.

Table (3.8): Application-oriented qualitative decision matrix for representative blockchain consensus mechanisms.

Consensus Algorithm	Most Suitable Deployment Context	Main Strength	Main Limitation
PoW	Open public blockchains operating in highly adversarial environments	Strong robustness and openness in permissionless settings	Very high energy consumption and limited scalability
PoS	Public blockchains seeking lower energy use with broad participation	Improved energy efficiency with strong economic incentive structure	Vulnerable to stake concentration and governance imbalance
DPoS	High-throughput decentralized applications and performance-sensitive public platforms	High throughput and low validation overhead	Greater exposure to delegate collusion and centralization
PoA	Enterprise systems, private blockchains, and permissioned IoT deployments	Fast confirmation and low resource consumption	Strong reliance on trusted authorities and limited openness
BFT	Small- to medium-scale permissioned systems with bounded trust	Strong fault tolerance under adversarial conditions	Communication overhead limits scalability
PBFT	Consortium blockchains requiring deterministic finality and strong consistency	High assurance and deterministic agreement	Poor scalability with large validator sets
PoET	Trusted-hardware-based enterprise or consortium environments	Low computational cost and efficient validator selection	Dependence on trusted execution environments
PoH	High-performance blockchain systems prioritizing ordering efficiency and throughput	Very high throughput and reduced coordination overhead	Greater architectural complexity and dependence on sequencing assumptions
PoSpace	Resource-conscious public blockchain systems seeking lower energy use than PoW	Lower energy burden compared with mining-intensive models	Dependence on storage resources and hardware-related constraints
PoB	Specialized public blockchain systems using economic sacrifice as a participation signal	Avoids continuous mining competition	Fairness and concentration concerns linked to asset destruction

PoR	Governance-controlled or reputation-sensitive blockchain environments	Rewards consistent historical behavior	Vulnerable to reputation manipulation and barriers to new entrants
PoC	Application-specific systems where useful contribution can be measured	Aligns validation influence with system contribution	Difficult contribution assessment and manipulation risk

Note: This matrix is intended as a qualitative decision-support summary derived from the comparative analysis presented in this survey. The listed deployment contexts are representative rather than exclusive, and the suitability of a given consensus mechanism may vary depending on validator composition, governance model, threat environment, performance targets, and sustainability requirements

3.8.2. Application-Oriented Selection of Consensus Algorithms

Because consensus mechanisms differ in trust model, validator structure, resource requirement, and finality, protocol selection must be driven by application requirements rather than by a single claim of 'best' performance. Table 3.8 maps the selected families to representative deployment contexts and summarizes their principal strengths and limitations (Wang et al., 2019; Yao et al., 2023).

The matrix provides contextual decision support by linking consensus assumptions to representative deployment requirements. It should not be read as a prescriptive rule: implementations within the same family can differ substantially in performance, security, and governance (Wang et al., 2019; Xiao et al., 2020; Yao et al., 2023).

For open adversarial public networks, PoW and some PoS designs provide permissionless participation under different resource assumptions. PBFT-style and PoA systems are generally more natural in governed validator sets; PoET additionally assumes trusted execution hardware; DPoS uses elected producers; and PoSpace substitutes storage commitment for continuous hashing. PoH is not treated here as an independent consensus family for deployment selection, but as an ordering/time primitive whose practical properties depend on the surrounding architecture (Xiao et al., 2020; Yakovenko, 2018).

Table 3.9 provides a high-level qualitative synthesis across deployment suitability, fault tolerance, energy demand, scalability tendency, and security profile. Because the underlying studies use different assumptions and evaluation methods, the table summarizes broad tendencies rather than measured effect sizes or universal performance rankings.

Table (3.9): Qualitative comparative summary of the selected blockchain consensus algorithms in terms of deployment suitability, fault tolerance, energy consumption, scalability, and security profile.

No.	Consensus Algorithm	Deployment Suitability	Fault Tolerance	Energy consumption	Scalability	Security Profile
1	BFT	Permissioned	Protocol-dependent; PBFT-style designs commonly assume <1/3 Byzantine	Low	Limited	Strong in bounded-trust settings
2	PBFT	Permissioned	f Byzantine faults among 3f+1 replicas	Low	Limited	Strong with deterministic finality
3	PoW	Permissionless	Resilient unless majority hash power is controlled	Very High	Limited	Strong in open adversarial environments
4	PoS	Primarily permissionless	Resilient unless majority stake is highly concentrated	Low	Good	Strong, but dependent on stake distribution and governance design
5	PoSpace	Permissionless	Resilient unless storage capacity is highly concentrated	Low to Moderate	Good	Moderate to strong, depending on storage distribution
6	PoB	Permissionless	Resilient unless economic influence becomes highly concentrated	Moderate	Good	Moderate, with economic-centralization concerns
7	DPoS	Permissioned / Permissionless	Dependent on delegate honesty and voting structure	Low	Very Good	Moderate to strong, but vulnerable to collusion and governance capture
8	PoA	Permissioned	Dependent on authority honesty	Very Low	Very Good	Strong in controlled environments, but weaker in openness
9	PoET	Permissioned	Dependent on trusted hardware and bounded trust	Very Low	Very Good	Moderate to strong, subject to trusted-execution-environment assumptions
10	PoH	Architecture-dependent; ordering/time primitive	Determined by surrounding consensus protocol	Not standalone	Architecture-dependent	Depends on surrounding consensus and sequencing assumptions
11	PoC	Permissioned / Application-specific	Dependent on contribution assessment integrity	Low	Good	Moderate, with measurement and manipulation concerns

12	PoR	Permissioned / Governance-controlled	Dependent on reputation integrity and governance rules	Very Low	Good to Very Good	Moderate to strong, but sensitive to reputation manipulation

Note: The entries in this table are qualitative comparative indicators rather than universally standardized measurements. Labels such as very low, low, moderate, good, and very good reflect broad tendencies reported in the reviewed literature and the architectural characteristics of each mechanism. Because consensus algorithms differ substantially in trust model, validator composition, implementation design, and deployment context, these labels should be interpreted as analytical summaries rather than as fixed quantitative benchmarks

3.8.3. Answers to the Research Questions

The comparative analysis answers the Chapter 3 research questions as follows.

RQ3.1: Which principal functional characteristics serve to differentiate blockchain consensus algorithms with respect to their validation model, finality, fault tolerance, throughput, and scalability? (Xiao et al., 2020)

Response to RQ3.1

Consensus mechanisms differ in the resource or authority used for validation, the method of choosing block producers or quorums, the finality model, the tolerated faults, and the communication pattern. PoW uses computational work, PoS uses stake, PBFT-style protocols use authenticated quorum agreement, and DPoS/PoA restrict block production to elected or governed validators. These choices directly affect latency, throughput, scalability, governance, and deployment suitability (Barinov et al., 2018; BitShares, n.d.; Castro & Liskov, 1999; King & Nadal, 2012; Xiao et al., 2020).

RQ3.2: In what ways do prominent consensus algorithms diverge in their resilience to security threats, including Byzantine faults, majority attacks, Sybil attacks, collusion, and pressures toward centralization? (Wang et al., 2019; Xiao et al., 2020)

Response to RQ3.2

Consensus families expose different attack surfaces because they rely on different security assumptions. PoW is sensitive to hash-power concentration, strategic mining, and network attacks (Gervais et al., 2016); PoS depends on stake distribution and protocol-specific accountability (Abellán Álvarez et al., 2024; Ge et al., 2022); PBFT-style protocols depend on explicit Byzantine thresholds and timing assumptions (Castro & Liskov, 1999); and DPoS/PoA gain efficiency through smaller validator sets at the cost of greater governance and collusion dependence (Barinov et al., 2018; BitShares, n.d.; Wang et al., 2022).

RQ3.3: How does energy consumption differ across the various consensus families, and which design decisions exert the greatest influence on the sustainability profile of blockchain systems? (Bhutta et al., 2021; Islam et al., 2023)

Response to RQ3.3

Energy demand differs primarily because consensus families use different resources and operating patterns. Competitive PoW requires sustained hashing, whereas PoS, DPoS, PoA, and many BFT-style deployments avoid network-wide hash competition and therefore generally use much less operational energy. Exact values remain implementation- and boundary-dependent; hardware, validator count, communication, workload, and infrastructure must be considered before making numerical comparisons (Platt et al., 2021; Sedlmeir et al., 2020).

RQ3.4: What trade-offs emerge among functionality, security, and energy efficiency in the selection of a consensus mechanism for a given blockchain application? (Wang et al., 2019; Xiao et al., 2020)

Response to RQ3.4

Trade-offs among functionality, security, governance, and energy are context dependent. A mechanism that reduces computation may do so by relying on economic stake, a smaller validator set, governed identities, trusted hardware, or another scarce resource. Protocol selection should therefore compare the full assumption set against the application's performance, threat, governance, and sustainability requirements rather than optimizing one metric in isolation (Wang et al., 2019; Xiao et al., 2020).

RQ3.5: Which consensus mechanisms are most appropriate for distinct deployment contexts, such as public blockchain platforms, consortium networks, enterprise environments, and resource-constrained systems? (Islam et al., 2023; Xiao et al., 2020)

Response to RQ3.5

No consensus mechanism is universally optimal. PoW is a natural fit for some open adversarial settings; PoS offers permissionless operation with lower continuous computation under stake-based assumptions; PBFT-style protocols fit bounded governed validator sets requiring deterministic finality; PoA and PoET fit identity- or hardware-governed environments; and PoR/PoC are appropriate only when reputation or contribution can be defined and verified credibly. Deployment suitability remains implementation specific (Xiao et al., 2020; Yao et al., 2023; Zhuang et al., 2019).

The main value of the comparative review is therefore to identify the structural conditions under which each family is more or less appropriate. Changes that improve energy, latency, or scalability often alter trust, governance, resource distribution, or attack exposure. These coupled effects should be evaluated explicitly rather than collapsed into a single ranking (Page, Moher, et al., 2021; Xiao et al., 2020).

3.9. Summary

This chapter examined twelve representative consensus mechanisms and supporting primitives using three analytical dimensions: functionality, security assumptions, and energy consumption. The review does not identify a universally superior mechanism. Instead, it shows that validation logic, trust model, fault tolerance, finality, governance, scalability, and resource demand interact differently across deployment contexts.

The review also demonstrates why direct cross-family numerical comparison is methodologically difficult. Terminology, hardware, network scale, workloads, trust assumptions, and reported metrics vary substantially across studies. The comparative tables should therefore be interpreted as transparent qualitative synthesis, not as universal quantitative benchmarks. This boundary motivates the thesis's later use of matched controls when evaluating PoCol.

Chapter 4
System Model and Problem Formulation

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System Model and Problem Formulation

4.1. Overview of the Proposed Setting

This chapter formalizes the system, threat assumptions, performance measures, work-accounting measures, energy model, and design objectives used to evaluate Proof-of-Collaboration (PoCol). PoCol is a PoW-derived coordinated design: it retains a hash-target condition and cumulative-work chain-selection objective, but it changes template formation, work ownership, round lifecycle, and reward allocation. The analysis distinguishes the permitted nonce domain from the SHA-256 output space and distinguishes exact-input work redundancy from electrical-energy consumption.

The system comprises miners, a peer-to-peer network, and clients that submit transactions. In the controlled same-template PoW comparator, miners instantiate candidate headers from the same nonce-free header template but choose nonce values independently. In PoCol, miners instantiate candidate headers from the same template using only nonce values within their assigned disjoint ranges. This comparator is intentionally artificial: it isolates coordinated versus uncoordinated nonce selection and does not claim that conventional Bitcoin miners normally hash identical headers (Nakamoto, 2008).

The primary work-allocation property of PoCol under the common-template comparator is the elimination of exact duplicate candidate-header inputs among honest miners when the assigned nonce ranges are disjoint. That property does not imply lower electricity consumption. Electrical energy is modeled from wall-clock residence in declared power states, so an energy reduction is licensed only when active power, active duration, or active participation is reduced. Chapters 6 and 7 report work, energy, service, and latency as separate quantities.

4.2. Network Model

$M = \{m_1, m_2, \dots, m_n\}$ is the set of miners participating in the blockchain network, where $n = |M|$, that is the number of miners. In this case, each miner m_i contributes a fraction of hashing power of the network h_i such that $\sum_{i=1}^{n} h_i = 1$ in the baseline PoW system. Practically, h_i reflects the computational capability of miner

m_i and is determined by the mining hardware of miner m_i (ASICs, GPUs, or special accelerators) and energy cost assigned to miner m_i. The power consumption of the network can be denoted by P (in W), and the process of generating a block is modeled as a Poisson process with the arrival rate of λ blocks in a second, where $\frac{1}{\lambda}$ is the expected delay time between blocks as in standard Bitcoin analysis.

The network can be modeled as a partially synchronous peer-to-peer network following the general approach in blockchain and BFT analysis. Messages, including transactions, block headers, complete blocks, and control messages, propagate using a gossip protocol. A bound Δ applies to message delay after the network's global stabilization time. Before stabilization, the network may behave asynchronously, while the stated safety properties are intended to hold. Each miner maintains a local view of the blockchain state. The blockchain is an append-only chain beginning at the genesis block, and the state-transition function applies the transactions in committed blocks to derive the current ledger state. Clients submit transactions that are propagated and stored in local mempools. In PoCol, the protocol aims to reconcile the eligible transaction set so that every honest miner derives the same canonical block template. That block template fixes the transaction list, Merkle root, previous-block hash, deterministic timestamp, target, and other consensus-critical fields. From it, every miner derives the same nonce-free header template. During a round, candidate headers are instantiated by inserting nonce values into the nonce field, no other consensus-critical field may vary. A candidate header satisfies the proof-of-work condition when $\mathrm{H}\big(\mathrm{Serialize}(\mathrm{header})\big) \leq T$, where T is the target threshold. The corresponding block is valid only if it also satisfies all non-PoW consensus rules (Nakamoto, 2008; Sapra et al., 2023).

In the controlled common-template PoW baseline used in this thesis, miner m_i evaluates candidate headers obtained by instantiating the same nonce-free header template with nonce values from the permitted nonce domain N. Let Q_i denote the number of block-header hash evaluations performed by miner m_i in a round, and let $Q_{\mathrm{total\_PoW}} = \sum_{i=1}^{n} Q_i$. A duplicated evaluation occurs only when two miners hash the same serialized candidate header. Reuse of the same numeric nonce is not a duplicate if any other header field differs. The baseline deliberately fixes a common template to isolate coordinated versus uncoordinated selection of nonce values; it does not claim that conventional Bitcoin miners normally hash identical headers. The SHA-256 hash-output space $\{0, 1\}^{256}$ is not partitioned among miners. PoCol partitions the nonce domain N. For a fixed header template, each assigned nonce range therefore determines a disjoint set of serialized candidate-header inputs. $Q_{\mathrm{total\_PoCol}}$ denotes the number of candidate-header hash evaluations performed by PoCol miners during the round.

Time is represented in discrete rounds or epochs. During each round, miners evaluate candidate headers generated from their assigned nonce ranges. When a miner obtains a header hash satisfying the target, it broadcasts the corresponding block, and nodes apply the protocol's cumulative-work chain-selection rule (Nakamoto, 2008). After a block is accepted, the block subsidy and transaction fees are distributed according to the protocol-defined reward function described later in the thesis (Lasla et al., 2022; Sethi et al., 2024).

4.3. Adversary and Threat Model

The adversary can be considered as a Probabilistic Polynomial-Time (PPT) entity capable of corrupting a fraction of miners to manipulate their behavior. Corrupted miners (called Byzantine miners) can deviate from the protocol in any manner, such as withholding blocks, publishing contradictory histories, trying to double-spend, and colluding in order to get majorities to get more rewards (Castro & Liskov, 1999; Lamport et al., 1982). Honest miners adhere to the specified protocol, such as nonce-domain assignment, hashing, block propagation and reporting their rewards.

We model the adversary as controlling a fraction β of the total network hash power. In Nakamoto-style PoW analyses, adversarial hash fractions below 0.5 are commonly considered within the corresponding security model (Del Monte et al., 2020; Nakamoto, 2008). This thesis does not assume that the same threshold transfers unchanged to PoCol, because deterministic nonce ownership, common-template coordination, and the PoCol round structure may alter the relevant adversarial conditions. Establishing the corresponding PoCol security threshold remains a subject for formal analysis. When discussing classical BFT-style reasoning separately, the standard fewer-than-one-third Byzantine bound applies under the corresponding $3f + 1$ replica model (Castro & Liskov, 1999; Lamport et al., 1982). The adversary may coordinate corrupted miners to execute certain attacks, including the following:

- Double spending. This involves broadcasting a transaction, getting it into a block, and attempting to revert the block by creating a longer private chain which excludes the transaction (applicable to classic PoW) (Del Monte et al., 2020; Nakamoto, 2008).
- Selfish mining or block withholding mining. This involves withholding found blocks to create forks which change rewards or reduce the effective contribution of the honest miners (Gervais et al., 2016; Sethi et al., 2024).
- Non-collaborative behavior in PoCol, which includes ignoring assigned nonce ranges, mining a modified block template that deviates from the canonical one, and copying nonce ranges from honest miners to claim unfair rewards, failing to report the number of hashes done or distributing block rewards amongst collaborators.
- Eclipse attacks or network partition attacks, which involve isolating groups of honest miners from the rest of the network to achieve their aims (Del Monte et al., 2020; Sapra et al., 2023).

It is also assumed that the adversary has partial control of the network channels of corrupted miners, which enables them to delay messages or drop them selectively. However, the adversary cannot overcome cryptographic primitives, such as hash functions or digital signatures, as well as forge messages from honest miners (Lamport et al., 1982; Nakamoto, 2008). The network is expected to be eventually synchronous with a bounded delay, Δ, thus ensuring liveness under the honest-majority assumption (Castro & Liskov, 1999; Lamport et al., 1982).

The protocol is designed to limit the value of such deviations, but this chapter does not assume that the listed security objectives have already been proved. In particular, permissionless Sybil resistance, contribution verification, template-grinding resistance, and PoW-equivalent double-spend probabilities require separate formal or empirical analysis.

4.4. Performance and Energy Metrics

The quantitative framework compares PoW and PoCol using four distinct classes of measures: service (accepted blocks and throughput), latency/finality proxies, physical work (total, unique, and duplicate candidate-header evaluations), and electrical energy derived from wall-clock power-state residence. Keeping these classes separate prevents a work-redundancy reduction from being reported as an electricity reduction without a power-time mechanism.

4.4.1. Throughput, latency, and finality

Throughput or transaction throughput refers to the total number of valid transactions confirmed by the blockchain in a given period of time, where it is measured as transactions per second. In order to calculate $N_{tx}\,(t_1, t_2)$ as the total number of transactions that have been included in the canonical chain in the time interval of $t_1$and t_2, the average throughput in this time interval can be calculated as

$$\Theta = \frac{N_{\text{tx}}(t_1, t_2)}{t_2 - t_1}$$

Latency reflects the time taken from the moment when the transaction is broadcasted until the moment when it is confirmed in the network. The time $t_{\text{broadcast}}$ used here is the point when the client broadcasts the transaction and t_{confirm} is the time of its inclusion in the block that forms part of the canonical chain to a sufficient depth k (for instance, it is 6 confirmations in Bitcoin). Therefore, latency is defined as $L = t_{\text{confirm}} - t_{\text{broadcast}}$, which is a function of the block interval $\frac{1}{\lambda}$ and network delays Δ as well as the number of confirmations k to be reached in order to be effective against attacks conducted by adversaries with a hashing power of β.

Finality refers to the point at which a transaction is treated as practically irreversible under a stated adversary model and confirmation policy. In probabilistic-finality systems, confidence generally increases as additional blocks extend the chain, but the exact rollback probability depends on the protocol, adversarial resource share, network model, and chain-growth behavior (Nakamoto, 2008). PoCol retains a cumulative-work chain-selection design and therefore targets probabilistic finality, but the Bitcoin confirmation analysis cannot be transferred unchanged: formal equivalence for common prefix, chain growth, chain quality, and k-confirmation rollback probability under deterministic nonce ownership has not been established. PBFT-style protocols instead provide deterministic finality after the required quorum conditions are met under their own assumptions.

4.4.2. Energy Metrics and Trade-offs

Power (P) represents the rate at which the network consumes electrical energy, measured in watts (joules per second). It aggregates the consumption of all miners' hardware, including compute, memory, networking, and cooling overheads (Koutra & Tenentes, 2024; Pham et al., 2020). For a given consensus protocol with block-generation rate λ, the expected energy consumed per block is

$$E_{\text{block}} = \frac{P}{\lambda} \tag{4.1}$$

This relation follows directly from modeling block generation as a Poisson process with a rate λ : in expectation, one block is produced every $\frac{1}{\lambda}$ seconds, during which the network consumes power P (Tereshchenko & Kyrychenko, 2024). If each block contains, on average, B_{tx} transactions, the expected energy per transaction is

$$E_{\text{tx}} = \frac{P}{\lambda B_{\text{tx}}} = \frac{P}{\lambda \cdot B_{\text{tx}}} \tag{4.2}$$

The notation below distinguishes hash evaluations from the domain of values being searched (Sapra et al., 2023; Tereshchenko & Kyrychenko, 2024). Under the common-template PoW baseline, $Q_{\text{total\_PoW}} = \sum_{i=1}^{n} Q_i$ is the total number of candidate-header hash evaluations performed across all miners in an epoch. Let $U_{\text{total\_PoW}}$ be the number of distinct serialized candidate-header inputs among those evaluations. The redundancy factor of PoW is then

$$R_{\text{PoW}} = \frac{Q_{\text{total\_PoW}}}{U_{\text{total\_PoW}}} \tag{4.3}$$

In the PoCol model, the canonical block and header templates are fixed for the round, and the permitted nonce domain is partitioned into disjoint ranges.

Consequently, every candidate-header input evaluated by an honest miner is distinct from those evaluated by other honest miners in that round:

$$Q_{\text{total\_PoCol}} = U_{\text{total\_PoCol}}, \qquad R_{\text{PoCol}} = 1 \tag{4.4}$$

Equations (4.3) and (4.4) define a work-redundancy measure, not an electrical-energy model. A lower redundancy factor means that a larger fraction of physical evaluations are unique under the declared common-template boundary, but it does not imply that the hardware consumed less energy. Electrical energy is instead computed from the wall-clock identity $E = \sum_i P_i t_i$, where P declared power states of miner i.

Accordingly, this thesis does not multiply aggregate network power by the duplicate-evaluation ratio. Work efficiency is reported directly from Q_{total} and U_{total}, whereas energy comparisons use measured or modeled power-state residence over the same horizon. Service-normalized quantities such as energy per accepted block are reported only when the denominator is defined.

This separation supersedes the earlier analytical intuition that attempted to convert candidate-evaluation redundancy into an energy-savings ratio. The authoritative energy accounting used in the final experiments is the wall-clock state-residency model implemented in Chapters 6 and 7.

As the number of miners n increases in the evaluated common-template baseline, duplicate candidate-header evaluations may increase, while PoCol maintains $R_{\text{PoCol}} = 1$ by construction. This property depends on two conditions: (i) all honest miners use the same immutable header template in a round, and (ii) their assigned nonce ranges are disjoint. If either condition is violated, duplicate serialized candidate-header inputs can reappear (Sapra et al., 2023; Zhang et al., 2023).

The energy-service trade-off must be evaluated with a declared horizon and service denominator. At fixed total power and a fixed observation horizon, total energy is fixed regardless of how many blocks are produced; block count changes energy per accepted block rather than the horizon total. When miners enter lower-power states, total energy may decrease, but block production and round latency can also change. For probabilistic-finality systems, confirmation depth and block interval must therefore be reported with the energy metric rather than treated as independent quantities (Del Monte et al., 2020; Nakamoto, 2008).

PoCol is designed to decouple exact-input duplication from coordinated work allocation, not to equate redundancy removal with electricity savings. Under one immutable template and honest disjoint assignments, PoCol eliminates exact duplicate candidate-header inputs. Calculated energy decreases only when the evaluated operating policy creates non-zero residence in a lower-power state or reduces active participation. Later chapters therefore report physical work, total energy, energy per accepted block, accepted blocks, and round latency together.

4.5. Problem Statement (Formal)

We now formalize the design problem addressed by the PoCol consensus algorithm. Consider a permissionless blockchain network with:

- A set of miners $M = \{m_1, \ldots, m_n\}$, where each miner m_i contributes hashing power fraction h_i and consumes power P_i, so that total network power is $P = \sum_{i=1}^{n} P_i$.
- A target block-generation rate λ (e.g., 1 block per 600 seconds) and an associated transaction throughput requirement Θ (transactions per second) determined by application needs (Del Monte et al., 2020; Nakamoto, 2008).
- An adversary that can corrupt a subset C ⊆ M of miners with total hashing power at most β (i.e., $\sum_{m_i \in C} h_i \leq \beta$), where β s a design parameter representing the adversarial share of network hash power. The relevant security threshold for PoCol is not assumed to be identical to the Nakamoto-style PoW threshold and remains to be established through formal analysis (Del Monte et al., 2020; Nakamoto, 2008)).
- A security threshold ε specifying the maximum acceptable probability that the adversary can successfully perform a double-spend attack against a k-confirmation transaction (e.g., $\varepsilon = 10^{-6}$) (Del Monte et al., 2020; Nakamoto, 2008).

Let Π denote the design space of PoW-derived coordinated protocols considered by this thesis. Members of Π retain the underlying hash function and a cumulative-work chain-selection objective, but may change canonical-template formation, round structure, work assignment, verification rules, miner-set handling, and reward distribution. This broader definition reflects the actual PoCol design rather than restricting Π to work assignment and rewards alone.

- A work-assignment function A_π that, in each epoch e, maps miner identities and possibly past history to disjoint (or minimally overlapping) subsets of the permitted nonce domain for honest miners.
- A verification mechanism V_π that verifies whether a proposed block conforms to the valid round assignment, including successful-nonce range membership, and may be extended with auditable evidence to verify the search contribution of non-winning miners claiming rewards. The latter contribution-verification capability is a design requirement and is not implemented in the completed PoCol evaluation.
- A reward function R_π that maps protocol-defined eligibility or, where available, verifiable contribution evidence to reward shares when a block is confirmed. In the baseline PoCol design evaluated in this thesis, reward eligibility is based on the registered miner set rather than on a completed proof of individual search contribution; contribution-weighted reward functions remain a design extension (Lasla et al., 2022; Sethi et al., 2024).

For a given protocol π, define the following quantities in steady state:

- $E_{block}(\pi)$: expected or run-normalized energy per accepted block, computed from the declared wall-clock power-state model and reported as undefined when no block is accepted.
- $E_{\text{tx}}(\pi)$: expected energy consumed per transaction, as per Equation (4.2).
- $L_{\text{avg}}(\pi)$: average transaction-confirmation latency for k confirmations, given λ, Δ, β, and the protocol's fork behavior (Del Monte et al., 2020; Nakamoto, 2008).
- $P_{\text{ds}}(\pi, k, \beta)$: probability that an adversary controlling hash fraction β can execute a successful double-spend attack against a k-confirmation transaction under protocol π (Del Monte et al., 2020; Nakamoto, 2008).
- $F_{\text{dev}}(\pi)$: a fairness metric capturing the deviation between miners' long-run reward shares and their hashing-power fractions, for example

$$F_{\text{dev}}(\pi) = \max_i |r_i(\pi) - h_i|,$$

where $r_i(\pi)$ is the long-run fraction of total rewards earned by miner m_i under π (Sapra et al., 2023; Xu et al., 2021).

The design problem can then be expressed as a constrained optimization problem:

Minimize $E_{\text{block}}(\pi)$.

over $\pi \in \Pi$

subject to

1. $\lambda(\pi) \approx \lambda(PoW)$ and $\Theta(\pi) \geq \Theta(PoW)$ (performance constraints),
2. $P_{\text{ds}}(\pi, k, \beta) \leq \varepsilon$ (security constraint),
3. $F_{\text{dev}}(\pi) \leq \delta$ (fairness constraint),
4. $E_{block}(\pi) \leq (1 - S_{min})E_{block}(PoW)$ under the same declared accounting boundary and with the associated service/latency constraints reported explicitly (energy-service target).
5. π is implementable without specialized trusted hardware and is compatible with standard blockchain data structures (deployability constraint).

These constraints define design targets rather than properties proved by the completed experiments. The frozen evaluation establishes exact-input work integrity and selected power-state energy reductions under simulator assumptions, but the preregistered service, useful-floor, fairness, incentive, and formal security objectives are not all satisfied.

The resulting design problem is therefore multi-objective: coordinate PoW-style candidate evaluation to avoid exact duplicate inputs, then determine whether lower-power operation produces a useful energy reduction without unacceptable loss of block production or latency. PoCol is evaluated as one candidate design; it is not claimed to preserve every deployed-PoW guarantee or to satisfy the full constraint set unconditionally.

4.6. Summary

This chapter defined the PoCol system, network, and adversary models and distinguished the permitted nonce domain from the SHA-256 output space. It also defined throughput, latency, finality, physical work, power, energy, and service-normalized metrics for controlled comparison. The common-template and disjoint-range mechanism establishes an exact-input-duplication property under honest execution, while electrical-energy claims require explicit wall-clock power-state accounting.

The resulting optimization framework is multi-objective: it treats safety, liveness, energy, service, fairness, deployability, and strategic behavior as separate requirements. Later chapters evaluate only a bounded subset of these objectives through simulation. Formal security equivalence, incentive compatibility, quantitative fairness, and complete distributed deployment remain future work.

Chapter 5
The Proof-of-Collaboration Consensus Algorithm

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The Proof-of-Collaboration Consensus Algorithm

5.1. Introduction and Design Goals

PoCol is motivated by the high operating energy of competitive PoW, but the protocol does not assume that duplicated nonce values are the principal cause of real-world Bitcoin energy use. In Bitcoin-style mining, miners can vary the coinbase extra nonce, Merkle root, timestamp, version bits, and other permitted fields, so the same numeric nonce can correspond to different serialized headers (Nakamoto, 2008). The PoCol evaluation therefore uses an intentionally controlled common-template baseline to isolate the effect of coordinated versus uncoordinated candidate selection.

In Bitcoin-style PoW, miners repeatedly hash serialized candidate block headers. The 32-bit header nonce is one mutable field, but exhausting its values does not exhaust the broader candidate-header input domain: a miner can change the coinbase extra nonce, thereby changing the Merkle root, and may update other permitted template fields before scanning the nonce domain again. Therefore, two miners using the same numeric nonce do not necessarily evaluate the same hash input. Exact duplication occurs only when the complete serialized headers are identical. The probability that a miner finds the next block depends primarily on its share of the total hash rate, and network energy consumption scales with aggregate mining activity and hardware efficiency (de Vries, 2018; Nakamoto, 2008). Empirical studies report substantial energy use by major PoW networks (Tereshchenko & Kyrychenko, 2024; Zhang et al., 2023). PoS- and BFT-style protocols reduce continuous block-header hashing but introduce different assumptions and mechanisms, including stake management, slashing, committee selection, and partial synchrony (Castro & Liskov, 1999; Lamport et al., 1982).

This thesis describes Proof of Collaboration (PoCol), which seeks to retain the hash-based proof-of-work structure while coordinating candidate-header evaluation. In each round, PoCol defines one canonical block template and one nonce-free header template. The permitted nonce domain is partitioned into non-overlapping ranges that are assigned deterministically and pseudo-randomly to registered miners. Each miner evaluates only candidate headers formed by inserting nonce values from its assigned range. A nonce value is successful only relative to a specific header template and target, namely when the resulting serialized header hashes to a value at or below the target. Under the common-template invariant, disjoint nonce ranges ensure that honest miners do not evaluate the same serialized candidate-header input during the round.

The design goals are: (1) eliminate exact duplicate serialized candidate-header inputs among honest miners under one common immutable template; (2) derive work assignments deterministically from public protocol state without a trusted range coordinator; (3) pre-commit a collaboration-aware reward transaction; and (4) expose explicit post-range operating policies so that work coverage and electrical-energy effects can be measured separately. These are protocol objectives, not claims of equivalence with Bitcoin's production mining workflow.

This chapter specifies the PoCol protocol architecture, including miner registration and lifecycle, common-template construction, nonce-domain assignment, collaborative search, block verification, reward distribution, and dynamic miner-set handling. The security discussion is deliberately preliminary: it identifies required properties and attack surfaces but does not constitute a formal proof of common prefix, chain growth, chain quality, Sybil resistance, contribution correctness, or incentive compatibility.

5.2. System and Threat Model

5.2.1. Network and Node Assumptions

The system model distinguishes validating nodes from miners. Validating nodes maintain the ledger and verify consensus rules; miners additionally participate in candidate-header evaluation and block production. Communication occurs over a peer-to-peer network that is modeled analytically as eventually or partially synchronous, while the current simulator implements only a subset of the distributed template-agreement and propagation behavior specified in this chapter.

Each honest node maintains a local ledger view and receives pending transactions through peer-to-peer propagation. Because local mempools can differ transiently, the protocol requires an explicit template-agreement procedure before mining begins. Agreement on the template is therefore an additional coordination requirement introduced by PoCol, not an assumption that follows automatically from gossip.

5.2.2. Miner Identities and Cryptographic Primitives

Each registered miner is represented by a long-term public key used for registration, deregistration, signatures, and the reward address. The protocol assumes a standard secure digital-signature scheme and a cryptographic hash function modeled, for the relevant analysis, as providing preimage resistance and unpredictable outputs. A public key identifies a protocol participant but does not by itself provide Sybil resistance; the cost or governance rule for obtaining mining identities remains a separate protocol-design requirement.

PoCol relies on the preimage resistance and output unpredictability of the hash function. For a fixed header template and target, miners are not assumed to have a shortcut that derives a nonce value whose instantiated candidate header hashes below the target, they must evaluate candidate headers according to the proof-of-work procedure. Collision resistance is relevant to preventing distinct inputs from being deliberately constructed with the same digest, but the mining condition itself is a target-preimage condition rather than a collision-finding problem.

5.3. PoCol State Representation on the Blockchain

5.3.1. Blockchain Ledger and Global State

The blockchain ledger maintained in PoCol consists of a series of blocks with B_0 being the genesis block created during the genesis round of PoCol. Each block consists of a header and a set of transactions. The global state of $S(B_h)$ for height h contains (a) balances of accounts along with any special data related to the application, (b) details of all the miners in M_{h}, and (c) other details needed by the reward distribution function.

The execution of transactions in block B_h follows state $S(B_{h-1})$. The standard transactions may alter the balances, deploy or invoke smart contracts, or make changes to application-specific states. Additionally, PoCol introduces miner registration and deregistration transactions in order to alter the set of miners through the processes discussed in Section 5.4.

5.3.2. Registered Miner Set

At each height h, M_h is a list of public keys of the miners who are eligible to participate in PoCol mining rounds in order to earn rewards. The list M_h at block h is derived from the historical record of miner registration and deregistration transactions recorded in blocks until the block height of h. In the simplest version, registration is permanent unless canceled, i.e., if a miner's registration transaction is confirmed, its public key will remain on the list of M_h until a deregistration transaction is executed.

5.3.3. Balance and Reward State

Balances are updated through standard transfer transactions and also by reward distribution transactions present in every block. Unlike PoW where one coinbase transaction pays the entire block reward, in PoCol, reward distribution transactions can have one output for each miner where the block reward amount can be distributed based on some reward rules. Therefore, PoCol ledger is the only official source for both user balances and miner earnings.

5.4. Miner Life-Cycle: Registration, Activity, and Leaving

5.4.1. Miner Registration Protocol

The miner's lifecycle in PoCol starts with their registration at genesis. During that phase, the first miner (Miner A), registers locally on the network and possibly also adds a genesis registration in B_0 to the register, depending on the deployment. Following that, the next miners join the network by making a miner registration transaction by providing their public key, network address, reward address, and possibly some optional information (e.g., declared hash power or jurisdiction). Afterward, this transaction is broadcast across the network in the same manner as other valid transactions and is included in a block by any miner. When the registration transaction is confirmed in block B_h, then the miner set of honest nodes is updated so that M_h now contains the new miner. From then on for the next PoCol round, starting from $h + 1$, the new miner is now eligible for nonce-range assignments and reward sharing. This design replicates how validator sets are updated in PoS and BFT-like blockchains where membership is only changed after having staked and governance transactions included (Castro & Liskov, 1999; Ge et al., 2022).

5.4.2. Explicit Miner Deregistration

Miners who do not want to participate in PoCol anymore (such as when they are decommissioning their equipment or transitioning to another network) are able to carry out a deregistration transaction. This transaction is signed by the miner with their public key and shows that the miner wants to be removed from their registered set. Nodes first check the validity of the signature and ensure that the miner currently belongs to the set M_h before recording the transaction in their mempools. After the deregistration transaction has been executed in a block, the miner is removed from M_h in the current state. Depending on the settings in the protocol, deregistration can take effect immediately with the next round of miners contributing to M_h or it can take place after k blocks to avoid abuse. After deregistration, the miner will no longer receive any nonce range or share of rewards. However, previously acquired rewards will still be recorded in the ledger.

5.4.3. Handling Silent Leaves and Inactive Miners

Due to the nature of open networks, it is not realistic to expect all miners to deregister as soon as they become inactive. It is necessary for PoCol to cover silent leave situations. In such cases, a miner can simply cease hashing or go offline without making a deregistration transaction. In this situation, an identifier of that miner remains in M_h, meaning that it no longer contributes towards consensus.

From the point of view of the nonce distribution function used in PoCol, inactive miners will continue to receive nonce ranges. However, these ranges will not be used at all. This situation is similar to the drop in total hashing power of classical PoW, during which certain miners stop using their hardware causing the effective hashing power of the network to decline. The action of PoCol will remain correct, but blocks in this case will be produced at a lower frequency until the correct difficulty setting is achieved.

The presence of inactive miners in M_h raises fairness concerns about the distribution of rewards. A simple policy for awarding miners who have registered is to provide them with rewards regardless of whether they participated in the recent round. More elaborate rules can include activity measurements and “proofs of hashing” to deny rewards to consistently inactive miners.

5.4.4. Optional Inactivity-Based Removal

In order to avoid having a number of inactive miners recorded in M_h, the protocol could be augmented with an inactivity-based removal feature. In this feature, for each miner, the protocol monitors the index of the last block in which that miner was actively involved (as a winner, co-signer of a heartbeat transaction, etc). If the miner remains inactive for more than T blocks, then any node in the network can suggest an inactivity removal transaction. As soon as this transaction has been included in the block, the miner is no longer considered to be registered.

This action resembles penalties for inactivity in PoS/BFT systems, however, PoCol allows implementing it without requiring the user to pay a penalty in stake. T can be adjusted so that either stability (no continuous removals of inactive miners from M_{h}) or responsiveness (quick removal of longtime inactive miners).

5.5. Overview of the PoCol Mining Round

5.5.1. Round Structure and Inputs

PoCol operates by having rounds, in which any round is meant to yield a single valid block of a specific height. Rounds are indexed to the blocks by having round r producing block B_r such that its predecessor is hash of the valid block of height B_{r-1}. Any round follows the same process and as such the miners will access consistent information on all the possible blocks produced from 0 to $r - 1$ and derive the required past state of the previous block’s hash ($prev_{\mathrm{hash}}$), set of active miners M_{r-1} as well as the current pending transactions (mempool).

Since all miners reference the same committed information from past states of the blockchain, PoCol requires them to derive the block template for B_r in an identical, deterministic manner. All miners participating in the same PoCol mining round must mine exactly the same block: they use a single canonical, immutable block template (Section 5.8.2), verify that they share the same template identifier $TemplateID_{\mathrm{r}}$, and may vary only the nonce during the active round. Identical templates are therefore not merely an expected behavior but a consensus rule, and the nonce distribution function of Section 5.6 is defined with respect to this common template.

5.5.2. High-Level Steps

At a high level, a PoCol round comprises four phases:

1) Canonical block template construction: all miners deterministically derive the same immutable block template for the round - containing the previous-block hash, the round number, the protocol version, the difficulty target (set by the protocol from the desired block interval and recent block production rate), a deterministic round timestamp, the ordered transaction list, its Merkle root, and the reward-distribution transaction - and verify that they share the same template identifier $TemplateID_{\mathrm{r}}$ (Section 5.8.2).
2) Nonce-domain partitioning: miners apply the PoCol allocation function to compute a pseudo-random ordering of miners and assign each miner a non-overlapping nonce range within the permitted nonce domain. The assignment is bound to the common block and header templates through $TemplateID_{\mathrm{r}}$ (Section 5.8.3).
3) Collaborative proof-of-work evaluation: each miner iterates only over its assigned nonce range. For each nonce n, the miner instantiates the common nonce-free header template to obtain a complete candidate header and evaluates its hash. If the header hash is at or below the target, the miner publishes the already committed block template together with the successful nonce value, the reward-distribution transaction is already included in that template.
4) Verification and commitment: upon receiving a proposed block, nodes verify its proof of work, confirm that its non-nonce consensus-critical fields match the committed canonical template, validate its transactions, verify that the successful nonce lies within the proposer's assigned range, and verify the predetermined reward distribution. If the block is valid and selected by the chain-selection rule, it is appended to the ledger and the round ends. If every assigned nonce range is exhausted without producing a header hash at or below the target, the round ends without a block, a new round and a new canonical template are then derived before mining resumes (Section 5.8.4).

The following sections formalize the nonce-domain distribution and the behaviors in genesis and normal rounds.

5.6. Deterministic Nonce-Domain Distribution

In PoCol, the goal of nonce-domain distribution is to ensure that, in each round r, the permitted nonce domain is partitioned into disjoint sub-ranges and each registered miner receives exactly one sub-range, without any trusted coordinator. All honest miners must be able to recompute the assignment locally and obtain identical results from public inputs only.

Let B_{r-1} be the last committed block at the start of round r, with header hash

$$h_{r-1} = H\big(\text{header}(B_{r-1})\big) \tag{5.1}$$

Let $M_r = \{m_0, m_1, \dots, m_{n-1}\}$ be the registered miner set extracted from the ledger state S_{r-1}, where each m_i is a unique miner identifier (e.g., a public key). Let $N = \{0, 1, \dots, N_{\max} - 1\}$ be the permitted nonce domain, with fixed size $N_{\max}$ (e.g., 2^{32} or 2^{64} depending on implementation constraints).

5.6.1. Round Seed

For each round $r \geq 0$, every miner computes the round seed as

$$\text{seed}_r = H(\text{“PoColNonce”} \parallel h_{r-1} \parallel r \parallel TemplateID_r) \tag{5.2}$$

The round seed binds work assignment to the previous committed block, the round index, and $TemplateID_r$. It is deterministic once those inputs are fixed. However, unpredictability and bias resistance are not guaranteed solely by hashing these values: because $TemplateID_r$ depends on template formation, any ability to choose among multiple admissible templates can create a grinding or assignment-bias surface. The present protocol therefore treats unbiasable seed derivation as a security requirement requiring further analysis.

5.6.2. Pseudo-Random Ordering of Miners

Given the miner set M_r and seed $seed_r$, PoCol defines a per-round ticket for each miner identifier $m_i \in M_r$ as

$$\text{ticket}_r(m_i) = H\big(\text{seed}_r \parallel \text{id}(m_i)\big) \tag{5.3}$$

where $id(m_i)$ is a canonical byte encoding of the miner's identifier (e.g., its public key). All miners then sort M_r by increasing ticket value and obtain a round-specific permutation

$$L_r = \left[m_{(0)}, m_{(1)}, \ldots, m_{(n-1)}\right] \quad (5.4)$$

such that

$$\text{ticket}_r\left(m_{(0)}\right) \le \text{ticket}_r\left(m_{(1)}\right) \le \cdots \le \text{ticket}_r\left(m_{(n-1)}\right)$$

The resulting list L_r is a deterministic round-specific ordering of the registered miner set for a fixed seed. Under a random-oracle abstraction, ticket values are uniformly distributed for an externally fixed seed, but this does not prove that a strategic participant cannot influence the seed itself through block or template choices. Range-ordering fairness is therefore conditional on an unbiased seed-generation process and is not claimed as a proved security or economic fairness property.

5.6.3. Partitioning the Permitted Nonce Domain

Let $n = |M_r|$. PoCol divides the permitted nonce domain N into n contiguous sub-ranges using simple integer partitioning:

$$\text{start}_r(i) = \left\lfloor \frac{i\, N_{max}}{n} \right\rfloor \quad (5.4a)$$

$$\text{end}_r(i) = \left\lfloor \frac{(i+1)\, N_{max}}{n} \right\rfloor - 1 \quad (5.4b)$$

for each position $i \in \{0, \ldots, n-1\}$. The nonce sub-range for miner $m_{(i)}$ in round r is then

$$R_r(m_i) = [\text{start}_r(i), \text{end}_r(i)] \quad (5.5)$$

By construction:

- $R_r\left(m_{(i)}\right) \cap R_r\left(m_{(j)}\right) = \emptyset$ for all $i \neq j$,
- $\bigcup_{i=0}^{n-1} R_r\left(m_{(i)}\right) = N$,
- $\left|R_{\mathrm{r}}\left(m_{(i)}\right)\right| \in \left\{\text{floor}\left(\frac{N_{\max}}{n}\right), \text{ceil}\left(\frac{N_{\max}}{n}\right)\right\}$.

Thus, each miner receives a disjoint portion of the nonce domain, and the assignment is uniquely determined by (M_{r}, h_{r-1}, r, $TemplateID_{\mathrm{r}}$), remains unchanged for the duration of the round, and is independently verifiable by every node. No coordination or message exchange is required for nonce-range assignment beyond the blockchain state and standard block propagation.

5.6.4. Range-Assignment Balance Across Rounds

For a fixed nonce-free header template and target, the position of a miner's assigned range within the nonce domain does not make individual nonce values intrinsically more likely to satisfy the target. A miner's probability of producing a successful candidate header depends on the number of candidate headers it evaluates within its assigned range and on the per-header success probability. The pseudo-random ordering changes the locations of assigned ranges across rounds, giving modeled range-assignment balance. Under the common-template invariant, disjoint assigned ranges also ensure that honest miners do not evaluate the same serialized candidate header in the same round. This is a modeled range-assignment property only; it is not a reward, incentive, participation, economic, Sybil-resistance, or proof-of-effort fairness claim.

5.6.5. Handling Unevaluated Nonce Ranges

When a registered miner is inactive, its assigned nonce range $R_r(m_i)$ is not evaluated. The set of nonce values actually examined in the round is therefore the union of the ranges assigned to active miners, and the effective aggregate hash rate of the active miner set is reduced.

In the worst case, all nonce values that would instantiate headers satisfying the target for the current template may lie inside ranges assigned to inactive miners. Active miners can then exhaust only their own assigned ranges without finding a successful candidate header, while part of the permitted nonce domain remains unevaluated. The round is therefore closed under the protocol's inactive-range or deadline policy and a new template and assignment are generated; this event must not be described as full-domain exhaustion.

Figure 5.1 illustrates a PoCol round in which Miner C is registered but inactive. All miners derive the same canonical block and header templates, verify the common $TemplateID_r$, and compute the disjoint nonce ranges locally. Miners A and B evaluate candidate headers using nonce values from their assigned ranges, while Miner C's assigned range is not evaluated. The result is a temporary reduction in effective aggregate hash rate. Under the baseline reward policy, Miner C may nevertheless remain eligible for a reward share while it remains registered, this reward-policy issue is distinct from verification of the successful nonce's range membership.

Figure (5.1): PoCol consensus round with an unevaluated nonce range caused by an inactive miner. All miners use the same immutable canonical block and nonce-free header templates, only the nonce value varies when candidate headers are instantiated.

5.6.6. Post-Range Operating Policies

PoCol distinguishes nonce-range allocation from the miner's operating state after completing the assigned search. The protocol may apply either an energy-saving idle policy or a continuous-performance policy. These policies are alternatives and must be identified explicitly for every experimental or deployment configuration.

5.6.6.1. Energy-Saving Idle Policy

Under the energy-saving policy, a miner searches only the nonce range assigned to it for the current immutable template. If no valid block is found before the range is exhausted, the miner stops active hashing and enters a low-power idle state until the end of the current collaborative mining round. The miner remains connected and may receive protocol messages, including notification that another miner has produced a valid block, but it performs no further mining hashes during the remaining round time.

The phrase "end of the current collaborative mining round" refers to the round deadline or target block interval defined by PoCol. It does not refer to a multi-block difficulty-adjustment epoch.

The energy benefit of this policy depends on the difference between active and idle power and on the duration of the idle interval. The policy may also reduce the active network hash rate as miners complete their ranges, therefore, its effects on block interval, range-exhaustion probability, liveness, and security must be evaluated together with its energy savings.

5.7. PoCol Genesis Round

5.7.1. Genesis Preconditions and First Miner Setup

The purpose of the PoCol genesis stage is to establish the initial blockchain state, create the genesis B_0 and define the initial registered miner set. In one possible deployment model, a bootstrapping authority or consortium specifies the initial protocol parameters, founding miner identities, and initial account balances required to initialize the ledger. The selected bootstrap procedure is a deployment choice rather than a consensus property of PoCol. Once the initial state and miner set are established, subsequent miner registration, nonce-domain allocation, and block-production rounds follow the protocol rules described in the following sections.

As for methods of bootstrapping, the PoCol refers to a more decentralized approach when the first miner self-registers independently from others only in order to have its transaction registered initially.

5.7.2. Genesis Miner Registration Sequence

In a typical situation, the process of registration in the genesis sequence goes as follows. Miner A, who desires to carry out local registration record, proceeds with executing genesis registration transaction possibly by disclosing its public key, network address, and reward address. However, after registration by Miner A, the two other miners – B and C – send registration requests to Miner A, which are validated by Miner A. As soon as Miner A has approved the requests, it registers them into the system.

No matter however many details exist concerning the manner of displaying genesis registration messages, there are two main ways to do this. The addressed message could be published as an explicit transaction or through configuration files. Nevertheless, it is essential that all the miners have to agree upon M_0 prior to start nonce domain distribution.

5.7.3. Genesis Round Nonce Distribution and Collaborative Search

In the genesis round ($r = 0$), miners read the genesis hash as $prev_{\text{hash}}$, deterministically derive the canonical genesis block template and its nonce-free header template, compute $TemplateID_0$ over the canonical template representation with the nonce value excluded or fixed to its canonical placeholder, and compute $seed_0 = \text{H}(\text{"PoColNonce"} \,||\, genesis_{\text{hash}} \,||\, 0 \,||\, TemplateID_0)$. They then derive the ordered miner list L_0 and assign nonce ranges R_{i} as in Section 5.6.

The canonical genesis block template is identical for every miner and includes the initial reward-distribution transaction and any bootstrap transactions. Every miner verifies $TemplateID_0$ before beginning proof-of-work evaluation. Each miner then instantiates candidate headers using nonce values from its assigned range. When a miner obtains a header hash at or below the target, it publishes B_0 by combining the committed template with the successful nonce value. Other miners verify the complete header, proof of work, transactions, and template commitment before accepting B_0 and $S(B_0)$.

5.7.4. Genesis Reward Distribution and Ledger Initialization

The reward distribution in the genesis block may follow a different policy than in normal rounds - for example, allocating initial balances to founding entities or to a treasury account. Nonetheless, the structure of the reward distribution transaction is the same: it is a standard transaction that creates outputs paying to specified addresses. After B_0 is committed, the ledger contains $S(B_0)$ with initialized balances and M_0, and PoCol proceeds with normal rounds for subsequent blocks.

Figure 5.2 summarizes the genesis-round workflow. Founding miner identities and initial balances are established by the selected bootstrap policy, after which participants derive the same genesis template, verify its $TemplateID$, compute the round seed and deterministic range assignment, and begin collaborative search. The figure is illustrative: the current simulator does not implement the complete distributed bootstrap and template-agreement exchange shown in the protocol specification.

Figure (5.2): PoCol consensus genesis round (round 0), including miner registration, transaction submission, collaborative nonce search, and commitment of the genesis block. All miners derive the same canonical genesis template ($TemplateID_0$) before the collaborative search.

5.8. PoCol Normal Rounds (r ≥ 1)

5.8.1. Reading State at the Start of Round r

At the beginning of a normal round $r \geq 1$, each miner reads the current state $S(B_{r-1})$ from its local blockchain view. This state provides: (i) the previous block hash $prev_{\text{hash}}$, (ii) the registered miner set M_r (derived from registration and deregistration transactions up to height $r-1$), and (iii) current account balances and application-specific data. Miners also maintain a mempool of pending transactions received since the last block.

5.8.2. Canonical Block Template Construction

In PoCol, round r does not involve independently constructed candidate blocks. Before nonce evaluation begins, all participating miners deterministically derive, or receive and verify, the same canonical block template $BlockTemplate_r$ from the committed state $S(B_{r-1})$ and the eligible transaction set. The block template fixes the transaction list, reward-distribution transaction, Merkle root, previous-block hash, round number, protocol version, target, deterministic timestamp, and every other consensus-critical field except the round's nonce value. From this block template, miners derive a nonce-free header template $HeaderTemplate_r$. Candidate block headers are produced only by inserting nonce values into the nonce field of $HeaderTemplate_r$.

The reference protocol defines a bounded transaction-collection window beginning after acceptance of the previous block. Transactions received before the cutoff and valid against the committed state are eligible for the round; later transactions remain pending for subsequent rounds. The window is intended to reduce divergent mempool views, but a bounded window alone does not prove that all honest miners have identical eligible sets under arbitrary delay or partition conditions.

From an agreed eligible set, the template is selected by a protocol-fixed deterministic rule. The reference rule orders transactions by decreasing fee density up to the block-size limit, with lexicographic transaction-ID tie breaking. This is a protocol specification rather than a feature of the current BlockSim implementation: the legacy prototype selects transactions after a winning event, has no complete tie-breaking and reconciliation exchange, and does not execute cross-miner template agreement end to end.

Residual mempool differences must therefore be resolved before a miner joins the round. $TemplateID_r$ acts as a consistency check: a miner whose locally derived identifier differs from the identifiers it observes must not begin the authorized search until the transaction view is reconciled. This rule detects disagreement, but the present thesis does not prove that peer-to-peer gossip alone always yields one globally agreed $TemplateID$ under partial synchrony, partitions, or strategic message withholding. A production protocol requires an explicit convergence or commitment mechanism.

The common-template mechanism should thus be understood as a specified coordination layer with four intended components: a deterministic round start, an eligibility cutoff, retrieval of missing transactions, and a $TemplateID$ agreement gate. The stages simulators implement the consequences of a common template for mining, but not the complete transaction-reconciliation and agreement protocol.

Each round is associated with a block-template identifier

$$TemplateID_{\mathrm{r}} = \mathrm{H}(\mathrm{Encode}(BlockTemplate_{\mathrm{r}}\ \text{without a round} - \text{specific nonce value}))$$

Every miner recomputes $TemplateID_{\mathrm{r}}$ locally and verifies that it refers to the same canonical block and nonce-free header templates before beginning nonce evaluation. During an active round, those templates are immutable, miners instantiate candidate headers only by varying the nonce value.

Common-Template Invariant (protocol target): for an authorized PoCol round r, honest miners that participate in the search must use the same immutable block template T_r and derived nonce-free header template. A candidate block is valid for that round only if all non-nonce consensus-critical fields match T_r. The simulator assumes this invariant; the distributed mechanism that establishes it is specified but not validated end to end.

5.8.3. Nonce-Domain Distribution in Round r

With the canonical block and nonce-free header templates fixed and $TemplateID_{\mathrm{r}}$ verified, miners compute the round seed $seed_{\mathrm{r}} = \mathrm{H}(\text{"PoColNonce"} \,||\, prev_{\mathrm{hash}} \,||\, r \,||\, TemplateID_{\mathrm{r}})$, derive the ordered list

L_r, and compute the nonce ranges R_i as described in Section 5.6. Including $TemplateID_r$ binds each nonce-range assignment to the canonical template. Each miner computes its own assigned range locally. The ranges are pairwise disjoint, collectively cover the permitted nonce domain, are independently verifiable from public inputs, and remain unchanged for the duration of the round.

Given an agreed nonce-free header template, disjoint nonce assignments eliminate exact duplicate serialized candidate-header inputs among honest miners because each nonce value maps to one serialized header under that fixed template. This statement is intentionally bounded to the common-template model. It does not imply that conventional Bitcoin miners normally perform duplicate work when they use the same numeric nonce, because other header fields may differ.

Disjoint-Evaluation Invariant: For every pair of distinct honest miners m_i and m_j in round r, $R_r(m_i) \cap R_r(m_j) = \emptyset$. Hence they never evaluate the same serialized candidate header $\mathrm{Header}_r(n)$.

5.8.4. Collaborative Proof-of-Work Search

Miners perform proof-of-work evaluation by iterating over nonce values $n \in R_i$. For each n, a miner constructs the complete candidate header $Header_r(n)$ by inserting n into the nonce field of the common nonce-free $HeaderTemplate_r$, then evaluates $H\left(Serialize\big(Header_r(n)\big)\right)$. The transaction list, Merkle root, previous-block hash, target, timestamp, protocol version, reward-distribution transaction, and all other consensus-critical fields remain fixed by $BlockTemplate_r$. Altering any such field produces a different template and is prohibited during the current round. If $H\left(Serialize\big(Header_r(n)\big)\right) \leq target$, the miner has produced a proof-of-work-valid header for B_r and broadcasts the committed block template instantiated with the successful nonce value.

If a miner exhausts its assigned range R_i without producing a header hash at or below the target, it marks that range as exhausted locally. If all assigned ranges are exhausted and no valid block has been received, the round ends without a block. No miner may alter a non-nonce field and continue under the same round identifier. Instead, the protocol creates a new round, derives a new canonical block and nonce-free header template, recomputes $TemplateID$, generates new nonce ranges, and resumes only after agreement on the new template. Thus, only the nonce value varies within one round, changes to timestamp, transactions, or other header fields require a new template and a new round.

5.8.5. Block Proposal, Verification, and Commitment

Upon receiving a proposed block B_r from some miner, each node verifies it as follows:

1) It checks that B_r extends the current longest valid chain (or a competing chain of comparable length) by verifying that $prev_{\text{hash}}$ matches the hash of B_{r-1} in its local ledger.
2) It recomputes the canonical block template, its nonce-free header template, and $TemplateID_{\mathrm{r}}$, then checks that every non-nonce consensus-critical field of the received block and header matches the committed template.
3) It checks that the transaction list and the Merkle root of the received block match the common template, and that the reward-distribution transaction is identical to the predetermined $T_{\text{reward}}(r)$ committed in the template (Section 5.9), so that the block contains no unauthorized modification of the template.
4) It recomputes $seed_{\mathrm{r}}$, L_{r}, and the nonce ranges R_{i}, and verifies that the successful nonce value lies within the proposing miner's assigned range. This verifies range membership of the successful candidate header; it does not prove exhaustive evaluation of other miners' ranges.
5) It validates all transactions in Br against $S(B_{r-1})$, ensuring no double spends or contract violations.
6) It verifies the reward distribution transaction according to the PoCol reward distribution function (Section 5.9).
7) It verifies that $H\left(Serialize\left(Header_r(n)\right)\right) \leq target$ for the received complete header and that the header's Merkle root and other committed fields are consistent with $BlockTemplate_{\mathrm{r}}$.
8) A header that satisfies the target but was constructed from different non-nonce fields is not a valid PoCol result for round r. Such a header derives from a different template, so its $TemplateID$ does not match the committed $TemplateID_{\mathrm{r}}$ and the corresponding block is rejected regardless of its proof of work.

If all validation checks pass, the block is eligible for the cumulative-work chain-selection rule. The verification procedure establishes template conformity, successful-nonce range membership, transaction validity, and proof of work. It does not prove that non-winning miners searched their entire ranges, and the interaction between these verification rules and full Nakamoto-style fork security remains to be proved formally (Garay et al., 2015; Nakamoto, 2008).

5.9. Reward Distribution Mechanism in PoCol

Let round r attempt to produce block B_{r} extending B_{r-1}. We denote by $M_{\mathrm{r}} = \{m_0, \dots, m_{n-1}\}$ the registered miner set used for round r. Let $R_{blk}(r)$ be the protocol-defined base block reward (subsidy) and $F_{tx}(r)$ the total transaction fees included in B_r. The total distributable reward in round r is

$$R_{tot}(r) = R_{blk}(r) + F_{tx}(r) \tag{5.6}$$

PoCol introduces a reward distribution function

$$f_r: M_r \to [0,1]$$

which maps each miner identifier to a reward share $s_i = f_r(m_i)$. The shares satisfy

$$\sum_{m_i \in M_r} f_r(m_i) = 1, \qquad s_i = f_r(m_i) \geq 0 \tag{5.7}$$

The monetary reward paid to miner m_{i} in round r is then

$$Reward_r(m_i) = s_i\, R_{tot}(r) \tag{5.8}$$

In the simplest equal-sharing baseline, PoCol sets

$$f_r(m_i) = \frac{1}{|M_r|}, \qquad \forall m_i \in M_r \tag{5.9}$$

In the simplest equal-sharing baseline, every registered miner receives the same fraction of the distributable reward. This can reduce reward variance mechanically, but it also creates an obvious free-riding and Sybil incentive because registration, not verified work, determines eligibility. Equal sharing is therefore retained only as a simple protocol baseline. A production reward rule requires an observable and auditable contribution measure and must be analyzed for identity splitting and incentive compatibility.

5.9.1. Reward Distribution Transaction

The reward distribution transaction $T_{reward}(r)$ generalizes the coinbase transaction in PoW systems (Del Monte et al., 2020). It is computed before mining from $S(B_{r-1})$ and the protocol rules, included in $BlockTemplate_r$, and committed through the Merkle root. The winning miner does not create or alter this transaction after producing a successful header, it broadcasts the committed block template instantiated with the successful nonce value and any required proof or signature. $T_{\text{reward}}(r)$ has:

- One output per miner $m_i \in M_r$, paying amount $Reward_r(m_i)$ to the on-chain reward address registered for m_i,
- Inputs that represent the newly created subsidy $R_{\text{blk}}(r)$ and the collected transaction fees $F_{\text{tx}}(r)$, according to the underlying UTXO or account model.

Conceptually, $T_{\text{reward}}(r)$ is a multi-recipient coinbase: instead of paying the entire reward to the single winner, PoCol encodes collaborative rewards as multiple outputs in a single transaction. For ease of parsing and verification, $T_{\text{reward}}(r)$ is placed as the first transaction in B_r, following the usual convention for coinbase transactions in PoW blockchains (Del Monte et al., 2020; Zhuang et al., 2019).

5.9.2. Verification Rules

When a node receives a proposed block B_r, it recomputes:

1) $R_{\text{tot}}(r)$ from the protocol rules and the included transactions,
2) The miner set M_r from the state S_{r-1},
3) The share vector $\{ s_i \}$ using the agreed function f_r and any on-chain metrics it depends on.

The node then verifies that:

- $Completeness$: $T_{\text{reward}}(r)$ contains exactly one output per miner in the eligible set (typically all $m_i \in M_r$, or a filtered subset if activity-based policies are used),
- Correctness: for each m_i, the payment amount equals $s_i \cdot R_{\text{tot}}(r)$ up to negligible rounding error,
- Conservation: the sum of all reward outputs equals $R_{\text{tot}}(r)$, no extra coins are created or destroyed,
- $Validity$: $T_{\text{reward}}(r)$ is well-formed under the base ledger rules (no double spends, non-negative output values, etc.).

If the pre-committed reward transaction violates the deterministic reward rule, the block is rejected. This prevents the winning miner from changing the committed payout after discovering a valid header. It does not, however, establish that the reward rule itself is fair or that non-winning participants earned their assigned shares; those questions depend on contribution auditing and identity policy.

5.9.3. Inactive Miners and Policy Options

Let $A_{\mathrm{r}} \subseteq M_{\mathrm{r}}$ denote the set of active miners in round r (e.g., those that are online and hashing). The baseline PoCol design uses the full M_{r} in (5.7)–(5.9), meaning inactive miners still receive rewards until they explicitly deregister or are removed by inactivity rules. This keeps the protocol simple but may be undesirable if large numbers of registered miners are persistently inactive.

A deployment may restrict rewards to miners satisfying an activity predicate, such as recent signed heartbeats or verifiable work evidence. Such a filter can reduce rewards to persistently inactive identities, but it is not equivalent to proving that an assigned nonce range was searched. The activity predicate, contribution proof, and Sybil cost must be specified together before a permissionless reward mechanism can be considered incentive compatible.

5.9.4. Interaction with Miner Set Dynamics

Because PoCol's reward mechanism is driven entirely by M_{r} and the deterministic function f_{r}, it naturally adapts to miner set changes. When a new miner joins (via registration) and becomes part of M_{r}, the next round automatically includes it in $T_{\mathrm{reward}}(r)$ with a non-zero share. When a miner deregisters or is removed due to inactivity, it disappears from M_{r} and thus receives no further rewards. There is no need for additional coordination or side agreements to determine baseline reward eligibility: the on-chain registered miner set and the protocol-defined reward function determine who is eligible to receive a share of $R_{\mathrm{tot}}(r)$ in each round. This eligibility determination does not by itself verify each miner's actual search contribution.

Figure 5.3 shows how rewards are distributed according to Section 5.9, illustrated as a scenario in which Miner B became the winning miner on arbitrary round r. Before mining begins, every miner reads the registered miner set M_{r} from its ledger view, applies the reward distribution function f_{r}, and computes the deterministic reward-distribution transaction committed in the canonical block template, with each registered miner's share drawn from the block subsidy plus the transaction fees. When the complete candidate header instantiated by Miner B's nonce value hashes at or below the target, Miner B does not create or modify any transaction, it publishes the already committed block template instantiated with the successful nonce value. The reward distribution transaction $T_{\mathrm{reward}}(r)$ has one output for each registered miner and is entered into the block before being transmitted. Verification rules described in Section 5.9.2 are followed, meaning that any receiving miner must confirm that the beneficiaries are equal to M_{r}, thus confirming that every share is equal to the prescribed value f_{r}, that the sum of outputs equals $R_{\mathrm{tot}}(r)$, and that neither value was created nor lost, prior to committing the block and updating balances accordingly.

Figure (5.3): PoCol reward distribution and verification in round r, from production of a proof-of-work-valid candidate header by the successful miner to balance updates at all registered miners. $T_{\text{reward}}(r)$ is computed before mining and committed in the canonical block template, the miner publishes that template instantiated with the successful nonce value.

5.10. Dynamic Miner Set Management in PoCol

In this section we reuse the notation from the system model in Chapter 4. In particular, let $M(r)$ denote the set of miners that are registered at the beginning of round r, with cardinality $n(r) = |M(r)|$. For each miner $m_{\mathrm{i}} \in M(r)$ we denote its hash rate by $H_i(r)$, and the aggregate network hash rate by

$$H_{tot}(r) = \sum_{m_i \in M(r)} H_i\,(r)$$

Chapter 4 defined the expected block-arrival rate $\lambda_{blk}(r)$ and the expected block interval $T_{blk}(r)$ as functions of $H_{tot}(r)$ and the difficulty parameter D_{r}, for example via

$$\lambda_{\mathrm{blk}}(r) = H_{\mathrm{tot}}(r)\, p_{\mathrm{succ}}(D_r)$$

$$T_{blk}(r) = \frac{1}{\lambda_{blk}(r)}$$

where $p_{\mathrm{succ}}(D_{\mathrm{r}})$ is the per-hash success probability at difficulty D_{r} (Garay et al., 2015; Nakamoto, 2008). Dynamic changes in the miner set $M(r)$ thus directly impact the performance and energy model derived in Chapters 3–4, because they modify both $H_{tot}(r)$ and the size of the registered set used by the PoCol nonce distribution function.

5.10.1. Joining New Miners

A new miner joins PoCol by issuing a registration transaction that is included in some block B_h. As in Section 5.3, this transaction carries the miner's public key, network address and reward address. Let $M(h)$ be the miner set after applying all transactions in B_h. Then, for any round $r > h$, the new miner appears in $M(r)$ and is consequently:

- included in the registered miner list used by the nonce-domain distribution function (Section 5.6), and
- included in the miner set used by the reward distribution function f_{r} (Section 5.9).

From the analytical point of view, adding a miner m_j at height h does two things:

1) It increases the total hash rate:

$$H_{tot}(r) \leftarrow H_{tot}(r) + H_j(r), \qquad \forall r > h$$

which shortens the expected block interval $T_{blk}(r)$ at fixed difficulty until the difficulty-adjustment mechanism responds (McDonald, 2021; Zhang et al., 2023).

2) It increases the size of the registered set $n(r)$, so the nonce domain $\{0, \dots, N_{max} - 1\}$ is partitioned into more sub-ranges (Equations (5.4a) - (5.5)). This does not change the aggregate probability of success for honest miners, but it refines the granularity of the PoCol assignment and slightly reduces the size of each individual miner's nonce range.

If the reward distribution function f_{r} uses equal sharing (Equation (5.9)), the arrival of a new miner also reduces the per-block share of all existing miners: each m_{i} receives $\frac{1}{|M(r)|}$ of the total reward in future rounds. If f_{r} is weighted by hash rate or stake, the impact depends on the chosen weighting scheme but remains completely determined by on-chain information and the formal definition of f_{r}.

5.10.2. Leaving Miners (Explicit and Silent)

Miners may leave the system explicitly or silently:

- In explicit leave, the miner issues a deregistration transaction that is confirmed in some block B_h. From that point on, the miner is removed from the registered set, so for all rounds $r > h$ it holds that

$$m_i \notin M(r), \qquad H_i(r) = 0$$

Consequently, the miner is no longer included in the nonce assignment (Section 5.6) nor in the reward distribution f_{r} (Section 5.9).

- In a silent leave, the miner simply stops hashing or goes offline without deregistering. In this case, its identifier remains in $M(r)$, and its nominal hash rate may still be modelled as $H_i(r) = 0$ in the analytical model of Chapter 4. Practically, the miner continues to appear in the PoCol miner list, but its assigned nonce range $R_r(m_i)$ is never explored.

In terms of the performance model, both explicit and silent leaves reduce the effective total hash rate from $H_{tot}(r)$ to

$$H_{eff}(r) = \sum_{m_i \in A(r)} H_i\,(r)$$

where $A_r \subseteq M(r)$ denotes the active subset of miners in round r. The block-arrival rate becomes

$$\lambda_{blk}^{eff}(r) = H_{eff}(r)\, p_{succ}(D_r)$$

which may be strictly smaller than the nominal $\lambda_{blk}(r)$ used in Chapter 4 if inactive miners are still counted in $M(r)$. This is analogous to hash-rate drops in classical PoW, where miners switching off hardware increase the actual block interval until the difficulty-adjustment algorithm reacts (Del Monte et al., 2020; Zhang et al., 2023).

From the incentive perspective, explicit leaving removes the miner from the reward function f_r, while silent leaving still allows the miner to collect rewards in the baseline equal-sharing design (because $m_i \in M(r)$ so long as it is not deregistered). The inactivity-aware variants of f_r discussed in Section 5.9.3 remove or down-weight such miners by conditioning on activity predicates, thereby aligning the economic model with the energy model.

In Figure 5.4, the explicit leaving procedure is illustrated according to sections 5.4.2 & 5.10.2. During this process, the miner who leaves (miner C) creates the deregistration transaction, with a signature, identifier (ID), network address, and timestamp on it, and sends it to the network. The receiving miners validate the transaction by ensuring that miner C's signature is valid and that miner C is still a member of the current registered miner set. Once validated, the transaction is accepted into a miner's memory pool (mempool) by each of those receiving miners. Once all the miners validate the original deregistration transaction, the deregistration transaction itself will be added to the next block created by collaborative mining. Once the block has been created and validated, all miners will now be able to derive the new registered miner set that includes only miner A and miner B. Therefore, miner C will no longer participate in any future rounds of the blockchain validation process or receive an assigned nonce range from the nonce-domain partitioning function. Additionally, the next invocation of the nonce distribution function will repartition the permitted nonce domain among the remaining miners, so no nonce range is assigned to Miner C after deregistration.

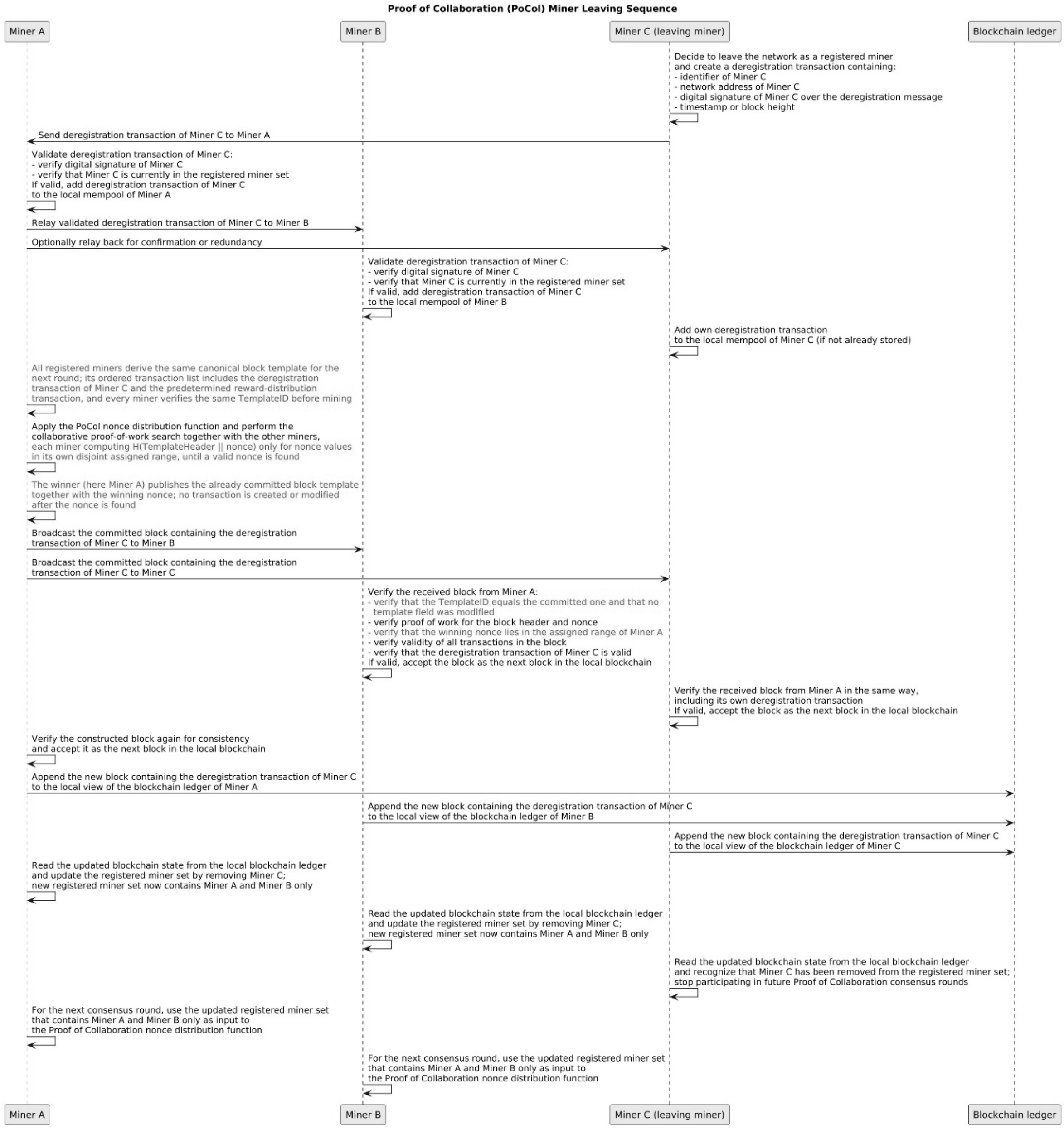

Figure (5.4): PoCol miner leaving sequence, showing explicit deregistration of Miner C and the resulting update of the registered miner set.

5.10.3. Long-Term Inactivity and Forced Removal

To avoid unbounded growth of $M(r)$ with many zero-hash-rate miners, PoCol can be extended with an inactivity-based removal rule. We introduce a function

$$\ell_i(h)$$

that records, for each miner m_i, the last block height at which its activity has been observed (e.g., as a block proposer, participant in a heartbeat mechanism, or contributor of verifiable work evidence-depending on the specific deployment).

Given an inactivity threshold $\Theta_{inact} > 0$ (in blocks), a miner is declared inactive if, at height h,

$$h - \ell_i(h) > \Theta_{inact}$$

Any honest node may then propose a forced-removal transaction that, once included in a block, updates the state so that $m_\mathrm{i} \notin M(r)$ for all subsequent rounds $r > h$. In the analytical model, this operation removes the miner from the sum defining $H_{tot}(r)$ and from the normalization in (5.7) - (5.9), all subsequent performance, energy and reward equations evolve with the new, reduced set $M(r)$.

This mechanism is conceptually similar to liveness penalties or slashing for non-participating validators in PoS and BFT-style protocols, but in PoCol it affects eligibility and reward sharing only, not stake balances (Deb et al., 2024; Saad et al., 2021). Tuning Θ_{inact} allows system designers to balance stability (rare changes in $M(r)$) against responsiveness (quick removal of long-term inactive miners), in line with the workload and energy constraints analysed in Chapters 3–4.

5.11. Summary

This chapter has presented Proof of Collaboration (PoCol), a PoW-inspired protocol in which all registered miners in a round use one immutable canonical block template and one derived nonce-free header template. The permitted nonce domain is deterministically partitioned into assigned ranges, enabling coordinated candidate-header evaluation and reward sharing without a trusted range coordinator.

The core of PoCol is its deterministic nonce-range allocation function. It computes a round-specific seed from the previous block hash, round number, and $TemplateID_\mathrm{r}$, derives a pseudo-random miner ordering, and assigns each miner a non-overlapping nonce range. Together with the Common-Template Invariant, this ensures that, for the fixed nonce-free $HeaderTemplate_\mathrm{r}$, no two honest miners evaluate the same serialized candidate header $Header_r(n)$ in a round. PoCol retains a hash-based proof-of-work condition and cumulative-work chain structure.

A proposed component of PoCol is its collaborative reward-distribution mechanism, which allocates rewards according to a configurable function. Equal sharing may reduce reward variance, but fairness, contribution verification, resistance to free riding, and Sybil robustness were not experimentally established. The security discussion is preliminary, and formal equivalence with PoW-style chain guarantees remains future work.

As shown in Figure 5.5 PoCol organizes block production as a sequence of rounds. At the start of each round r, the PoCol protocol requires participating miners to converge on and verify a common block template and a nonce-free header template without relying on a trusted range coordinator. The current simulator assumes the resulting common-template invariant rather than implementing the full distributed template-convergence process end to end. A round seed derived from the previous block hash, the round index, the fixed domain-separation label "PoColNonce", and the template identifier deterministically orders the registered miners and assigns each miner a disjoint sub-range of the permitted nonce domain.

Miners then search only within their assigned ranges. If any miner finds a nonce that produces a header at or below the target, it publishes the block containing the pre-committed reward distribution transaction, which allocates the block subsidy and transaction fees among all eligible miners according to the agreed reward rule f_r (e.g., equal sharing or a weighted rule). All nodes independently verify both the proof-of-work and the reward transaction before accepting the block and updating the state.

Miner set dynamics are handled on-chain through registration, explicit deregistration, and forced removal of long-term inactive miners. These updates automatically affect nonce distribution, reward sharing, and subsequent protocol state. They support auditability and deterministic recomputation from public protocol information, but do not by themselves establish decentralization, Sybil resistance, contribution correctness, or incentive compatibility.

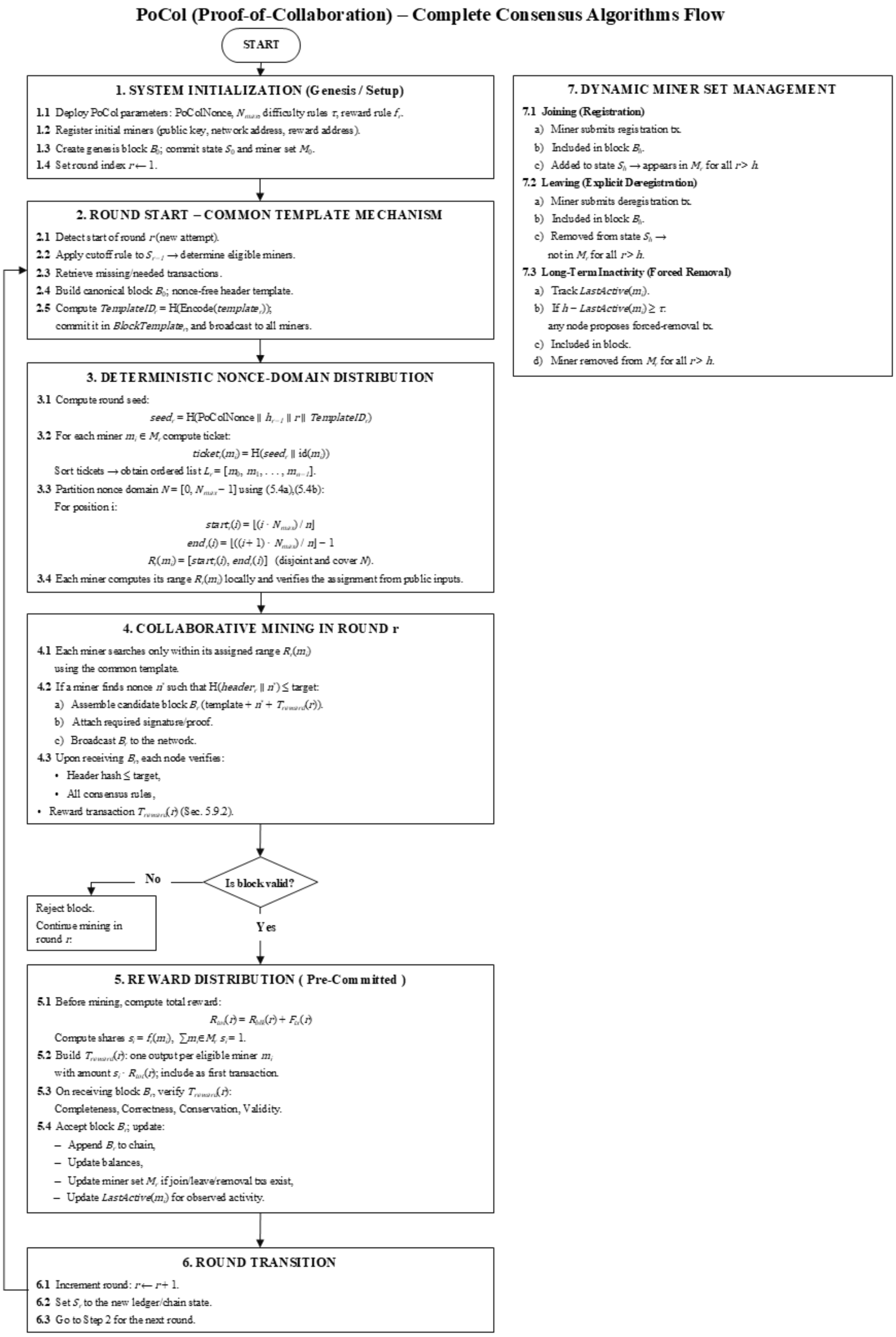

Figure (5.5): Complete PoCol protocol workflow, illustrating system initialization, common-template formation, deterministic nonce-domain distribution, collaborative mining and block validation, pre-committed reward distribution, round transition, and dynamic miner-set management without a trusted coordinator.

Chapter 6
Implementation and Experimental Setup

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6.1. Chapter Scope and Objectives

This chapter outlines the execution of the Proof-of-Collaboration (PoCol) mechanism and how its associated experiments were designed, executed, recorded and analyzed. The chapter does not evaluate the results; this will be done in the Evaluation chapter. Instead, this chapter documents the implementation of the PoCol mechanism and the environmental conditions under which the output files for this thesis were produced. The framework for the simulation workflow used in the PoCol implementation conforms to the precedent set within the blockchain research community regarding the controlled and repeatable testing of blockchain networks and behaviors using configurable network parameters and consensus parameters (Alharby & van Moorsel, 2019; Paulavičius et al., 2021).

BlockSim, an event driven blockchain simulator, was used for the implementation of PoCol. BlockSim has been utilized extensively in previous research as a tool for simulating the operation of Bitcoin-style networks and extending consensus and incentive calculation methods (Alharby & van Moorsel, 2019; Faria & Correia, 2019). The decision to utilize this simulator was based upon the findings of previous research that simulation is an acceptable method for conducting blockchain network evaluations when direct experimentation of large, geographically dispersed blockchain networks is impractical, difficult to control and cannot be replicated (Paulavičius et al., 2021; Smetanin et al., 2020). The outputs of the BlockSim simulations were exported into formatted log workbooks, which serve as the basis of this chapter, as well as the comparative evaluations presented in Chapter 7.

6.2. Simulator Platform and Baseline Model

The implementation platform used in this thesis is based on BlockSim, a discrete-event simulation framework for modeling blockchain functionality. Mining, block propagation, block reception, and chain-selection processes are explicitly represented within the simulation model. Prior research has demonstrated BlockSim's effectiveness in investigating network propagation, mining behavior, and miner incentives, while also supporting the implementation and evaluation of custom blockchain protocols beyond Bitcoin-specific configurations (Faria & Correia, 2019). Similarly, BlockSim has been identified as an extensible blockchain simulation framework suitable for research on blockchain architecture and consensus-protocol design (Paulavičius et al., 2021).

For sustainability-oriented evaluation, Rasheed and AbuSamra (2026) extended BlockSim by integrating energy-consumption and carbon-footprint modeling directly into its discrete-event execution framework. The extension instruments relevant simulation events to estimate computational and communication energy consumption at both node and network levels and converts the resulting electricity demand into CO_2 emissions using configurable carbon-emission factors. It further provides aggregate and normalized sustainability metrics, including energy consumption and CO_2 emissions per block and per transaction, thereby enabling energy and environmental assessment to be conducted alongside conventional blockchain performance evaluation (Rasheed & AbuSamra, 2026).

The energy-evaluation methodology adopted in this thesis builds on this extended BlockSim framework. This provides a consistent basis for associating simulated mining activity with its corresponding energy expenditure while retaining the underlying event-driven representation of blockchain operation. Within the present study, the energy-accounting layer is further aligned with the specific experimental requirements of Proof-of-Collaboration (PoCol), allowing energy measurements to be interpreted together with physical hashing work, block production, round latency, throughput, and other protocol-level outcomes.

Chapter 6 distinguishes the implementation layers used for the final evidence. The macro-scale Stage-1 engine supports the accounting and diagnostic analyses, while the later target-coupled core supports the Stage-2 and Stage-3 confirmatory experiments.

6.3. PoCol-Specific Implementation in BlockSim

The original PoCol BlockSim extension added collaborative round state, nonce-range assignments, event logging, energy and carbon accounting, and canonical-chain records. This path generated the macro-scale Stage-1 scientific matrix.

For the common-template property, the Stage-2 and Stage-3 engines assume that the participating miners have already converged on the same canonical template for the round. They evaluate the consequences of that invariant but do not implement the full distributed transaction-reconciliation and template-agreement protocol specified in Chapter 5. The target-coupled core records each causally committed physical evaluation and distinguishes total, unique, duplicate, and post-round work. It also tracks active, low-power listen, reserve standby, waking, and activated-reserve residence for wall-clock energy accounting.

Stage-3 added an independent matched same-template PoW control that reuses the same template stream, SHA-256 primitive, target, nonce domain, horizon, rates, powers, seeds, acceptance point, and stale-work cancellation discipline. The intended treatment difference is that matched-PoW miners search the full nonce domain independently, whereas PoCol uses coordinated disjoint assignments.

All implementation layers preserve detailed run identities, configuration records, deterministic event ordering, and machine-checkable output artifacts. The evidence hierarchy and exact source files for each claim are stated explicitly so that each result is tied to its appropriate engine, comparator, and experimental family (Paulavičius et al., 2021; Smetanin et al., 2020).

6.4. Experimental Design

The completed evidence comprises three frozen experiment families. Stage-1 contains 63 experimental configurations × 30 master seeds = 1,890 physical executions, including continuous and idle-policy configurations over miner counts 100-500. Stage-2 contains four PoCol scenarios × 12 fresh paired seeds = 48 runs at 20 miners, a 300-second horizon, a 1,600-nonce domain, and difficulty 1000. Stage-3 contains five scenarios × 12 seeds = 60 runs under the same reduced scale, including

W00 and W01 same-template PoW controls and three PoCol arms. Each family uses its own declared seed registry; cross-stage comparisons are descriptive only.

Parameters were held fixed within each comparison family, not across every experiment in the thesis. Stage-1 fixed an aggregate hash rate of 141 TH/s (nominal hash rate of the BITMAIN Antminer S19 XP; BITMAIN, 2022), an efficiency of 21.5 J/TH (nominal on-wall efficiency of the BITMAIN Antminer S19 XP at 25°C; BITMAIN, 2022), a 10,000-second horizon (the fixed Stage-1 simulation horizon specified in the frozen experimental configuration/InputConfig), a 600-second target interval (Bitcoin's nominal 10-minute target block interval; Nakamoto, 2008), and a 0.42-second propagation delay (the Bitcoin block-propagation delay parameter adopted from the BlockSim baseline; Alharby & van Moorsel, 2019), while varying declared scientific factors including miner count, rate distribution, allocation policy, inactivity, idle-power ratio, and delay. Stages-2 and Stage-3 used the preregistered reduced-scale parameters stated above and heterogeneous integer miner rates. Every claim is tied to its relevant parameter set. Table 6.1 summarizes the fixed parameters and workbook outputs for the macro-scale Stage-1 experiment.

Table (6.1): Fixed parameters and workbook outputs for the macro-scale Stage-1 experiment.

Parameter	Value
Consensus models	Stage-1 PoW/PoCol experimental configurations (B0-C2)
Miner populations	100, 200, 300, 400, and 500 miners
Target block interval	600 s
Block propagation delay	0.42 s
Simulation time	10,000 s
Total network hash rate	141 × 10^12 H/s
Miner efficiency	21.5 J/TH

Within each family, paired scenarios share the same seeds and all non-treatment parameters. This design supports causal interpretation of the declared treatment contrast while preventing results from different engines, scales, or seed registries from being combined as if they were one homogeneous experiment (Paulavičius et al., 2021; Smetanin et al., 2020).

Stage-1 matrix size and replication budget. The 63 Stage-1 experimental configurations are the distinct rows produced by the frozen scenario matrix after applying the declared factor combinations; they are not an independently chosen statistical sample size. The executed counts comprise five configurations each for B0 (independent-template PoW baseline, in which miners search their own block templates), B1 (common-template unpartitioned search, in which all miners start from nonce zero, representing the intentionally worst-case redundant-search condition), and B2 (common-template randomized-start search, in which miners begin from different random nonce positions, reducing but not eliminating overlap); 29 configurations for the shared B3/C1 physical dataset (continuous disjoint-range search with no post-completion idling: B3 represents the coordinated partitioned-search baseline, whereas C1 represents the same physical execution interpreted under PoCol metadata and reward semantics); and 19 configurations for C2 (PoCol disjoint-range search with explicit active-to-idle transitions after a miner completes its assigned local range, used to evaluate idle-state energy effects). With 30 master seeds per configuration, this yields 1,890 physical runs. The 30 seeds provide replicated random realizations and are reused across matched contrasts so that comparisons can be made within the same seed. The value 30 is a planned reproducibility and uncertainty-estimation budget, not a claimed universal minimum or an optimally powered sample size (Paulavičius et al., 2021; Rasheed & AbuSamra, 2026; Smetanin et al., 2020).

Hardware-calibrated hash-rate and efficiency anchor. The aggregate hash rate of 141 TH/s and nominal efficiency of 21.5 J/TH were adopted as a hardware-calibrated reference corresponding to the published specifications of the BITMAIN Antminer S19 XP. They are applied to the simulated network as aggregate quantities, not as the capacity and power of every logical miner. Holding the aggregate rate fixed while varying the miner population from 100 to 500 isolates the effects of miner count, rate distribution, allocation policy, inactivity, and idle residence from the confounding effect of increasing total computational capacity. The implied aggregate power is $P = 141\ \text{TH/s} \times 21.5\ \text{J/TH} = 3031.5\ \text{W}$, which gives 8.420833333 kWh over 10,000 s under continuous full participation. These are nominal on-wall specifications used for model calibration; they are not direct electrical measurements of the simulated runs and do not include site-level cooling or auxiliary losses (BITMAIN, 2022).

Reduced-scale miner population. The Stage-2 and Stage-3 population of 20 logical miners was selected to expose the operating-policy mechanisms at a fully auditable scale. The population comprises 16 active primary miners and four reserve miners, giving a reserve fraction of 0.20 and permitting reserve standby, wake-up, activation, reassignment, and handoff behavior to occur within the same run. This size is large enough to exercise coordination and heterogeneous-rate effects while remaining small enough to retain an evaluation-level ledger. It is a mechanism-test population rather than a claim that 20 nodes represent a production permissionless network.

Reduced-scale time horizon. The 300-second horizon was chosen as a bounded confirmatory window that generates repeated rounds, accepted blocks, range completions, low-power residence, reserve decisions, and handoff opportunities while controlling the size and cost of the physical-evaluation log. Unlike the macro-scale event abstraction, the Stage-2/Stage-3 engine records causally committed SHA-256 evaluations and detailed power-state residence. A substantially longer horizon would increase the evaluation ledger and execution cost across all paired scenarios and seeds. The 300-second window therefore supports controlled policy comparison, but it is not intended to estimate long-duration production availability, daily energy use, or steady-state network performance.

Finite nonce domain and target difficulty. The 1,600-nonce domain was selected to be finite, exhaustible, and completely auditable. With 16 active primary miners, equal initial allocation gives 100 nonce values per primary; with the preregistered batch size of 25, each initial range contains four traceable batches. This structure permits exact verification of disjoint ownership, frontier progress, reassignment, duplicate evaluations, post-round work, and domain exhaustion. Difficulty 1000 sets the per-evaluation success probability to approximately 1/1000 under the implemented target rule. Consequently, a complete 1,600-nonce scan contains an expected 1.6 valid candidates, the approximate probability of at least one valid candidate is 0.798 (computed as one minus the 1,600th power of 1 - 1/1000), and the complementary no-solution probability is approximately 0.202. The paired values therefore generate both successful and exhausted rounds, which are needed to test template refresh and operating-policy behavior. They are simulator-scale parameters and are not numerically comparable with the deployed Bitcoin difficulty or its effective header search space.

Fresh paired seeds in the confirmatory experiments. Each Stage-2 and Stage-3 scenario uses 12 fresh paired seeds. Fresh registries separate the confirmatory experiments from earlier exploratory and macro-scale runs and reduce the risk of reusing random realizations on which policies were developed or interpreted. Pairing means that all arms within an experiment use the same seed-specific random realization, so the statistical analysis is based on within-seed differences rather than unrelated samples. The physical run remains the statistical unit. Twelve pairs constitute a preregistered, resource-bounded confirmatory design rather than a universal minimum; a practical consequence is that the exact sign-permutation analysis can enumerate all 4,096 possible sign assignments (2 raised to the power 12), avoiding a large-sample approximation and permitting a smallest one-sided exact probability of 1/4,096 = 0.000244. No formal prospective power analysis is claimed for the choice of 12.

Interpretive boundary of the parameter choices. Taken together, these settings prioritize mechanism isolation, matched comparison, complete work accounting, and deterministic reproducibility. The hardware-derived values anchor the macro-scale power calculation, whereas the reduced-scale values deliberately make nonce ownership, exhaustion, power-state transitions, and handoffs observable and computationally tractable. The settings must therefore be interpreted as controlled experimental design choices. They support conclusions about the behavior of the modeled PoCol mechanisms under the frozen configurations, but they do not by themselves establish external validity for arbitrary miner populations, modern hardware generations, long-running public networks, or a complete distributed implementation.

The operating policy must be stated explicitly for each PoCol experiment. A complete comparative design distinguishes among: (1) a continuously operating PoW baseline, (2) PoCol under continuous operation, and (3) PoCol under the post-range idle policy. Energy accounting for the idle scenario must separately record active mining time, idle time, active power, idle power, and coordination overhead.

Parallel PoW, Green-PoW, Proof of Team Sprint, Collaborative Proof-of-Work, StrongChain, and APoW are treated as conceptual and architectural comparisons unless their complete algorithms are independently implemented under identical simulation parameters. Numerical superiority must not be claimed from cross-paper results obtained using different hardware, network sizes, workloads, difficulty settings, or energy models.

6.5. Scenario Matrix and Traceability of Executed Runs

Table 6.2 summarizes the three frozen evidence families used in the final statistical evaluation.

No result is interpreted in this setup section. Chapter 7 reports the macro-scale accounting identity, within-PoCol operating-policy outcomes, and the matched same-template PoW comparison separately.

Table (6.2): Frozen experiment families used in the thesis.

Evidence family	Purpose and scenarios	Scale	Analysis
Stage-1 macro-scale	Accounting invariant, duplicate-input scenarios B0–B3/C1, idle policy C2	63 groups × 30 seeds = 1,890 runs; N=100–500; 10,000 s	Seed-matched summaries; cluster bootstrap/permutation; Holm where declared
Stage-2 within-PoCol	S00–S03 operating policies; throughput, useful floor, churn, power-null energy	4 scenarios × 12 paired seeds = 48 runs; 20 miners; 300 s	Exact sign assignments; paired bootstrap; preregistered Holm family
Stage-3 matched comparison	W00/W01 matched PoW and P00–P02 PoCol; energy, service, physical work	5 scenarios × 12 paired seeds = 60 runs; 20 miners; 300 s	Exact sign assignments; paired bootstrap; preregistered Holm family; integrity gates

6.6. Reproducibility Procedure

Reproduction requires checking out the recorded source commit, verifying the relevant checksum manifest, loading the frozen seed registry and scenario matrix, and running the family-specific harness. The analysis scripts consume only the frozen datasets and regenerate decision tables and figure data deterministically. The workbook-based procedure remains applicable only to the Stage-1 path.

Reproducibility therefore depends on preserving code, environment records, seeds, event semantics, raw run records, analysis roots, and transformations. The Stage-2 and Stage-3 packages additionally verify byte-identical regeneration of tables, captions, metadata, and figures (Paulavičius et al., 2021; Smetanin et al., 2020).

6.7. Summary

This chapter documented the macro-scale Stage-1 engine, the target-coupled Stage-2/Stage-3 core, and the matched same-template PoW control. It identified the exact parameters, datasets, logging formats, statistical units, decision rules, and integrity checks associated with each evidence family.

The completed evaluation is not a ten-run descriptive study: it comprises 1,890 Stage-1 executions, 48 Stage-2 runs, and 60 Stage-3 runs, with family-specific frozen seeds and machine-verifiable provenance.

The resulting evidence supports bounded claims about exact-input duplication, power-state energy accounting, block-production retention, matched-control energy and work efficiency, and service latency. It does not constitute a full distributed deployment, hardware measurement, fairness evaluation, or formal security proof.

Chapter 7
Experimental Results and Evaluation

Chapter 7
Experimental Results and Evaluation

7.1. Introduction

This chapter integrates three evidence layers: the macro-scale Stage-1 accounting matrix, the Stage-2 within-PoCol operating-policy experiment, and the Stage-3 matched same-template PoW comparison.

Every result identifies its comparator. Stage-1 compares experimental configurations under fixed aggregate power; Stage-2 compares PoCol operating policies with a no-floor PoCol control and a within-run power-null reference; Stage-3 compares PoCol with population- and active-capacity-matched same-template PoW controls.

7.2. Evaluation Objectives

The evaluation addresses four bounded questions: 1) whether continuous fixed-horizon energy is invariant under matched power; 2) whether disjoint common-template assignments remove exact duplicate inputs; 3) whether low-power operating policies retain useful block production while reducing calculated energy; and 4) how PoCol compares with a matched same-template PoW control in energy, physical work, accepted blocks, and round latency.

Stage-1 spans miner populations of 100–500 over 10,000 seconds. Stages-2 and Stage-3 use 20 miners, a 300-second horizon, a 1600-nonce domain, difficulty 1000, and 12 paired seeds per scenario. The distinct scales and controls answer different questions and are not merged into one pooled comparison.

Table 7.1 provides a consolidated interpretation of the principal findings obtained from the frozen Stage-1 experimental matrix comprising 63 configurations, each executed with 30 master seeds, for a total of 1,890 physical simulation runs. Rather than presenting all raw observations, the table identifies the main quantities investigated, summarizes the corresponding experimental result, and assigns an evidential disposition to each finding. In the Disposition column, ACCOUNTING

IDENTITY identifies a result that follows directly from the defined accounting model and fixed experimental parameters rather than representing an empirical or statistical discovery. DESIGN INVARIANT VERIFIED indicates that the reported property follows from the declared protocol construction and that the simulation confirmed that the implemented path preserved that invariant. ACCOUNTING IDENTITY VERIFIED identifies an accounting relationship that follows from the defined energy model and was additionally verified against the recorded simulation outputs. SUPPORTED WITH TRADE-OFF indicates that the observed relationship is supported by the experimental evidence, while its practical interpretation involves a measurable trade-off between competing operational outcomes rather than an unconditional improvement.

The first and most important result concerns A1, total energy consumption. Across all configurations in which all miners remained continuously active for the entire fixed simulation horizon, the total energy consumption was exactly 8.420833333 kWh, apart from negligible numerical floating-point error. This value follows directly from the fixed aggregate hash rate of 141 TH/s, the mining-device efficiency of 21.5 J/TH, and the 10,000-s simulation duration. These parameters imply an aggregate mining power of 3,031.5 W, and therefore the corresponding fixed-horizon energy consumption is (3031.5 \times 10000 / 3.6\times10^6 = 8.420833333) kWh. Consequently, changing the number of miners or partitioning the nonce-search space does not, by itself, reduce total energy when aggregate computational power and active mining duration remain unchanged. This result is therefore classified as an ACCOUNTING IDENTITY, because it follows directly from the energy-accounting equation, rather than from a statistical difference observed between algorithms.

The H1 duplicate-evaluation result serves primarily as an implementation-integrity validation of the disjoint-evaluation invariant and as a matched comparison against less structured search configurations. At the protocol level, zero exact-input duplication among honest miners follows directly from the combination of one immutable common template and pairwise-disjoint nonce assignments. The observed ordering was B1 > B2 > B3/C1, with the B3/C1 configuration producing zero duplicate serialized candidate-header identities among honest miners. This progression demonstrates the effect of increasingly structured nonce allocation. When miners search overlapping portions of the nonce domain, duplicate evaluations can occur because two or more miners may independently evaluate the same serialized candidate-header input. As the allocation becomes more strongly partitioned, this overlap decreases. Under the fully disjoint B3/C1 allocation, each miner owns a distinct portion of the nonce domain, so honest miners no longer evaluate the same serialized candidate-header input. The zero-duplicate result is therefore the expected operational consequence of enforcing the common-template and disjoint-evaluation invariants, while the simulation confirms that the implemented path preserves this property.

The H1 comparison was not based on isolated simulation runs. The experiment used matched random realizations so that configurations could be compared under the same master seeds. The table reports 150 physical seed-matched pairs arranged within 30 independent seed clusters. Reusing a master seed across a matched contrast means that the compared configurations experience corresponding stochastic realizations wherever the experimental design permits, thereby making the comparison less dependent on unrelated random variation. The statistical comparison remained significant after Holm adjustment, with an adjusted probability on the order of (2e-4). Accordingly, the zero-duplicate B3/C1 outcome is classified as DESIGN INVARIANT VERIFIED, because the property follows from the common-template and pairwise-disjoint assignment rules, while the simulation confirms that the implemented path preserved this invariant. Importantly, however, this result establishes elimination of redundant evaluations, not an automatic reduction in fixed-horizon energy consumption.

The H3 idle-energy identity explains the circumstances under which PoCol can convert its operating behavior into an actual energy reduction. Under the energy-saving idle policy, a miner that completes its assigned nonce range does not necessarily continue consuming full active mining power while waiting for the remainder of the current search process. Instead, its energy contribution during the completed period can be represented using an idle power state. The resulting energy saving is calculated from the accumulated idle duration and the difference between active and idle power.

This identity was verified across all 570 relevant C2 physical runs. The measured energy difference agreed with the energy predicted from idle time and the active-to-idle power difference in every tested run, with a maximum absolute residual of approximately (7.1e-15) kWh and no failures. Such a residual is numerically negligible and is attributable to finite-precision floating-point computation rather than a physically meaningful discrepancy. H3 is therefore classified as ACCOUNTING IDENTITY VERIFIED, because the energy-saving relationship follows directly from the defined power-state accounting model and was confirmed against the recorded simulation outputs across all 570 relevant C2 runs. The importance of this result is that it provides a direct accounting link between the simulator's operational state transitions and measured energy consumption: whenever a miner is moved from the active state to the defined lower-power idle state, the resulting energy reduction can be explained exactly by the reduction in power-time product.

The combined H4/H5 allocation results demonstrate that the way in which nonce ranges are distributed among miners affects completion-time dispersion, idle duration, and consequently the opportunity for energy saving. Under homogeneous mining conditions, where miners have identical computational capacities and receive equal portions of the search domain, their expected range-completion times are aligned. The observed dispersion therefore becomes effectively zero, and no miner systematically finishes substantially earlier than the others. In this case, there is no meaningful idle period created by unequal completion times, and energy remains at the fixed continuous-participation anchor of 8.420833333 kWh.

The situation changes when miners are heterogeneous. If miners with different hash rates are nevertheless assigned equal-sized nonce ranges, faster miners can complete their assigned ranges earlier than slower miners. The resulting difference in completion times creates finish-time dispersion and may produce periods during which completed miners remain idle. Under the energy-saving idle policy, those periods reduce raw energy consumption because completed miners are no longer consuming full active mining power.

This H4/H5 finding reveals an important operational trade-off. Better load balancing improves synchronization of range completion and reduces unequal waiting behavior, but the same balancing can eliminate the idle periods that were responsible for an apparent reduction in raw energy. Therefore, an allocation policy cannot be evaluated solely by asking whether it minimizes energy. It must also be considered in terms of balanced work distribution, utilization, search progress, and block-production behavior. For this reason, the table classifies H4/H5 as SUPPORTED WITH TRADE-OFF: the underlying allocation effect is clearly supported, but its practical interpretation involves competing operational objectives rather than a simple monotonic improvement.

Accordingly, the Stage-1 evidence should be interpreted as establishing three distinct but related layers: redundancy elimination, operational allocation behavior, and energy accounting. The first is directly supported by the zero-duplicate B3/C1 result; the second explains how allocation, heterogeneity, idle states, participation, and search-space management affect system behavior; and the third establishes that energy reduction occurs only when these mechanisms reduce the active power-time term in the energy equation.

Table (7.1): Stage-1 summary of the frozen 1,890-run evaluation (replacing the obsolete PoW-versus-PoCol energy/carbon summary). Total fixed-horizon energy is invariant across the compared continuous configurations; energy reductions are idle- or participation-driven only. B3 and C1 are one physical dataset.

Id / quantity	Result (Stage-1)	Disposition
A1 energy	Total fixed-horizon energy = 8.420833333 kWh for all continuous full-participation configs and all N (max rel. dev. 3.6e-16); partitioning alone does not reduce energy	ACCOUNTING IDENTITY
H1 duplicate rate	B1 > B2 > B3/C1; B3/C1 = 0 duplicate identities; 150 physical seed-matched pairs; 30 independent seed clusters; Holm p ~ 2e-4	DESIGN INVARIANT VERIFIED
H3 idle identity	saving = Sum idle_time*(P_active-P_idle)/3.6e6 verified for 570/570 C2 runs; max abs residual 7.1e-15 kWh; 0 failures	ACCOUNTING IDENTITY VERIFIED
H4/H5 allocation	homogeneous+equal: dispersion 0, idle 0; weighted removes dispersion and idle and returns C2 energy to the anchor	SUPPORTED WITH TRADE-OFF

7.3. Energy Consumption Results

7.3.1. Stage-1 Energy and Work-Efficiency Results

7.3.1.1. Results figures (Stage-1)

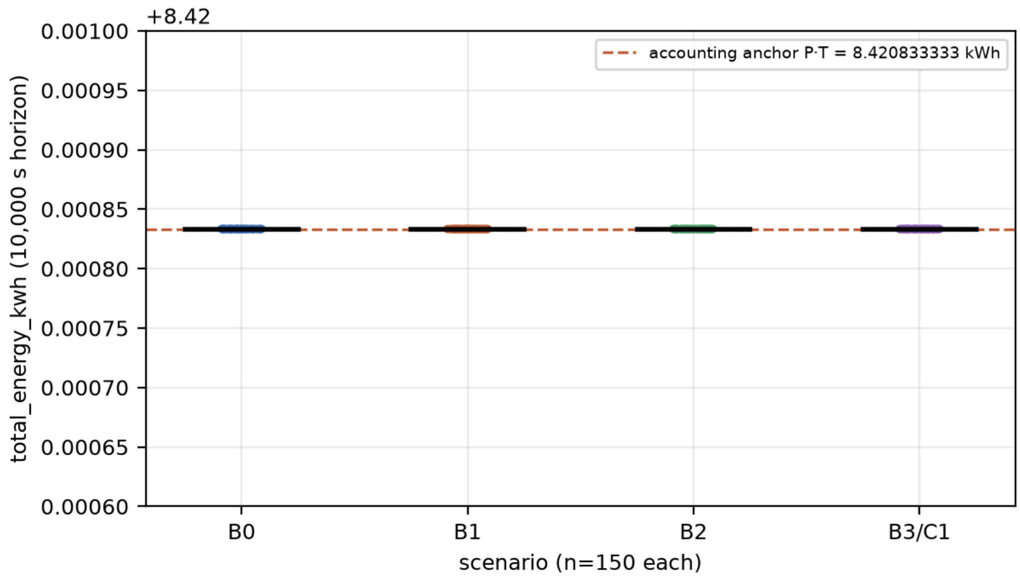

Figure (7.1): A1 accounting invariant - total fixed-horizon energy is 8.420833333 kWh for every continuous full-participation configuration and every miner count.

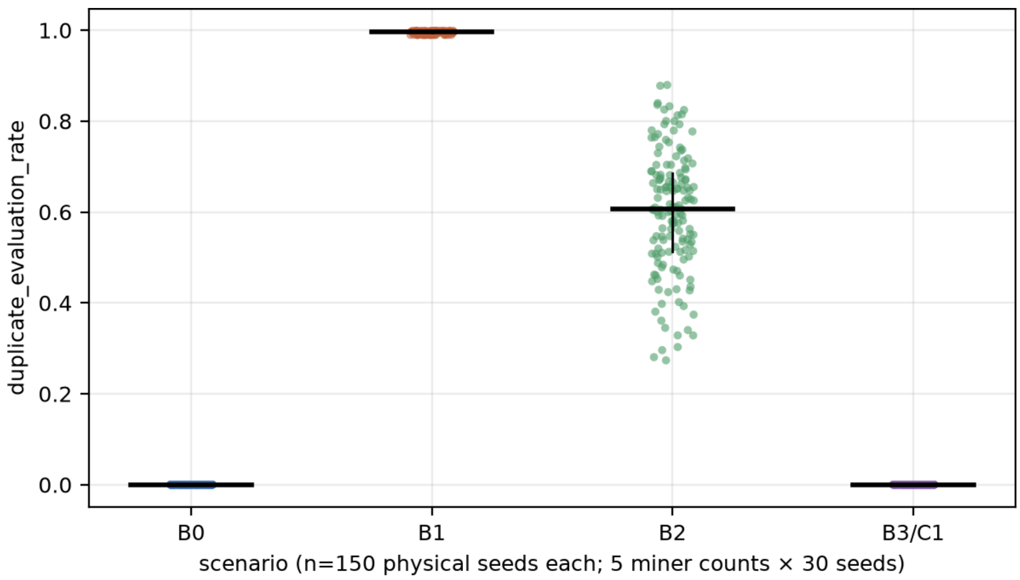

Figure (7.2): H1 duplicate candidate-header evaluation rate by scenario, ordering B1 > B2 > B3/C1 under the modeled immutable-common-template and disjoint-range assumptions (150 physical seed-matched pairs; uncertainty from 30 independent seed clusters).

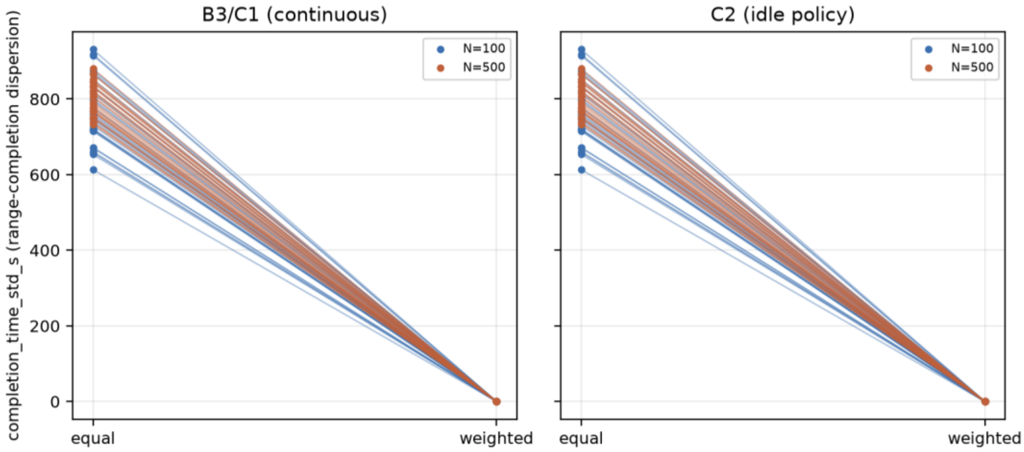

Figure (7.3): H5 heterogeneous-rate completion-time dispersion - hash-rate-weighted ranges remove modeled range-completion dispersion (seed-matched pairs).

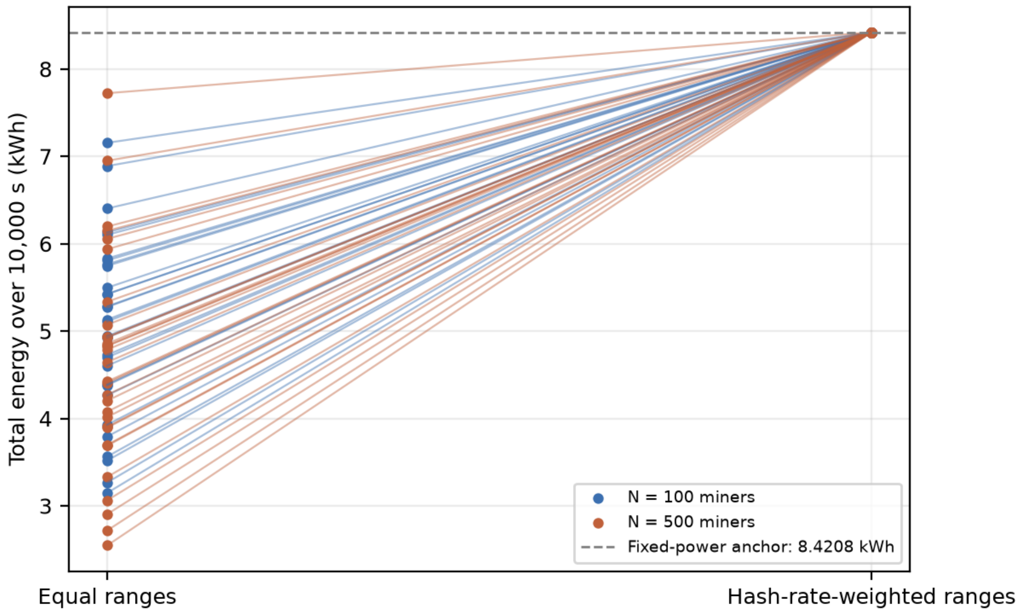

Figure (7.4): H5 C2 idle-energy trade-off under heterogeneous hash rates: weighted ranges remove modeled idle and return energy to the fixed-power anchor. Each line connects seed-matched runs; equal-size ranges create idle opportunity under heterogeneity; weighted ranges reduce completion-time dispersion and remove that idle opportunity - a completion-balance versus idle-energy trade-off, not a reward or incentive result.

7.3.2. Energy–performance trade-offs

A simultaneous energy-and-service claim must be evaluated against its preregistered gates. Stage-2 retained 90.39% of the no-floor PoCol block production and achieved a 55.05% energy reduction relative to the within-run power-null reference; however, the preregistered useful-floor and churn criteria were not both met under this configuration. Stage-3 achieved a 44.95% energy reduction relative to W01; however, the preregistered accepted-block, median-duration, and useful-floor criteria were not simultaneously met under this configuration. The correct interpretation is therefore a measured energy-service trade-off rather than joint operational superiority.

The energy-saving and continuous policies represent different operating objectives. Stage-1 verifies the per-miner idle-energy identity and the fixed-power anchor. Stage-2 and Stage-3 extend this accounting to reserve standby, waking, reassigned work, and matched controls. Energy, accepted blocks, round duration, and coordination churn are reported together so that lower energy caused by reduced useful service is not misclassified as efficiency.

A. Work and integrity properties.

Under the honest common-template configuration PoCol performs no duplicate physical evaluations, whereas the uncoordinated matched PoW control duplicates most of its work. All deterministic integrity gates passed in every one of the 60 Stage-3 runs (zero duplicate nonces, zero post-round evaluations, zero physical-frontier rewinds, and zero nonterminal handoff, activation, lease or reassignment records).

Table (7.2): Physical-evaluation efficiency (means over 12 seeds)

Scenario	Total evaluations	Unique evaluations	Duplicate evaluations	Blocks per million evaluations
PoW Control (Population-Matched)	1,447,489	219,374	1,228,115	172.4
PoW Control (Active-Capacity-Matched)	1,158,562	216,190	942,371	208.7
PoCol (No Floor)	152,980	152,980	0	1,166.1
PoCol (Coarse Reassignment)	140,645	140,645	0	1,172.2
PoCol (Single Handoff)	139,712	139,712	0	1,173.2

Note: The PoW controls in this table are matched same-template experimental controls. They use the same template stream, SHA-256 primitive, target, nonce domain, horizon, rates, powers, seeds, acceptance point, and stale-work cancellation discipline as the corresponding PoCol arms; they are not intended to reproduce the full candidate-generation behavior of deployed Bitcoin mining.

B. Best within-PoCol energy/throughput result (Stage-2).

PoCol retained 90.39% of the accepted-block output of the no-floor PoCol control while producing a mean calculated low-power-state energy reduction of 55.05%.

Block production is relative to the no-floor PoCol control (S00). Energy reduction is relative to the within-run power-null reference on the same event path (95% CI 54.95%-55.15%).

Table (7.3): Stage-2 within-PoCol results (means over 12 seeds). S03 is the best throughput-oriented policy. Source: Author's simulation results.

Scenario	Accepted blocks	Closed rounds	Rel. energy reduction	Below useful floor (s)	Reassign./ round
S00 (no floor)	177.9	223.8	0.5383	0.00	0.0000
S01 (static floor)	150.2	189.8	0.5722	42.25	0.0000
S02 (useful floor)	150.2	189.8	0.5722	42.25	0.0000
S03 (coarse reassign.)	160.8	203.7	0.5505	28.71	1.7482

Note: "Rel. energy reduction" is calculated independently for each scenario relative to its own within-run power-null reference; therefore, S00 is not the zero-reference point for this column.

Table (7.4): Stage-2 S03 energy attribution relative to the within-run power-null. Positive entries are avoided energy; activated-reserve hashing is an added cost. Reserve and wake components are reported separately from range-idle. Source: Author's simulation results.

Component	Energy attribution relative to power-null (kWh)
Primary range-idle saving	0.004421
Reserve-standby saving	0.004724
Wake-transient saving	0.000863
Activated-reserve hashing cost	0.000190

C. Matched same-template PoW versus PoCol comparison (Stage-3).

Relative to the active-capacity-matched same-template PoW control, PoCol used 44.95% less total energy and approximately 19.3% less energy per accepted block, with approximately 87.9% fewer physical evaluations and zero exact-input duplicates. PoCol produced 67.70% of the control's accepted blocks and had a 5.219-times larger median round duration. Accordingly, the evaluated single-handoff configuration supports an energy-service trade-off interpretation rather than a joint superiority claim across energy and service metrics.

7.4. Summary

This chapter presented the three frozen evidence layers without merging their comparators. Stage 1 established the 8.420833333 kWh fixed-horizon invariant for continuous full-participation configurations and the exact-input-duplication effect. Stage-2 retained 90.39% of the no-floor PoCol control's accepted-block output while achieving a 55.05% calculated energy reduction relative to the within-run power-null reference, although the preregistered useful-floor and churn criteria were not both met under this configuration. Stage-3 used 44.95% less total modeled energy than W01 and substantially fewer physical evaluations, while producing 67.70% of W01's accepted blocks at a 5.219-times larger median round duration, thereby demonstrating an energy-service trade-off under the evaluated configuration.

Chapter 8
Conclusion and Future Work

Chapter 8
Conclusion and Future Work

8.1. Summary of Research Work

This thesis developed Proof-of-Collaboration (PoCol), a coordinated proof-of-work protocol in which honest miners derive one immutable template and evaluate disjoint nonce ranges. The design separates two questions that were often conflated in earlier drafts: whether coordination eliminates exact duplicate candidate inputs, and whether an operating policy creates real residence in lower-power states. The first is a work-integrity property; the second is an energy-accounting and service trade-off.

The research combined a structured comparative review, a PoCol protocol and threat specification, a common-template coordination design, an extended BlockSim evaluation framework, a macro-scale accounting matrix, and two preregistered reduced-scale experiments. The implemented evidence covers the collaborative-search and operating-policy abstraction; it does not include an end-to-end distributed implementation of transaction-set reconciliation and template agreement.

The final outcomes are deliberately conditional. Stage-1 shows that continuous fixed-horizon energy is invariant at 8.420833333 kWh under matched aggregate power, so nonce partitioning alone does not save energy. Stage-2 showed that the best throughput-oriented PoCol policy retained 90.39% of the no-floor PoCol control's accepted-block output while achieving a 55.05% calculated energy reduction relative to the within-run power-null reference, although the preregistered useful-floor and churn criteria were not both met under this configuration. Stage-3 showed that the single-handoff PoCol policy used 44.95% less total modeled energy and approximately 19.3% less energy per accepted block than the active-capacity-matched same-template PoW control (W01), with approximately 87.9% fewer physical evaluations and zero exact-input duplicates. Under the same comparison, PoCol produced 67.70% of W01's accepted blocks and had a 5.219-times larger median round duration, demonstrating an energy-service trade-off under the evaluated configuration.

8.2. Revisited Research Questions and How They Were Answered

Table 8.1 answers the four refined research questions using the frozen evidence and keeps every answer bounded by its comparator, experimental scale, and unmet criteria.

The refined research questions (RQ1–RQ4) are answered by the frozen evidence as follows.

Table (8.1): Research-question answer matrix. Source: Author's synthesis of the frozen simulation evidence.

RQ	Answer (bounded)
RQ1	Yes, conditionally. Under one common immutable template and honest pairwise-disjoint nonce assignments, exact duplicate serialized candidate-header inputs are eliminated by construction. The Stage-1 and Stage-3 simulations verified that the implemented paths preserved this invariant, including zero duplicate physical evaluations in the relevant PoCol configurations.
RQ2	Substantial - a 55.05% calculated low-power-state energy reduction relative to the within-run power-null (Stage-2), attributable to range-idle, reserve-standby, and wake-transient savings, net of activated-reserve hashing cost; partitioning alone does not reduce energy.
RQ3	The best throughput-oriented policy retained 90.39% of the no-floor PoCol control's accepted-block output under the evaluated Stage-2 configuration.
RQ4	PoCol used 44.95% less total modeled energy and approximately 19.3% less energy per accepted block, with approximately 87.9% fewer physical evaluations than the active-capacity-matched same-template PoW control. Under the same comparison, PoCol produced 67.70% of the control's accepted blocks and had a 5.219-times larger median round duration, demonstrating an energy-service trade-off under the evaluated configuration.

8.3. Contributions Revisited

The contributions are conceptual, methodological, and empirical. Conceptually, PoCol integrates immutable-template agreement, deterministic disjoint nonce ownership, collaborative reward rules, and explicit post-range operating policies. The contribution is coordinated exact-input evaluation and its measurable trade-offs.

Methodologically, the thesis provides a sustainability-aware discrete-event framework with wall-clock energy accounting, physical-evaluation ledgers, power-state residency, deterministic provenance, paired seed registries, preregistered decision rules, and matched controls. It also documents the boundaries of the simulator abstractions used for direct comparison.

8.4. Recommendations for Practitioners

Empirically, the thesis establishes four bounded findings: exact duplicate candidate-header inputs are zero under honest common-template disjoint execution;

continuous fixed-power energy is invariant; explicit low-power policies can reduce modeled energy while retaining substantial but not complete block production; and the matched same-template PoW control closes rounds faster and produces more blocks while performing substantially more physical evaluations.

8.5. Future Research Directions

The next step for future study is the comparison of PoCol's performance with that of other consensus classes besides proof-of-work (PoW), for example, with Proof-of-stake (PoS), Practical Byzantine Fault Tolerance (PBFT), Proof-of-Authority (PoA) or Hybrid Protocols. The present work already includes an initial conceptual placement of PoCol against these other classes, but additional empirical benchmarking will provide greater support for where collaborative PoW is most competitive, and where alternative consensus forms (e.g., non-PoW) continue to be superior (Castro & Liskov, 1999; Ge et al., 2022). Specifically, future studies should compare throughput and energy, as well as governance assumptions, security thresholds, and operating costs.

A second priority is a complete permissionless identity and contribution-verification model. Range membership proves only that the winning nonce belongs to the proposing miner; it does not show that non-winning miners searched their assigned ranges. Future work should therefore evaluate auditable work evidence, Sybil-cost mechanisms, and incentive-compatible reward functions jointly, including the possibility of identity splitting, free riding, strategic idling, and block withholding.

Finally, PoCol should be compared empirically with additional consensus families only through explicitly matched implementations and declared measurement boundaries. Cross-paper comparisons can motivate hypotheses, but numerical superiority over PoS, PBFT, PoA, or other protocols should not be claimed unless workload, hardware, network, security assumptions, and service denominators are made comparable.

Formal analysis of PoCol's common-prefix, chain-growth, chain-quality, and confirmation-security properties under its coordinated nonce-assignment and common-template model remains an important direction for future research.

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